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1 PURPOSE AND SCOPE

1.1 SCOPE OF DOCUMENT

An IPTV (Internet Protocol Television) system installed in an FPSO vessel is a digital television distribution system that delivers live TV channels, recorded content, and information services over the vessel's IP network infrastructure. Unlike traditional coaxial TV distribution, IPTV uses the onboard Ethernet/LAN network to transmit video streams from a centralized headend to multiple user endpoints such as crew cabins, control rooms, and offices.

This Installation, Operation, and Maintenance (IOM) Manual provides guidelines for the safe installation, operation, and maintenance of the IP TV Radio Communication System installed onboard. The document is intended for use by commissioning engineers, operations personnel, and maintenance technicians.

1.2 SYSTEM DESCRIPTION

The system consists of the following major equipment components:

ITEM	EQUIPMENT DESCRIPTION	MAKE	MODEL	NOS
01	IPTV / TVRO ANTENNA	SEATEL	120TV	01
02	ANTENNA CONTROLLER	SEATEL	LMXP	01
03	IPTV SET-TOP BOX	CLEARTV	CLEAR H98MUV3	17
04	24P NTEWORK SWITCH	HUAWEI	S5735-L24T4S-A-V	01
05	24P NTEWORK SWITCH	CISCO	C1300-24FP-4X	01
06	COAXIAL MULTI SWITCH	TRIAX	TMP5x24	01
07	IPTV SERVER WITH VOD	DELL	PowerEdge R740	01
08	8 WAY ENCODER	CLEAR	H3118AV (8-WAY)	01
09	4 WAY ENCODER	CLEAR	H3114AV (4-WAY)	01
10	KVM	ATEN	CL5708	01
11	SMART TV	PHILIPS	32HFL3014	17
13	TV OUTLET SOCKET	COMMSCOPE	2153449-4	57

Each of the main components of the system is described in more detail below.

1.3 EQUIPMENT DESCRIPTION

1.3.1 IPTV / TVRO ANTENNA SEATEL 120TV

The SEATEL 120TV is a high-performance satellite TV reception antenna specifically designed for marine and offshore environments, providing reliable reception of television broadcast signals for the vessel's IPTV / TVRO (Television Receive Only) system. Constructed to withstand harsh offshore conditions, this stabilized TV antenna ensures consistent and high-quality satellite signal acquisition even with vessel motion and sea state variations.

The antenna features a precision-engineered reflector and feed assembly optimized for reception in the C-band and Ku-band satellite frequencies typically used for direct-to-home (DTH) television transmissions as well as marine broadcast services. Its robust mechanical design integrates a stabilized platform that compensates for roll and pitch, maintaining accurate satellite alignment and reducing signal loss due to vessel movement.

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The SEATEL 120TV operates in conjunction with a TVRO receiver and the onboard IPTV headend, delivering satellite television channels to the IPTV distribution system. It connects via coaxial feed cables (50 Ω) to the receiver demodulators, which decode the incoming RF satellite signals into digital video streams. These streams are then encoded and distributed over the FPSO’s IP network to multiple endpoints such as crew cabins, lounges, and common areas.

IPTV / TVRO ANTENNA SEATEL 120TV installed at the following location:

- COMPASS DECK STARBOAR

1.3.2 ANTENNA CONTROLLER SEATEL LMXP

The Sea Tel LMXP Antenna Controller is installed as part of the onboard satellite TV system supporting the vessel’s IPTV distribution network. In this configuration, the controller manages the stabilized satellite antenna that receives television broadcast signals, which are then distributed internally through the IPTV system.

The LMXP continuously controls antenna positioning by processing vessel motion data such as heading, roll, and pitch to maintain precise alignment with the selected broadcast satellite. This ensures uninterrupted satellite signal reception even during vessel movement and rough sea conditions. Once the satellite TV signal is received, it is passed to the TVRO and IPTV headend equipment for decoding and distribution across the ship’s internal network to cabins and common areas.

The controller provides system monitoring, satellite selection, tracking status, and alarm functions. It also allows configuration and diagnostic access to maintain stable performance. By ensuring accurate satellite tracking and consistent RF signal quality, the Sea Tel LMXP plays a critical role in maintaining reliable IPTV service onboard.

ANTENNA CONTROLLER SEATEL LMXP installed at the following location:

- A-DECK TELECOM ROOM IPTV SYSTEM CABINET (86-RX-009)

1.3.3 IPTV SET-TOP BOX CLEARTV CLEAR H98MUV3

The Clear TV Clear H98MUV3 IPTV Set-Top Box is a digital receiver used onboard to decode and display television content distributed through the vessel’s IPTV network. It functions as the end-user interface between the ship’s internal IP television system and the television display units installed in cabins or common areas.

The Clear H98MUV3 receives compressed digital video streams transmitted over the vessel’s LAN network from the IPTV headend system. It processes and decodes the incoming IP-based television signals and converts them into HDMI audio-visual output compatible with standard television screens. The unit supports high-definition video playback and provides user navigation through channel selection menus, program guides, and system settings.

Designed for continuous operation, the set-top box connects to the onboard network via Ethernet and is powered locally at each installation point. It ensures stable reception of live TV broadcasts and other IPTV services distributed from the satellite TVRO system and IPTV server infrastructure. Within the ship’s communication system, the Clear H98MUV3 acts as the final delivery component that enables reliable television viewing for crew and operational areas.

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IPTV SET-TOP BOX CLEARTV CLEAR H98MUV3 installed at the following location:

- A-DECK MEETING ROOM (86-RE-201)
- A-DECK MESS ROOM (86-RE-202)
- A-DECK RECREATION ROOM (86-RE-203)
- RAISED DECK DOCTOR'S OFFICE (86-RE-204)
- RAISED DECK GYM ROOM (86-RE-205)
- RAISED DECK HOSPITAL (86-RE-206)
- C-DECK MAINTENANCE SUPERINTENDENT OFFICE (86-RE-207)
- C-DECK MEETING ROOM (86-RE-241)
- C-DECK VIP-P1 (86-RE-243)
- C-DECK NUPRC REP BEDROOM (86-RE-244)
- C-DECK GYM (86-RE-246)
- C-DECK VIP-S1 (86-RE-255)
- C-DECK SAFETY OFFICER (86-RE-256)
- C-DECK OIM BEDROOM (86-RE-257)
- C-DECK PRODUCTION SUPERVISOR BEDROOM (86-RE-258)
- NAVIGATION DECK HELI BRIEFING ROOM (86-RE-247)

1.3.4 24P NTEWORK SWITCH HUAWEI S5735-L24T4S-A-V

The Huawei S5735-L24T4S-A-V is a managed Gigabit Ethernet access switch designed for enterprise and industrial network environments. It provides twenty-four 10/100/1000Base-T Ethernet ports for device connectivity along with four Gigabit SFP uplink ports for fibre backbone integration. This model supports Layer 2 switching functions with basic Layer 3 routing capability, making it suitable for structured LAN architectures where traffic segmentation and controlled data distribution are required.

In an FPSO or offshore installation, the switch typically serves as an access layer device within the communication rack. It distributes IP-based services such as IPTV, VoIP, CCTV, automation systems, and general data connectivity to various user endpoints. The SFP uplink ports are generally connected to a fibre backbone or core switch, enabling stable high-bandwidth communication between different decks, control rooms, or E-houses.

The switch supports VLAN configuration, Quality of Service (QoS), Access Control Lists (ACL), and port security features, allowing network segmentation and traffic prioritization according to operational requirements. It also supports IPv4 and IPv6 dual stack operation and can be managed through a web interface, command line interface (CLI), or SNMP-based network management systems.

When installed inside a protected rack cabinet within a controlled communication room, the Huawei S5735-L24T4S-A-V provides reliable and stable performance suitable for offshore network infrastructure where continuous data availability is essential.

24P NTEWORK SWITCH HUAWEI S5735-L24T4S-A-V installed at the following location:

- A-DECK TELECOM ROOM IPTV SYSTEM CABINET (86-RX-009)

1.3.5 24P NTEWORK SWITCH CISCO C1300-24FP-4X

The Cisco C1300-24FP-4X is a managed Gigabit Ethernet access switch designed for reliable enterprise and industrial network applications. It provides 24 Gigabit Ethernet ports with full Power over Ethernet (PoE) capability, allowing connected devices such as IP phones, CCTV cameras, wireless access points, and IPTV receivers to receive both data

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and power through a single Ethernet cable. The switch also includes four 10-Gigabit SFP+ uplink ports for high-speed fibre connection to a core switch or backbone network.

The unit supports Layer 2 switching features with basic Layer 3 functionality, including VLAN configuration, Quality of Service (QoS), and multicast handling. In offshore or FPSO installations, it is typically installed in a communication rack to distribute IP services across cabins, control rooms, and operational areas while maintaining secure and stable network performance.

24P NTEWORK SWITCH CISCO C1300-24FP-4X installed at the following location:

- A-DECK TELECOM ROOM IPTV SYSTEM CABINET (86-RX-009)

1.3.6 COAXIAL MULTI SWITCH TRIAX TMP5X24

The TRIAX TMP 5×24 Coaxial Multi-Switch is an RF signal distribution component designed for satellite and terrestrial signal distribution systems, providing 5 input feeds and 24 subscriber outputs with integrated power supply functionality. It supports combined satellite intermediate frequency (SAT-IF) and terrestrial (TER) signal distribution across a wide frequency range typically from 47 MHz to 2150 MHz (terrestrial and satellite bands respectively), enabling efficient distribution of broadcast signals to multiple receivers. The unit features a compact, installer-friendly design with colour-coded F-connector input and output ports and adjustable gain and slope controls to compensate for cable losses and varying subscriber line lengths. It provides low insertion loss, high isolation, and return loss performance, ensuring minimal signal degradation between inputs and outputs, and supports standard satellite switching commands such as 13 V/18 V and 22 kHz tone control for LNB polarity switching. The Mult switch includes an internal switch-mode power supply to feed LNBS where required, and LED indicators for power status, making it suitable for medium- to large-scale RF distribution installations in IPTV, MATV, SMATV, or multi-user satellite systems.

COAXIAL MULTI SWITCH TRIAX TMP5X24 installed at the following location:

- A-DECK TELECOM ROOM IPTV SYSTEM CABINET (86-RX-009)

1.3.7 IPTV SERVER DELL POWEREDGE R740

The Dell PowerEdge R740 is a versatile 2U enterprise rack server engineered to handle compute-intensive and scalable workloads including virtualization, VDI, cloud infrastructure, and database applications. It supports up to two Intel Xeon Scalable processors and large DDR4 memory capacity, enabling high core density and fast data processing. The server offers flexible storage configurations with support for SAS, SATA, and NVMe drives, allowing optimization for performance or capacity-focused deployments.

Designed for modern data centres, the R740 includes multiple PCIe expansion slots for GPU or accelerator support, making it suitable for AI inference and high-performance computing scenarios. Integrated iDRAC9 provides advanced remote monitoring, automation, and lifecycle management, while redundant power supplies and hot-swappable components ensure high availability and reliability. Its balance of scalability, performance, and manageability makes it ideal for both enterprise IT environments and edge deployments.

IPTV SERVER installed at the following location:

- A-DECK TELECOM ROOM IPTV SYSTEM CABINET (86-RX-009)

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1.3.8 CLEAR ENCODER 4 WAY H3114AV & 8 WAY H3118AV

The Clear is a multi-channel audio/video encoder designed for IPTV headend applications. The unit supports up to eight independent AV input channels and converts analog or HDMI video sources into compressed digital IP streams suitable for distribution over an Ethernet network.

Within an FPSO IPTV system, the encoder is typically installed in the communication rack as part of the IPTV headend. It receives video inputs from satellite receivers, media players, or other broadcast sources and encodes them into MPEG transport streams. These streams are then transmitted via multicast over the LAN to the IPTV server or directly to network switches for distribution to set-top boxes and smart TVs.

The H3118AV supports standard compression formats such as H.264 and generates stable, synchronized IP streams optimized for bandwidth efficiency. Each channel operates independently, allowing multiple television programs or video sources to be encoded simultaneously.

This encoder forms a critical interface between traditional AV broadcast equipment and the IP-based distribution network, enabling centralized and scalable television services onboard the vessel.

CLEAR ENCODER 4 WAY H3114AV & 8 WAY H3118AV installed at the following location:

- A-DECK TELECOM ROOM IPTV SYSTEM CABINET (86-RX-009)

1.3.9 KVM ATEN CL5708

The ATEN CL5708 is an integrated LCD KVM (Keyboard, Video, Mouse) console designed to provide centralized control of multiple servers from a single rack-mounted unit. It combines a 17-inch LCD monitor, keyboard, touchpad, and an 8-port KVM switch within a 1U slide-out rack housing, making it suitable for communication and IT cabinets where space optimization is essential.

The CL5708 allows an operator to access and manage up to eight connected servers or network devices directly from the front panel console. Each connected device is interfaced using dedicated KVM cables such as the ATEN 2L-5201U, which carry video, keyboard, and mouse signals between the console and the target equipment.

In an FPSO or offshore installation, the KVM console is typically installed in the main communication rack to provide local access to IPTV servers, network switches, firewalls, or other IT equipment. This setup eliminates the need for separate monitors and keyboards for each device, enabling efficient maintenance, configuration, and troubleshooting from a single control point.

The ATEN CL5708 supports high-resolution video output and secure switching between connected systems, ensuring reliable and organized management of critical onboard network infrastructure.

KVM ATEN CL5708 installed at the following location:

- A-DECK TELECOM ROOM IPTV SYSTEM CABINET (86-RX-009)

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1.3.10 SMART TV PHILIPS 32HFL3014

The Philips 32HFL3014 is a 32-inch hospitality-grade LED Smart television designed for commercial and marine accommodation environments. This model is specifically built for controlled installations such as hotels, offshore vessels, and FPSO living quarters where centralized management and reliable operation are required.

The television supports Full HD resolution and is equipped with multiple input interfaces including HDMI, USB, and RF antenna input, allowing connection to IPTV set-top boxes, coaxial TV systems, or media devices. In an FPSO IPTV installation, the Philips 32HFL3014 typically operates in conjunction with the onboard IPTV server and set-top box, receiving IP-based television streams through the Ethernet network and displaying live TV channels or video-on-demand content.

As a hospitality model, the unit includes features such as volume limitation, channel control, cloning capability for bulk configuration, and restricted user settings to prevent unauthorized changes. This makes it suitable for crew cabins where standardized configuration and ease of maintenance are important.

The Philips 32HFL3014 forms the end-user display component of the IPTV distribution system, providing reliable television service for crew welfare and onboard information broadcasting.

IPTV SET-TOP BOX CLEARTV CLEAR H98MUV3 installed at the following location:

- A-DECK MEETING ROOM (86-RE-301)
- A-DECK MESS ROOM (86-RE-302)
- A-DECK RECREATION ROOM (86-RE-303)
- RAISED DECK DOCTOR'S OFFICE (86-RE-304)
- RAISED DECK GYM ROOM (86-RE-305)
- RAISED DECK HOSPITAL (86-RE-306)
- C-DECK MAINTENANCE SUPERINTENDENT OFFICE (86-RE-307)
- C-DECK MEETING ROOM (86-RE-341)
- C-DECK VIP-P1 (86-RE-343)
- C-DECK NUPRC REP BEDROOM (86-RE-344)
- C-DECK GYM (86-RE-346)
- C-DECK VIP-S1 (86-RE-355)
- C-DECK SAFETY OFFICER (86-RE-356)
- C-DECK OIM BEDROOM (86-RE-357)
- C-DECK PRODUCTION SUPERVISOR BEDROOM (86-RE-358)
- NAVIGATION DECK HELI BRIEFING ROOM (86-RE-347)

1.3.11 TV OUTLET SOCKET COMMScope 2153449-4

The CommScope 2153449-4 TV outlet socket is a coaxial wall outlet designed for television signal termination in structured cabling systems. It serves as the final connection point between the coaxial distribution network and the television or set-top box installed in cabins, lounges, or common areas.

This outlet is typically used in SMATV or TVRO-based systems where RF television signals are distributed via coaxial cable from a centralized headend or satellite receiver. The socket provides a secure and impedance-matched connection, ensuring minimal signal loss and maintaining proper signal quality at the user end.

In an FPSO or offshore installation, the TV outlet socket is mounted in accommodation areas and connected to the vessel's coaxial backbone. It ensures organized termination of RF cables within the wall panel and supports reliable television reception as part of the overall IPTV or conventional TV distribution system.

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TV OUTLET SOCKET COMMSCOPE 2153449-4 installed at the following location:

- A-DECK MEETING ROOM (86-RE-101)
- A-DECK MESS ROOM (86-RE-102)
- A-DECK RECREATION ROOM (86-RE-103)
- RAISED DECK DOCTOR'S OFFICE (86-RE-104)
- RAISED DECK GYM ROOM (86-RE-105)
- RAISED DECK HOSPITAL (86-RE-106)
- C-DECK MAINTENANCE SUPERINTENDENT OFFICE (86-RE-107)
- C-DECK MEETING ROOM (86-RE-141)
- C-DECK VIP-P1 (86-RE-143)
- C-DECK NUPRC REP BEDROOM (86-RE-144)
- C-DECK GYM (86-RE-146)
- C-DECK VIP-S1 (86-RE-155)
- C-DECK SAFETY OFFICER (86-RE-156)
- C-DECK OIM BEDROOM (86-RE-157)
- C-DECK PRODUCTION SUPERVISOR BEDROOM (86-RE-158)

2 REFERENCE DRAWINGS

The following reference drawings shall be consulted prior to installation, operation, or maintenance of the equipment.

No	Document No.	Rev	Title or Description
1	WCPOKWTP-450-TE-(VAE)BF-010	Z	Telecom Cable Block Diagram - IPTV System
2	WCPOKWTP-450-TE-(VAE)LY-030	0	Telecom Cable Tray Layout - IPTV SYSTEM
3	WCPOKWTP-450-TE-(VAE)LY-011	Z	Telecom Device Location Layout Drawing- IPTV System
4	WCPOKWTP-450-TE-(VAE)GA-004	C	Telecom GA Drawing - IPTV System
5	WCPOKWTP-450-TE-(VAE)DG-004	C	Telecom Internal Wiring Drawing- IPTV System
6	WCPOKWTP-450-TE-(VAE)DG-032	0	Telecom Termination Drawing - IPTV System
7	WCPOKWTP-450-TE-(VAE)DG-014	0	Telecom Typical Installation Drawing - IPTV System

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3 MANUFACTURERS' MANUALS

This section includes the manufacturers' manuals for the installation, operation, and maintenance of the equipment.

3.1 INSTALLATION INSTRUCTIONS

3.1.1 IPTV / TVRO ANTENNA SEATEL 120TV

The Sea Tel 120 TVHD is a maritime stabilized satellite TVRO antenna system designed to provide reliable satellite television reception onboard vessels. The system features a fully enclosed radome antenna with advanced three-axis stabilization, enabling continuous satellite tracking and stable signal reception even under vessel motion and rough sea conditions. It supports DVB-S and DVB-S2 broadcasts and integrates an automatic satellite acquisition and tracking system for uninterrupted TV service.

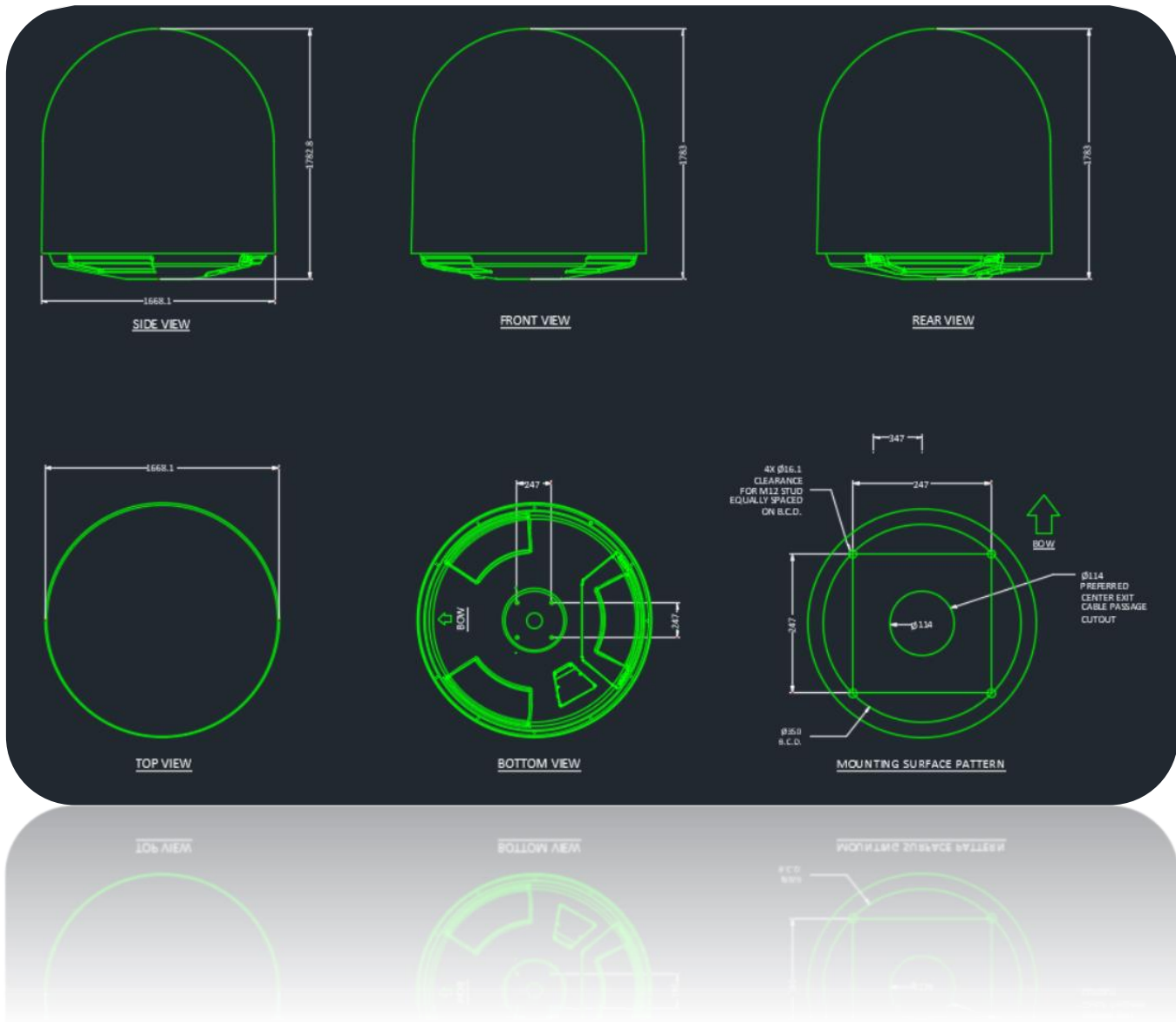
The antenna is engineered for marine environments, offering high pointing accuracy and robust mechanical construction suitable for continuous operation at sea. With its integrated control and stabilization technology, the Sea Tel 120 TVHD ensures dependable onboard entertainment by maintaining consistent satellite lock and high-quality signal reception across global coverage areas.



SPECIFICATIONS	
Min. EIRP Ku-Band	41 dBW
Min. EIRP Ka-Band	44 dBW
Dish Diameter	129.54 cm; 51 in
Radome Diameter	166.88 cm; 65.7 in
Radome Height	178.31 cm; 70.2 in
ADE Weight	117.93 kg; 260 lbs.
Operating Temperature Range	-20°C to 55°C
Storage Temperature Range	-40°C to 70°C
Pedestal Type	3 Axis (Azimuth, Elevation, Cross Level)
Ships Motion	Roll +/- 25°; Pitch +/- 15°
Turning Rate	15°deg /sec @ 5°deg /sec2
Azimuth Range	680°

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Elevation Range	-15° to 115°
Polarization	Linear/Circular Selectable
Stabilization Accuracy	0.2° Peak error under specified ship's motion
Skew Contro	Automatic
LNB Type	Dual Band Ku/Ka Programmable
Frequency Range	Ku-Band (10.7 - 12.75 GHz)
	Ka-Band (18.3 - 18.8 GHz) (19.7 - 20.2 GHz)



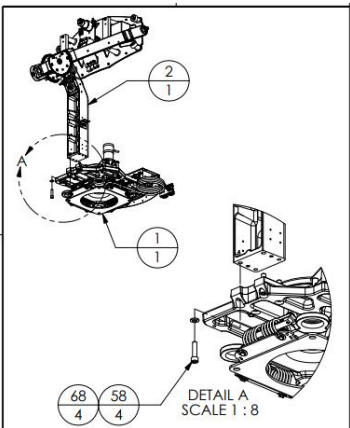
 WORLD CARRIER CORPORATION  HBA Offshore (Int) Pte. Ltd  VIVO ASIA ENERGY & TRADING	EMEM FPSO	WCPOKWTP-450-TE-(VAE) MN-010		 ORIENTAL OKWOK  ADDAX PETROLEUM
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- WHAT'S IN THE BOX

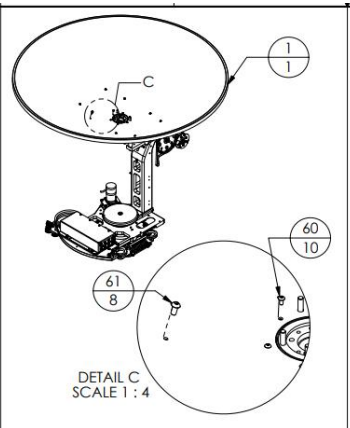
Find No	QTY	Description
0	1	ASSEMBLY DRAWING, NGTV SYSTEM
0	1	Software Assembly General Release TVRO
1	1	BASE SPINDLE & TURNTABLE ASSY, TVRO, 1.2M
2	1	ANTENNA ASS'Y, KU/KA, 1.2M
3	1	RADOME ASSY, GA INSTALL, 1.64 M, TVRO, NCS WHITE
4	1	LMXP, TVRO
8	1	CUSTOMER DOC PACKET, TVRO, 1M
9	1	DECAL KIT, 66-81 IN RADOME, SEA TEL
10	1	BALANCE WEIGHT KIT, BASIC, MEDIUM SYSTEMS
15	1	HARNESS ASSY, UPPER, TVRO, KIT
16	1	HARNESS ASSY, MOTOR EXTENSION, 30 IN
18	1	HARNESS ASSY, MOTOR TO ELEVATION, 72 IN
20	1	CABLE ASSY, INTERFACE, DE15P- DE15S,7 S/UTP,96 IN
22	4	BASE STOP TOP, DIA 60, T&T
24	4	BASE STOP BOTTOM, DIA 60
26	4	BASE STOP STUD, DIA 10 X 87MM
28	8	RUBBER PAD, BASE STOP, DIA 60 - SHORE A 60, T&T
46	4	SCREW, FLAT HD, SKT DRV, M8 X 16, SS
51	1	SCREW, HEX HD M6X90
53	1	SCREW, SOCKET HD, M3 X 8, SS.
55	3	SCREW. SOCKET HD, M4 X 8, SS.
58	4	SCREW, SOCKET HD, M10 X 40, SS.
62	1	WASHER, FLAT, M3, SS.
64	3	WASHER, FLAT, M4, SS
66	2	WASHER, FLAT, M6, SS.
68	4	WASHER, FLAT, M10, SMALL PATTERN, SS
70	1	SPACER, 1/4 X .50 OD X 1.50, NYLON
72	1	CABLE CLAMP, NYLON, .25 DIA, #8 MTG HOLE
74	2	CABLE CLAMP, NYLON, .375 DIA, #8 MTG HOLE
76	2	INSULATED P-CLAMP 5/8 IN
77	1	PACKAGING, ARBITRATOR
80	1	DECAL, MODEL/PART/SERIAL, LARGE

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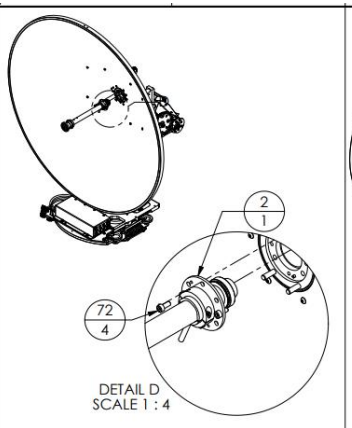
•
 INSTALATON INSTRUCTIONS



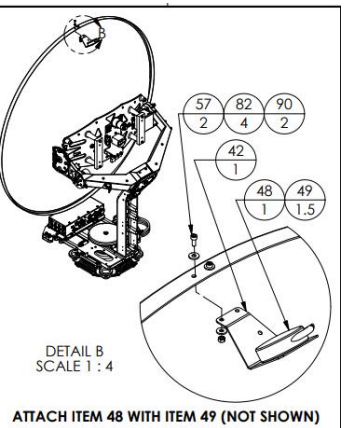
DETAIL A
SCALE 1 : 8



DETAIL C
SCALE 1 : 4

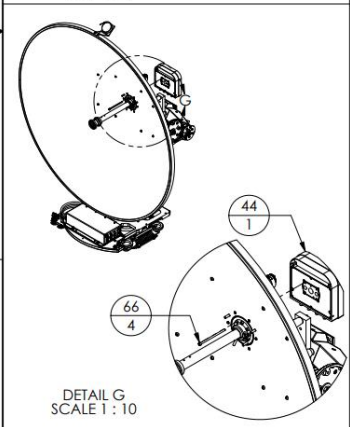


DETAIL D
SCALE 1 : 4

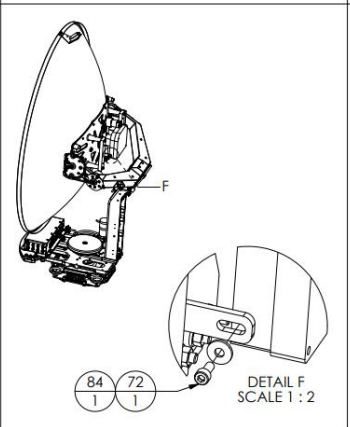


DETAIL B
SCALE 1 : 4

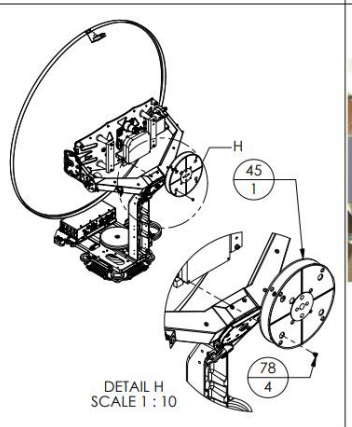
ATTACH ITEM 48 WITH ITEM 49 (NOT SHOWN)



DETAIL G
SCALE 1 : 10



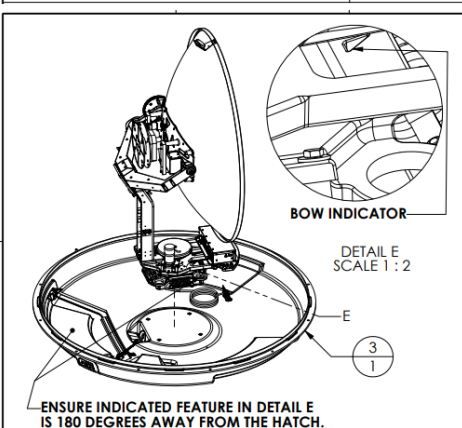
DETAIL F
SCALE 1 : 2



DETAIL H
SCALE 1 : 10



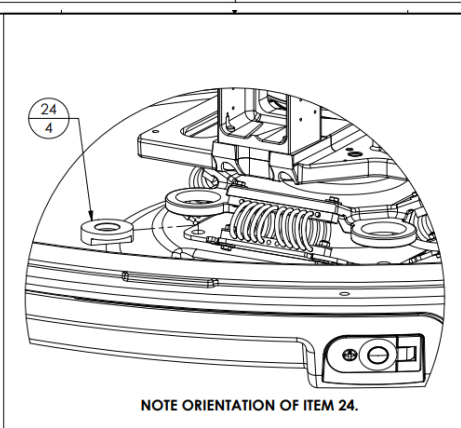
ATTACH ITEM 80 TO THE LEVEL BEAM APPROX. WHERE SHOWN.



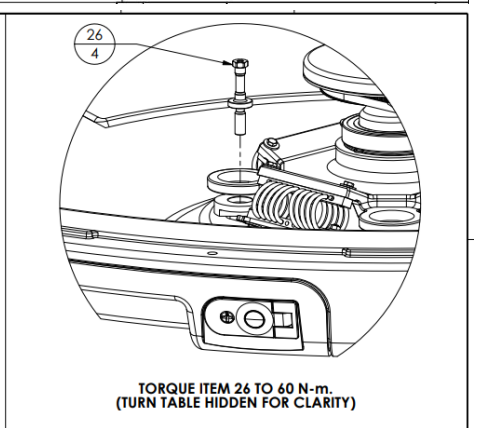
BOW INDICATOR

DETAIL E
SCALE 1 : 2

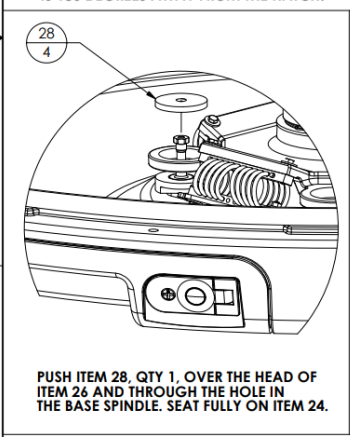
ENSURE INDICATED FEATURE IN DETAIL E IS 180 DEGREES AWAY FROM THE HATCH.



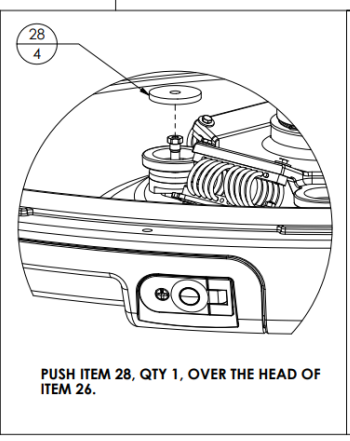
NOTE ORIENTATION OF ITEM 24.



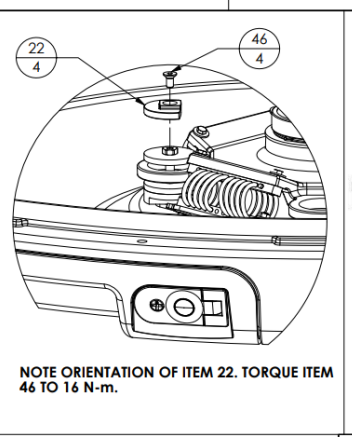
TORQUE ITEM 26 TO 60 N-m. (TURN TABLE HIDDEN FOR CLARITY)



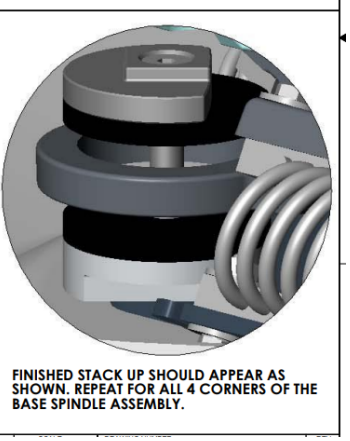
PUSH ITEM 28, QTY 1, OVER THE HEAD OF ITEM 26 AND THROUGH THE HOLE IN THE BASE SPINDLE. SEAT FULLY ON ITEM 24.



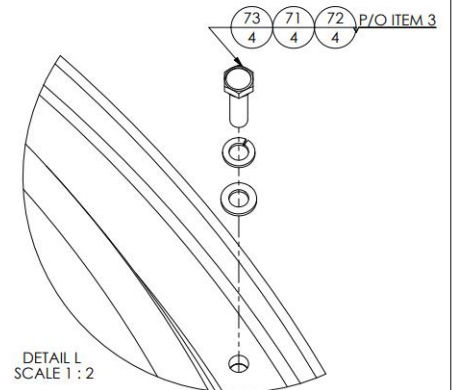
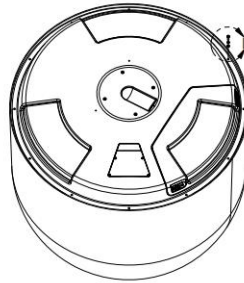
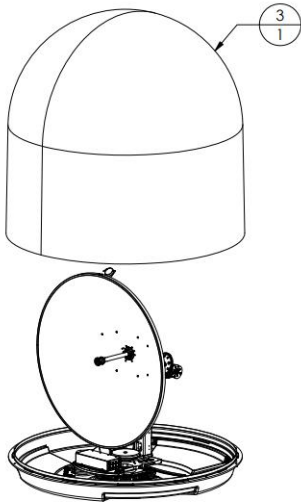
PUSH ITEM 28, QTY 1, OVER THE HEAD OF ITEM 26.



NOTE ORIENTATION OF ITEM 22. TORQUE ITEM 46 TO 16 N-m.



FINISHED STACK UP SHOULD APPEAR AS SHOWN. REPEAT FOR ALL 4 CORNERS OF THE BASE SPINDLE ASSEMBLY.



ATTACH RADOME TOP WITH 4X M10X40 BOLTS WITH LOCK WASHERS AND FLAT WASHERS.



CONNECT THE POWER CABLE AND THE 4 COAX CONNECTORS TO THE LNB. NOTICE THE COLOR CODED HEAT SHRINK ON EACH. WRAP HARNESS AROUND SPOOLER TWO (2) FULL ROTATIONS.



72 64 55
1 1 1



CONNECT THE 8 PIN IDC CONNECTOR TO THE MOTOR TERM PCB ON THE POLANG ASSEMBLY.

53 62
1 1



70 76 66 51
1 1 1 1



ROUTE HARNESS TO THE RIGHT SIDE OF THE CROSS LEVEL AS SHOWN. SECURE WITH TIE WRAPS.

76 66 56
1 1 1



WRAP EL MOTOR CABLE AROUND LEVEL BEAM AND SECURE WITH TIE WRAP.



PRESS TIE WRAPS INTO PULLEY IN INDICATED LOCATIONS. SECURE CABLES.



PRESS TIE WRAPS INTO POST AND TURN TABLE IN INDICATED LOCATIONS. SECURE CABLES.



APPLY TORQUE SEAL TO ALL COAX CONNECTIONS AS SHOWN.



PRESS TIE WRAPS INTO POST IN INDICATED LOCATIONS. SECURE CABLES.



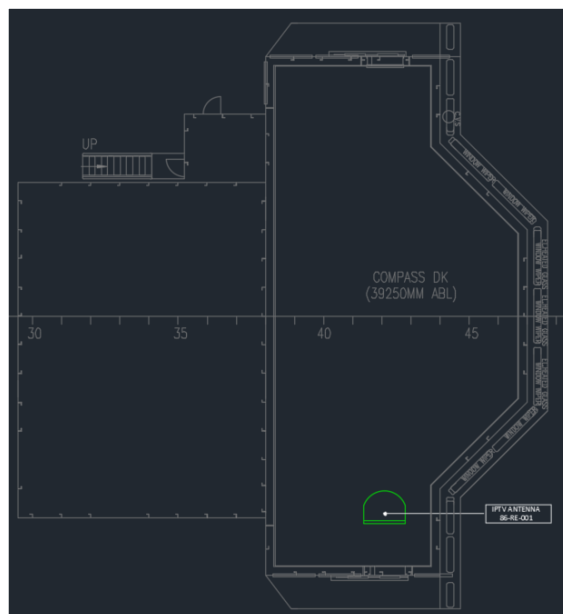
55 64 74
2 2 2



USE HOT GLUE TO ATTACH CONNECTOR BRACKET TO RADOME IN APPROX. LOCATION. USE ROTALOC, .50" CABLE CLAMP, AND M4 HARDWARE TO SECURE CABLE LOOP TO RADOME BASE.

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- MOUNTING LOACTION

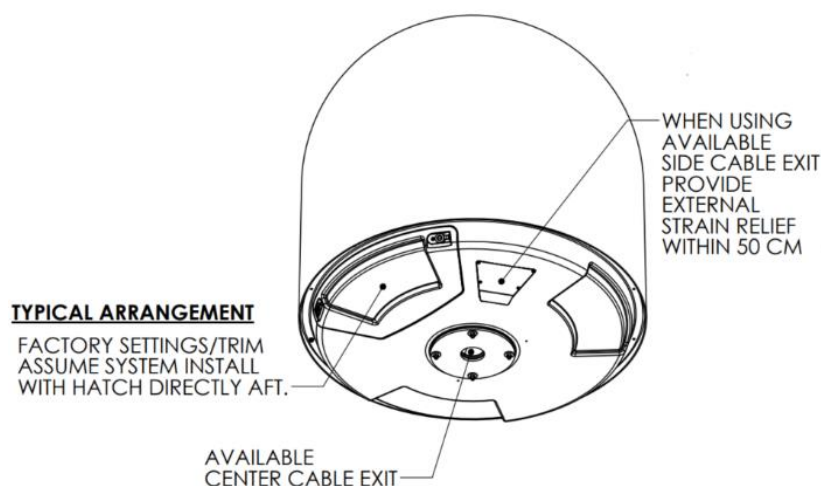


The stabilized TVRO antenna is installed on the Compass Deck at the designated mounting location shown in the layout drawing to ensure an unobstructed 360° satellite visibility and minimum shadowing from nearby structures. The antenna is mounted on a dedicated steel pole foundation fabricated and installed by the yard, providing sufficient elevation above deck level to maintain clear line-of-sight and reduce multipath interference.

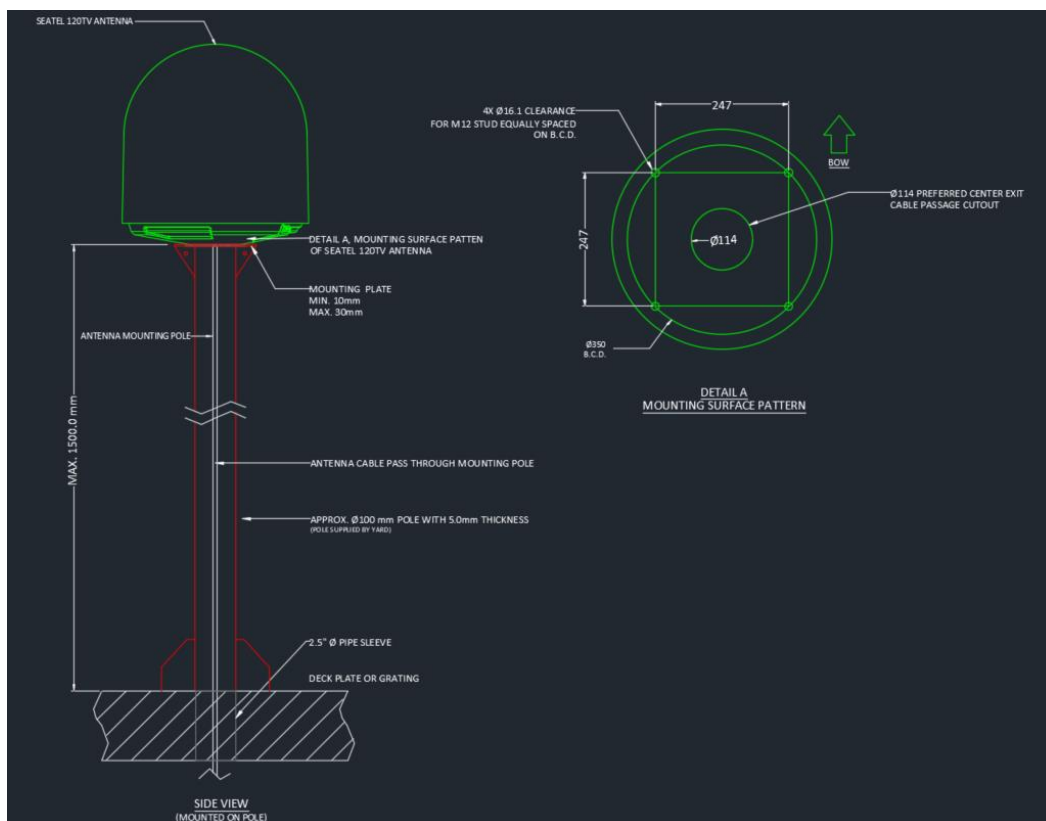
The antenna pedestal is secured to a circular mounting plate using a four-bolt fixing arrangement based on the manufacturer's mounting pattern. The mounting interface incorporates a central cable entry cut-out to allow RF and control cables to pass internally through the pole.

A vertical mounting pole of approximately 100 mm diameter with adequate wall thickness is used to support the antenna dome. The RF and control cables are routed internally through the pole and deck sleeve to protect against environmental exposure and mechanical damage. A sealed pipe sleeve is installed at the deck penetration point to ensure watertight integrity and corrosion protection.

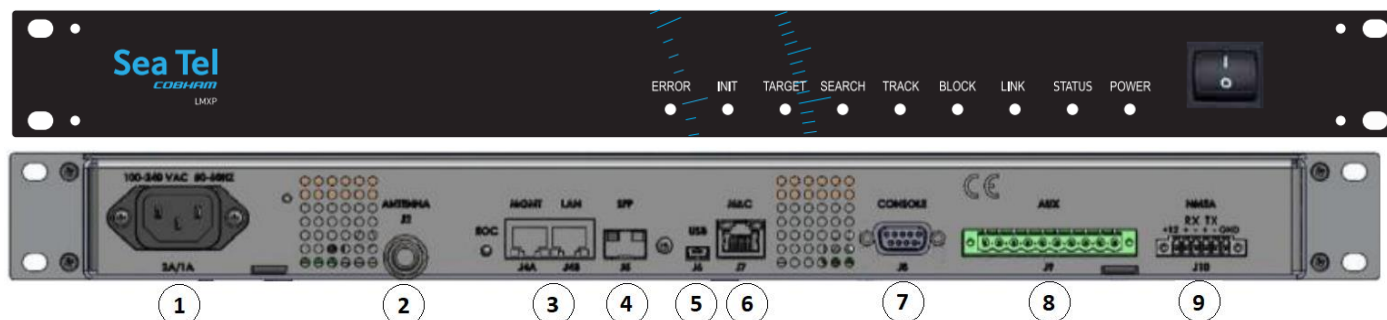
The mounting arrangement is aligned with vessel bow reference as required by the antenna stabilization system to ensure correct azimuth referencing during satellite tracking. All installation practices comply with marine standards for vibration resistance, grounding, and corrosion protection, ensuring reliable long-term operation of the TVRO system in offshore environments.



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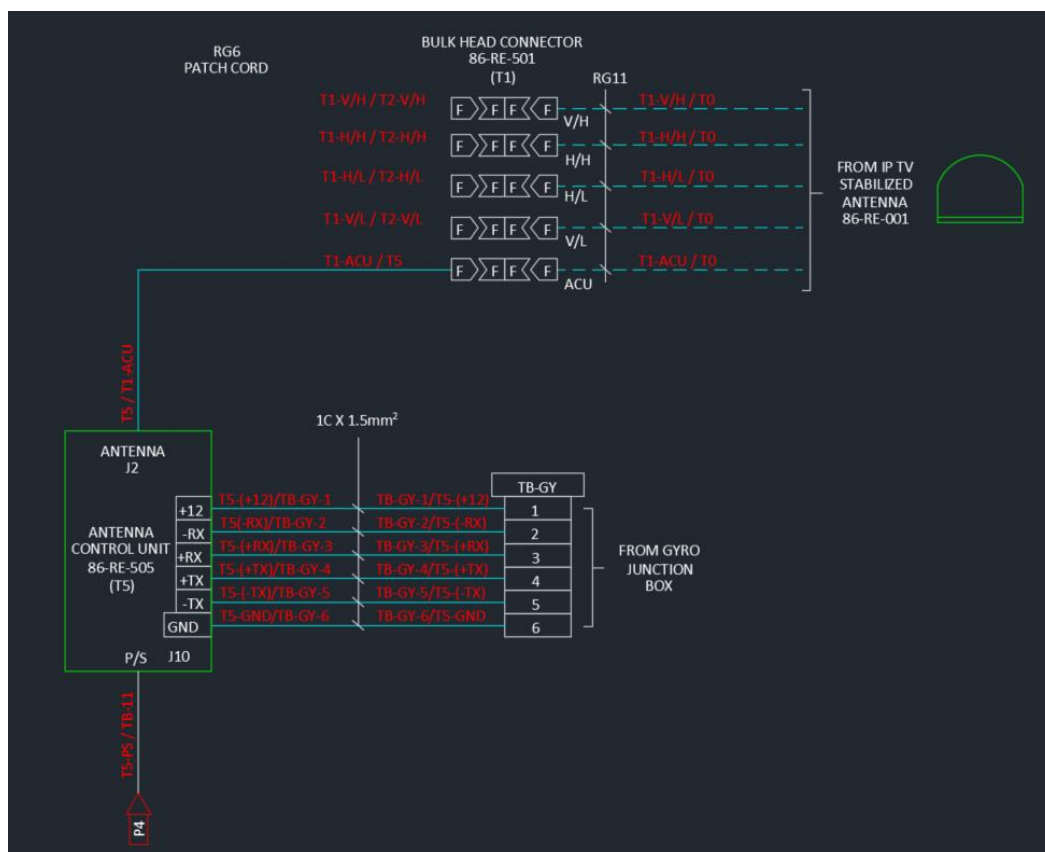


3.1.2 ANTENNA CONTROLLER SEATEL LMXP



Key	Description
1	Ships AC Mains -Connect the power cord from the rear panel of the LMXP to AC voltage power source
2	J2 Antenna -Connect the Red antenna control coax cable to this connector on the rear panel of the LMXP
3	J4 A/B & J4 A/B - Ethernet 2 Port 10/100 switch -Ethernet connections to computer or LAN devices as desired
4	J5 SFP Fiber Interface -SFP Gigabit Ethernet connection.
5	J6 Mini-USB Computer M&C Connection -Mini-USB Antenna M&C connection, if desired.
6	J7 M&C Host-Not connected -Future development
7	J8 Console -Antenna M&C Serial connections
8	J9 AUX -Computer RJ-45 Serial M&C connections
9	J10 NMEA 0183 Gyro Compass -NMEA 0183 I/O connections.

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The satellite television distribution system is centred around the SEATEL 120TV stabilized antenna, which receives satellite signals while compensating for vessel motion. The antenna is electrically and logically controlled by the SEATEL LMXP, which acts as the Antenna Control Unit (ACU), providing power, stabilization control, and communication between the antenna pedestal and the vessel's gyro system.

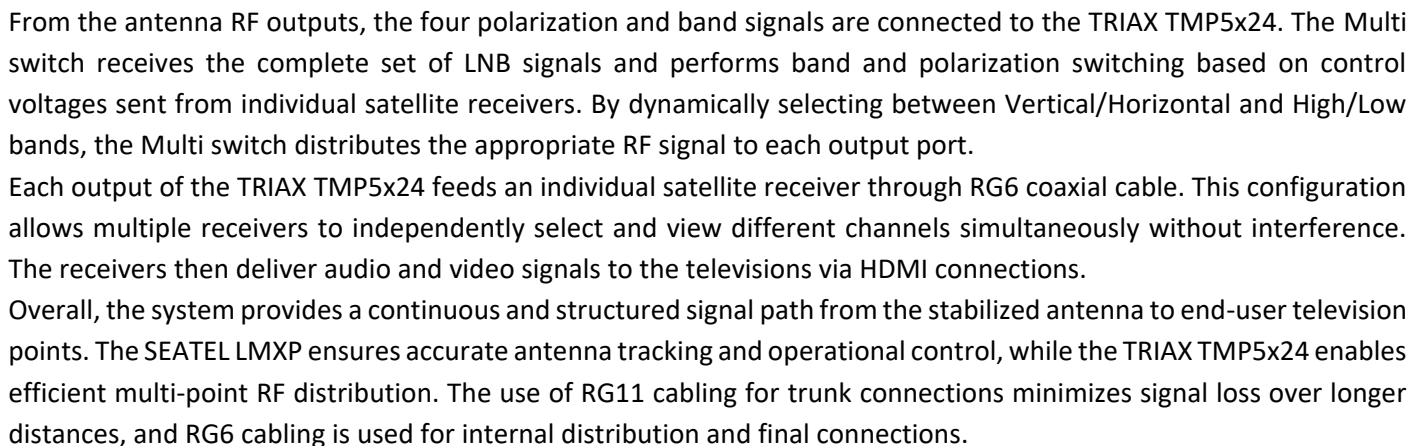
The SEATEL 120TV antenna delivers four LNB RF outputs corresponding to Vertical High (V/H), Horizontal High (H/H), Horizontal Low (H/L), and Vertical Low (V/L) frequency bands. These signals are routed via RG11 coaxial cables to bulkhead connectors and then directed toward the distribution system. Simultaneously, the SEATEL LMXP (ACU) supplies DC power, tracking commands, and receives feedback signals from the antenna. It interfaces with the vessel's gyro through a multi-core control cable to obtain heading and stabilization data, ensuring accurate satellite tracking during vessel movement.

3.1.3 COAXIAL MULTI SWITCH TRIAX TMP5X24



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Specifications	
Input polarity gain control TER	15.0 dB
Input polarity gain control SAT	4 X 10.0 dB
Max. output level SAT @ -35 dB IMD3	100 dBμV
Max. output level TER @ -60dB IMD3	95 dBμV
Output level control - long cable	0 dB
Output level control - medium cable	6 dB
Output level control - short cable	12 dB
LNB current max.	600 mA
Switch commands	13V/18V - 13V/22 kHz - 18V/22 kHz
Frequency	
Frequency range TER	47-862 MHz
Frequency range SAT	950-2150 MHz
Loss	
Insertion loss - trunkline SAT	2.0 dB
Insertion loss - trunkline TER	0.0 dB
Isolation	
Isolation SAT to TER	32 dB
Isolation TER to SAT	32 dB
Isolation cross polarisation H/V	30 dB
Isolation out-out SAT	30 dB
Isolation out-out TER	30 dB
Return Loss	
Return loss SAT inputs	10 dB
Return loss SAT outputs	8 dB
Return loss TER inputs	10 dB
Return loss TER outputs	8 dB
Operational	
LINE power DC voltage (max.)	12.0 VDC
LINE power current (max.)	50.0 mA
AC Supply voltage	180...264 VAC
DC Operating voltage	18.0 VDC
DC Current consumption	1500 (47-63 HZ) mA
Temperature - operating	0 - 55 °C
Connectors	
Connector Type	F-female
Number of inputs	4 SAT + 1 TER
Number of trunk inputs	4 SAT, 1 TER
Subscriber outputs	24
Colour coding of IF and TER inputs	Yes
Mechanical	
Product Size (H x D x W)	51 x 157 x 340mm
Net Weight	1,592 kg



A black, rack-mountable video encoder unit. The front panel features a series of horizontal ventilation slats. A small label on the left side of the front panel reads "VIDEO ENCODER". The unit is designed for installation in a standard 19-inch equipment rack, as evidenced by the mounting ears on the sides.

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This encoder features embedded hardware encoding designed for professional IPTV and video distribution systems. It supports multiple input formats including HDMI and CVBS video with independent audio channels, enabling flexible integration with cameras, media players, and AV sources. The unit supports HD resolutions up to 1080p and advanced compression formats such as H.265 and H.264 profiles, ensuring efficient bandwidth utilization while maintaining high video quality.

The encoder provides Gigabit Ethernet connectivity with full-duplex transmission and supports multiple streaming protocols including RTSP, RTMP, HTTP, and UDP (unicast and multicast). It is compatible with standard streaming servers such as Wowza and FMS, allowing seamless live broadcasting and network distribution. Dual-stream functionality enables simultaneous main and sub-stream output using different protocols and bitrates.

Additional features include adjustable bitrate control (CBR/VBR), customizable resolution output, on-screen watermark overlay, and remote management via web interface. The system supports DHCP network configuration, firmware upgrades, and remote maintenance, making it suitable for continuous operation in IPTV, surveillance, and onboard multimedia distribution environments.

Specifications	
Model	H3114AV 4 Road HDMI HD encoder
Input	4 HDMI signal input, 8 CVBS signal input, 123.5 audio input
Output	The 4-way IP signal output
Video	
input resolution	1920x1080p / 1920x1080i / 1280x720p / 720 * 576i / 720 * 576p / 640 * 480i
Coding standard	H.264/H.265
Encoding level	H.265 MAIN PROFILE /H.264 BASELINE PROFILE/ H.264 MAIN PROFILE /H.264 HIGH PROFILE
After coding size	640 * 360~1920 * 1080
Video code rate	The CBR / VBR code rate was controlled by 16 KBIT / S ~12MBIT / S
Video frame rate	At 30 frames / 60 frames
OSD	Text
Audio	
Input	4-way HDMI embedded audio, 12-way 3.5MM standalone audio
Audio coding	AAC/MP3
Audio code rate	24Kb/s~256Kb/s
sampling rate	32000、44100
IP	

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IP output	4 RJ45 1000M adaptive Ethernet ports, TS standard flow, and ONVIF protocol
System	
Networking protocol	HTTP, RTSP, RTMP, UDP, multicast, and unicast are supported
Configuration management	Supports IE configuration and remote upgrade
Common	
Power	220V
consumption	<28W
Size	430mmx250mmx45mm (LxWxH)
Working temperature	-20°C+ 60°C
Working humidity	<90%
Net Weight	3KG

3.1.5 8 WAY ENCODER CLEAR-H3118AV (8-WAY)



video distribution applications. It supports up to eight HDMI or AV video inputs along with independent audio acquisition, enabling simultaneous encoding of multiple sources within a single compact unit. The encoder outputs dual H.264 video streams with MP3 or AAC audio, allowing flexible bitrate control and optimized bandwidth usage while maintaining high-quality video performance.

The system supports a wide range of streaming protocols including RTSP, RTP, RTMP, HTTP, HLS, and UDP, making it compatible with most streaming platforms and media servers. Dual-stream functionality enables primary and secondary streams with configurable encoding profiles and bitrates (CBR/VBR), supporting both unicast and multicast

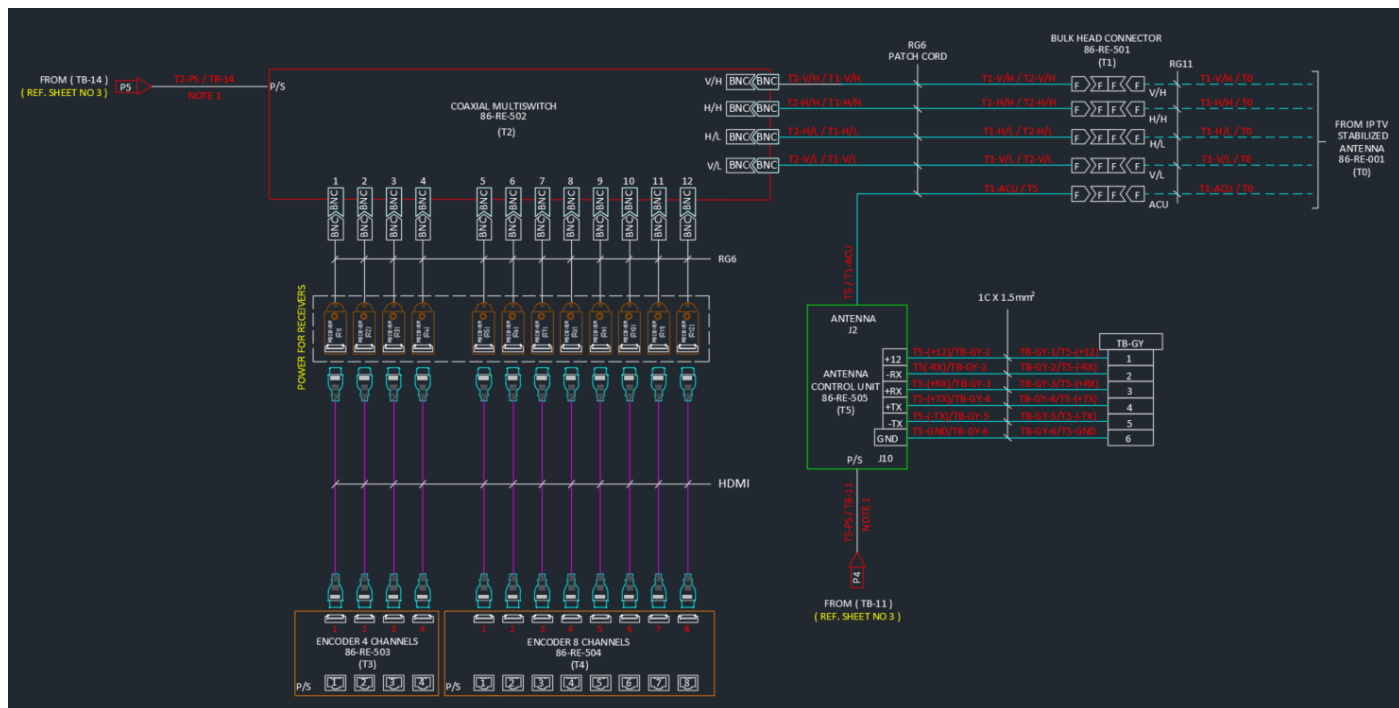
 WORLD CARRIER CORPORATION HBA Offshore (Int) Pte. Ltd.  VIVO ASIA ENGINEERING & TRADING	EMEM FPSO	WCPOKWTP-450-TE-(VAE) MN-010		 ORIENTAL OKWOK  ADDAX PETROLEUM
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network transmission over Gigabit Ethernet. It also supports multiple packaging methods such as FFMPEG and VLC for enhanced integration flexibility.

Designed for reliable continuous operation, the encoder features web-based remote management, DHCP support, firmware upgrades, and on-screen display (OSD) customization including logos and text overlays. Additional capabilities include adjustable video parameters, audio gain control, and support for no-signal encoding. With low power consumption and broad OS compatibility, the H3118AV is well suited for IPTV headend systems, surveillance networks, and onboard multimedia broadcasting environments.

Specifications	
Model	H3114AV 4 Road HDMI HD encoder
Input	8-way HDMI / AV signal input and 8-way 3.5 audio input
Output	The 8-way IP signal output
Video	
input resolution	1920x1080p/1920x1080i/1280x720p/720*576i/720*57 6p /640*480i
Coding standard	H.264
Encoding level	H.264BASELINEPROFILE / H.264MAINPROFILE /H.264HIGHPROFILE
After coding size	640*360~1920*1080
Video code rate	TheCBR/VBRcoderatewascontrolledby16KBIT /S~12MBIT/S
Video frame rate	30Frames
OSD	Text/picture
Audio	
Input	8hdmiiembeddedaudio,83.5independentaudio
Audio coding	AAC/MP3
Audio code rate	64Kb/s~384Kb/s
sampling rate	32000、44100、48000
IP	
IP output	8RJ100M/1000MadaptiveEthernetport, TS standard stream, dual code stream ONVIF protocol
Networking protocol	Supports IE configuration and remote upgrade
Configuration management	Supports IE configuration and remote upgrade
Common	
Power	220V
consumption	<40W
Size	440mmx300mmx43mm (LxWxH)
Working temperature	-20°C+60°C
Working humidity	<90%
Net Weight	3.5KG

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The coaxial Multi switch functions as the RF distribution hub between the stabilized satellite antenna system and the IPTV encoding equipment. It receives multiple RF feeds from the antenna via RG6 coaxial inputs routed through the bulkhead connector. These feeds typically represent different polarization and band selections (Vertical/Horizontal and High/Low), along with the antenna control (ACU) signal path.

Inside the cabinet, the Multi switch processes and distributes the incoming satellite IF signals to multiple independent RF outputs. Each output provides a selectable satellite feed to a dedicated receiver module associated with the encoder channels. This enables simultaneous access to different satellite transponders without signal conflict, allowing multiple TV programs to be received in parallel.

The RF outputs from the Multi switch are connected via coaxial jumpers to the satellite receivers (integrated or external), which demodulate the DVB-S/S2 signals and convert them into baseband audio/video outputs. These outputs are then fed into the HDMI inputs of the multi-channel encoders. The encoders compress and convert the video streams into IP-based transport streams for IPTV distribution over the vessel's network.

In summary, the coaxial Multi switch provides controlled RF signal splitting and polarization management, ensuring stable multi-channel satellite reception and enabling the encoders to generate multiple IPTV streams from a single stabilized antenna source.

3.1.6 24P NETWORK SWITCH HUAWEI S5735-L24T4S-A-V



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Key	Description
1	Twenty-four 10/100/1000BASE-T ports
2	Four 1000BASE-X ports
3	One console port
4	One ETH management port
5	One USB port
6	One PNP button
7	Ground screw
8	Jack for AC power cable locking strap
9	AC socket

The Huawei S5735-L24T4S-A-V is a managed Gigabit Ethernet switch designed for enterprise and industrial access-layer deployments. It provides 24 × 10/100/1000BASE-T RJ45 ports and 4 × Gigabit SFP uplink ports, enabling high-speed connectivity between network devices and aggregation layers. The switch is housed in a compact 1U rack-mount chassis and supports reliable continuous operation in communication cabinets.

Based on Huawei's S5735 series platform, the switch supports advanced Layer 2 features and basic Layer 3 functions including VLANs, QoS, static routing, and security controls. It supports flexible management via CLI, web interface, and SNMP-based monitoring. In onboard communication systems, the S5735-L24T4S-A-V is typically used as an access or distribution switch to interconnect servers, IP communication equipment, and network infrastructure while ensuring stable and efficient data transmission.

3.1.7 24P NETWORK SWITCH CISCO C1300-24FP-4X



Specification	
Model	C1300-24FP-4X
Brand	Cisco
Capacity	95.23
Switching Capacity (Gbps)	128.0
Power Dedicated to PoE	375W
Number of PoE Ports	24
System Power Consumption	110V: 40.9W; 220V: 40.9W
Power Consumption with PoE	110V: 454.2W; 220V: 441.2W

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Heat Dissipation (BTU/hr)	1548.8
Idle Power	110V: 21.9W; 220V: 22.5W
Total System Ports	24 x Gigabit Ethernet + 4 x 10G
RJ-45 Ports	24 x Gigabit Etherne
Combo Ports (SFP+)	4 x SFP+
Packet Buffer	1.5 MB
Unit Dimensions (W x D x H)	445 x 345 x 44 mm (17.5 x 13.78 x 1.73 in)
Fan (Number)	1
Acoustic Noise (25°C)	39 dBA
MTBF at 25°C (hours)	389,607

The Cisco C1300-24FP-4X is a managed Gigabit Ethernet switch designed for enterprise and industrial network deployments requiring high port density and PoE capability. It provides 24 x 1-Gigabit RJ45 ports with integrated PoE+ power delivery and 4 x 10-Gigabit SFP+ uplink ports for high-speed backbone connectivity. The switch supports a PoE power budget of approximately 375 W, enabling reliable operation of IP phones, wireless access points, cameras, and other powered network devices.

Built on the Cisco Catalyst 1300 platform, the unit offers Layer 2 switching features, VLAN segmentation, spanning tree protocols, and secure network management functions. With a switching capacity of around 128 Gbps, it supports stable traffic forwarding in high-utilization environments.

In communication or marine systems, the C1300-24FP-4X is typically used as a core or distribution switch to interconnect IP radios, servers, IPTV equipment, and network infrastructure while also supplying PoE power to field devices.

3.1.8 KVM ATEN CL5708

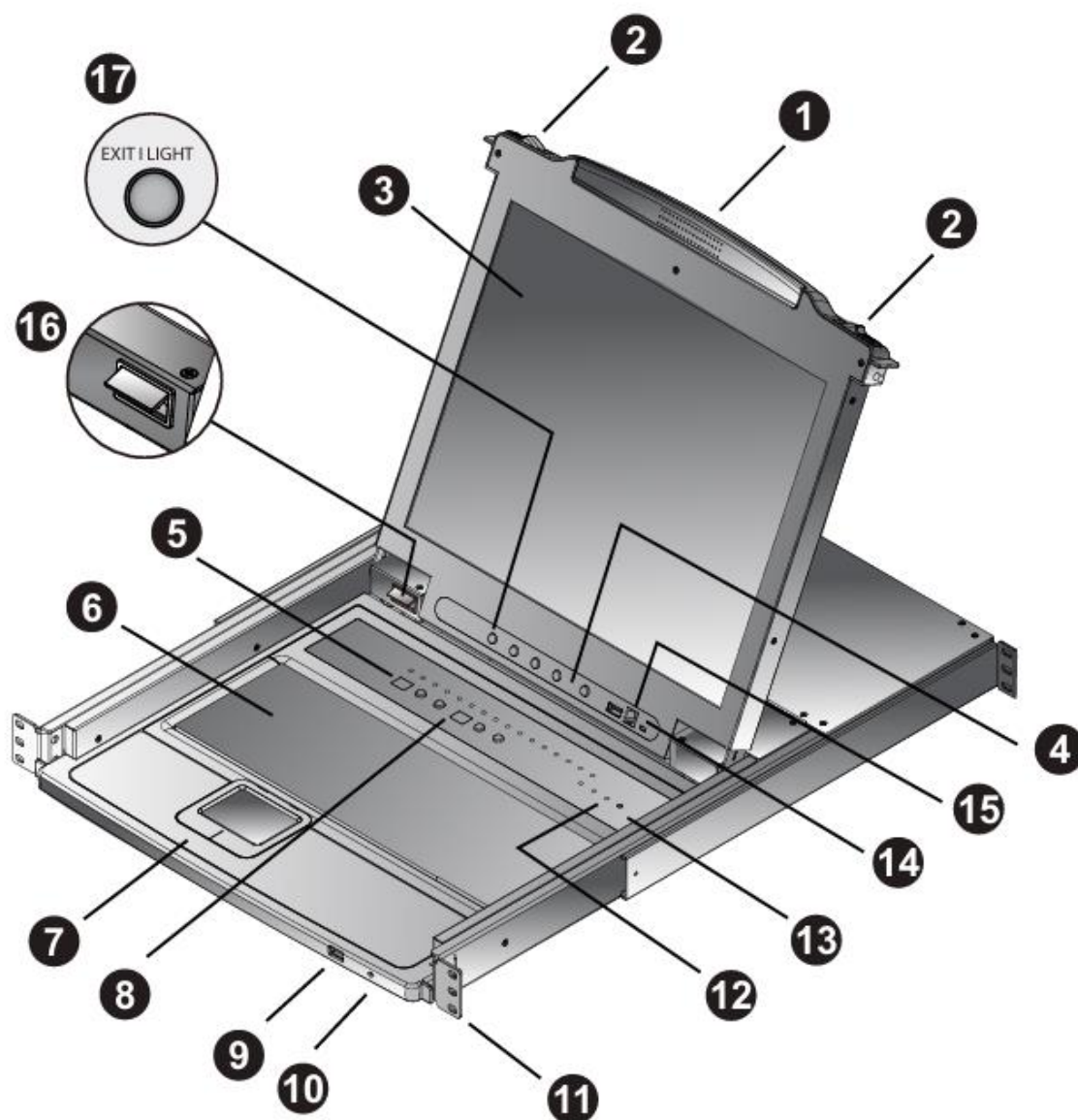


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The CL5708 Slideaway LCD KVM Switch is an integrated control unit that enables centralized management of multiple computers from a single keyboard, video, and mouse (KVM) console. A single unit supports control of up to 8 or 16 computers, and additional switches can be daisy-chained to expand control capacity up to 256 systems from one console.

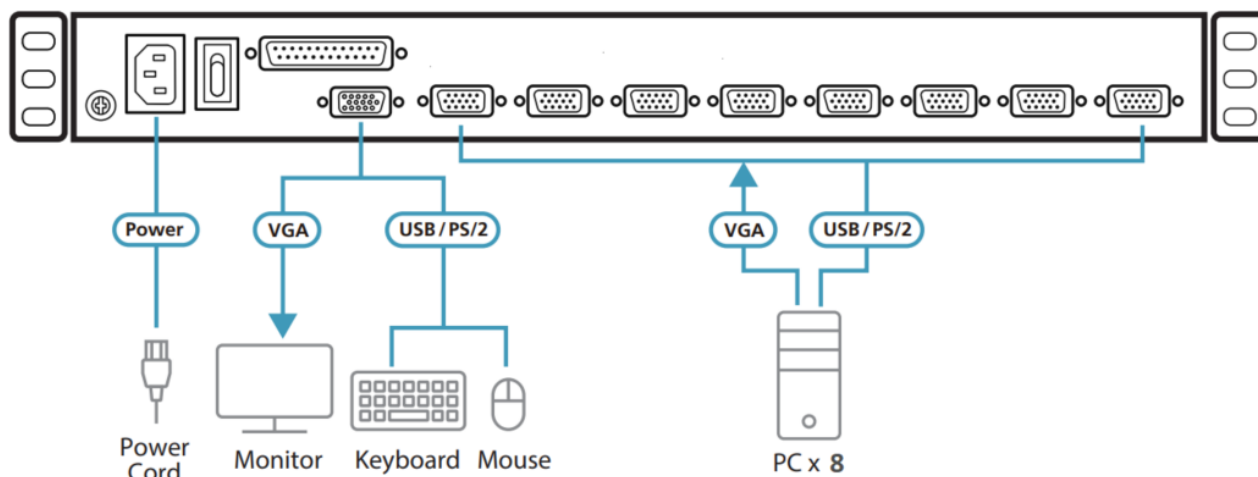
The device features a space-saving 1U rack-mountable slideways design that integrates a keyboard, touchpad, and LED-backlit LCD display. The Full HD display provides clear video output for monitoring and control tasks. For enhanced security, the system optionally supports fingerprint authentication to restrict access to both the KVM switch and connected servers.

Built with a modular architecture, the KVM console section can be separated from the switch module, allowing easier maintenance, servicing, and component replacement without removing the entire unit from the rack.

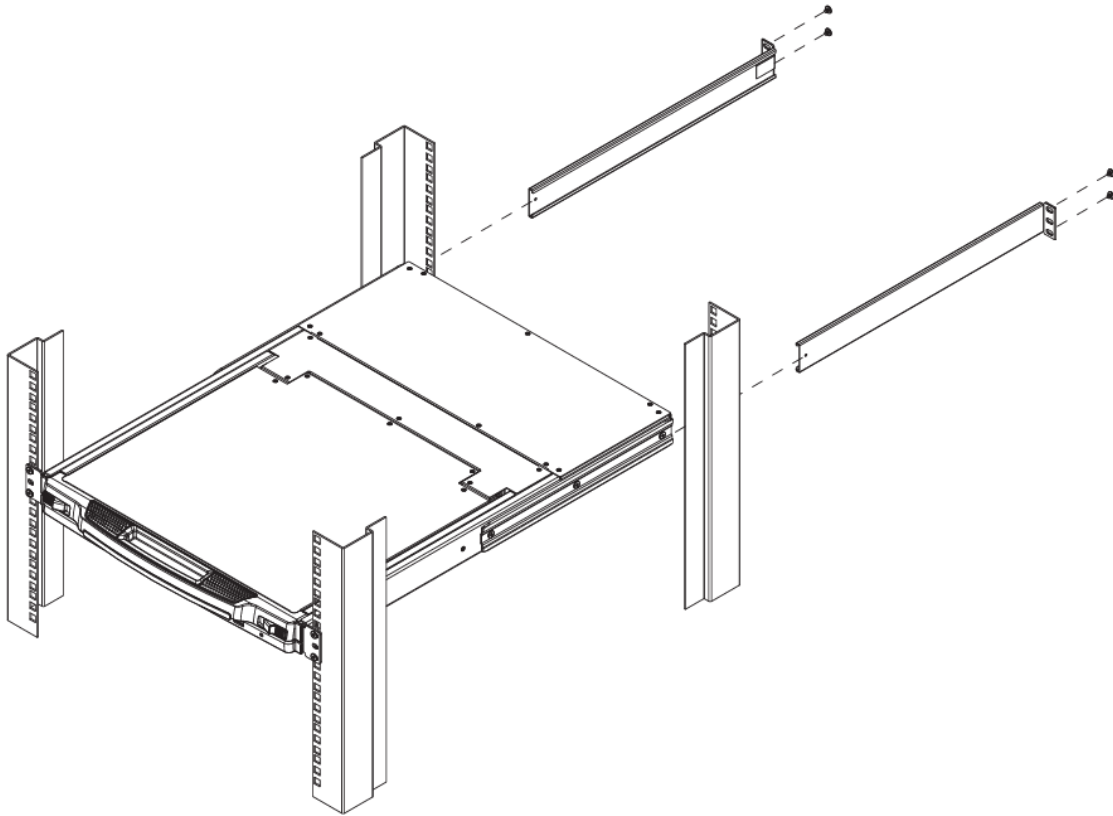


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Key	Component	Description
1	handle	Pull to slide the KVM module out; push to slide the module in
2	slide release	To slide the console out, you must first release it by sliding these tabs to the inside. See page 21 for details on sliding the console in and out.
3	LCD display	After sliding the KVM module out, flip up the cover to access the LCD monitor.
4	LCD controls	The LCD On / Off switch is located here, as well as buttons to control the position and picture settings of the LCD display. See page 25 for details.
5	port LEDs	An ON LINE LED lights to indicate that the computer attached to its corresponding port is up and running.
6	keyboard	Standard 105-key keyboard
7	touchpad	Standard mouse touchpad
8	station / port switches	Press the UP or DOWN buttons for the port ID and station ID to bring the KVM focus to the computer attached to the corresponding port.
9	external mouse port	A USB mouse port is provided for users who prefer to use an external mouse.
10	power LED	Lights to indicate that the unit is receiving power.
11	rack mount brackets	The rack mount brackets located at each corner of the unit secure the chassis to a system rack.
12	lock LEDs	Num Lock, Caps Lock, Scroll Lock LEDs are located here.
13	reset button	Located to the right of the Lock LEDs. Press this button in with a small object to perform a system reset.
14	firmware upgrade section	- Firmware upgrade port: The firmware upgrade cable that transfers the firmware upgrade data from the administrator's computer to the CL5708 plugs into this RJ 11 connector. - Firmware upgrade switch: During normal operation this switch should be in the NORMAL position.
15	USB port	The USB port is available to connect a USB 1.1 peripheral device to the switch.
16	LED illumination light	Illuminates the keyboard and touchpad to allow visibility in low-light conditions.
17	Exit / Light pushbutton	Press the Exit / Light push button for two seconds to turn the LED illumination light on or off. (Default: On) During the LCD OSD mode, this pushbutton has more functions,



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Install the unit using two people: one person should hold the unit in position in the rack while the other secures the front brackets. Then slide the left and right L-shaped brackets into the side mounting brackets from the rear and fasten them with the provided screws. Once all brackets are aligned, tighten all screws securely. Ensure adequate ventilation by leaving at least 5.1 cm clearance on each side and 12.7 cm at the rear for airflow, power, and cable routing.

3.1.9 DELL POWEREDGE R740



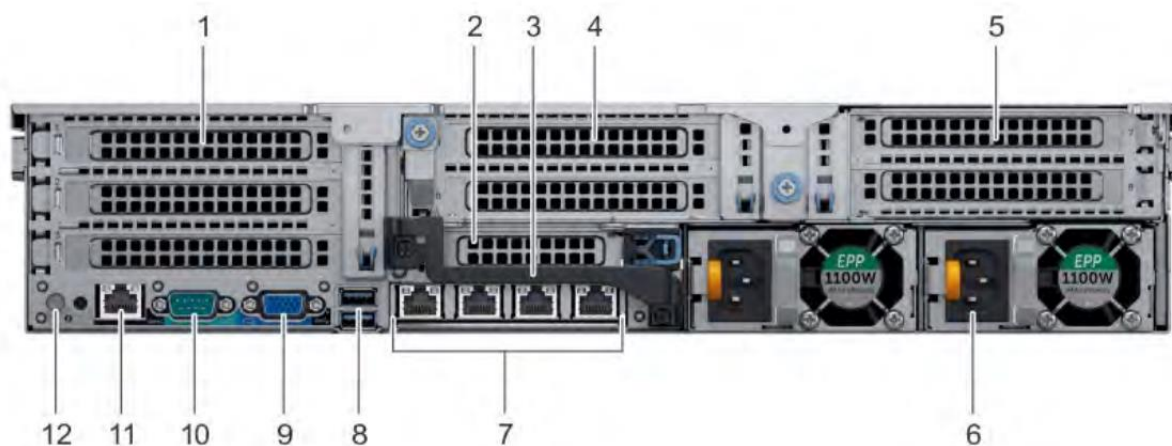
The Dell PowerEdge R740 is deployed as the central server platform for the IPTV system, providing processing, storage, and management functions for onboard television and media distribution services. It operates as the core middleware and streaming server, controlling live TV channels, VOD content, user access management, and network-based video delivery.

Built on a dual-processor architecture with scalable memory capacity, the R740 delivers stable performance for handling multiple concurrent IPTV streams across the vessel. The server supports flexible storage configurations to host media files and system data, with RAID protection ensuring operational reliability and data security.

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Integrated high-speed Ethernet connectivity enables seamless integration with IPTV encoders, network switches, and smart TVs distributed throughout the FPSO. Redundant power supplies and enterprise-grade hardware design ensure continuous 24/7 operation, which is essential for crew welfare services.

Remote management features allow monitoring, configuration, and troubleshooting without direct physical access, supporting efficient maintenance in offshore environments. Overall, the PowerEdge R740 provides a reliable and scalable platform for delivering consistent IPTV services onboard.



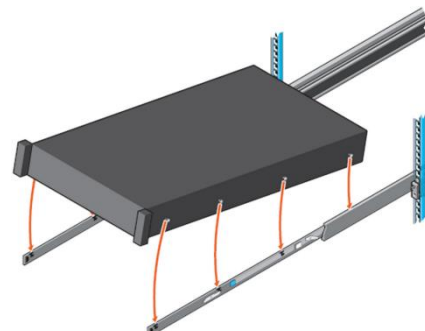
Key	Component	Description
1	Full-height PCIe expansion card slot	The PCIe expansion card slot (riser 1) connects up to three full-height PCIe expansion cards to the system. For more information, see the Expansion card installation guidelines section.
2	Half-height PCIe expansion card slot	The PCIe expansion card slot (riser 2) connects one half-height PCIe expansion cards to the system
3	Rear handle	The rear handle can be removed to enable any external cabling of PCIe cards that are installed in the PCIe expansion card slot
4	Full-height PCIe expansion card slot	The PCIe expansion card slot (riser 2) connects up to two full-height PCIe expansion cards to the system
5	Full-height PCIe expansion card slot	The PCIe expansion card slot (riser 3) connects up to two full-height PCIe expansion cards to the system. For more information
6	Power supply unit	
7	NIC ports	The NIC ports that are integrated on the network daughter card (NDC) provide network connectivity. For more information about the supported configuration
8	USB port	The USB ports are 9-pin and 3.0-compliant. These ports enable you to connect USB devices to the system
9	VGA port	Enables you to connect a display device to the system. For more information
10	Serial port	Enables you to connect a serial device to the system. For more information
11	iDRAC9 dedicated port	Enables you to remotely access iDRAC
12	System identification button	The System Identification (ID) button is available on the front and back of the systems. Press the button to identify a system in a rack by turning on PowerEdge R740 system overview 17 Item Panels, ports and slots Icon Description the system ID button. You can also use the system ID button to reset iDRAC and to access BIOS using the step through mode

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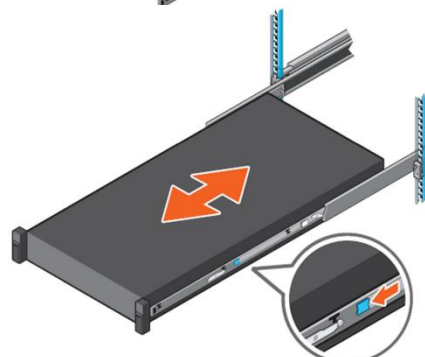
- Pull the inner rails out of the rack until they lock into place.



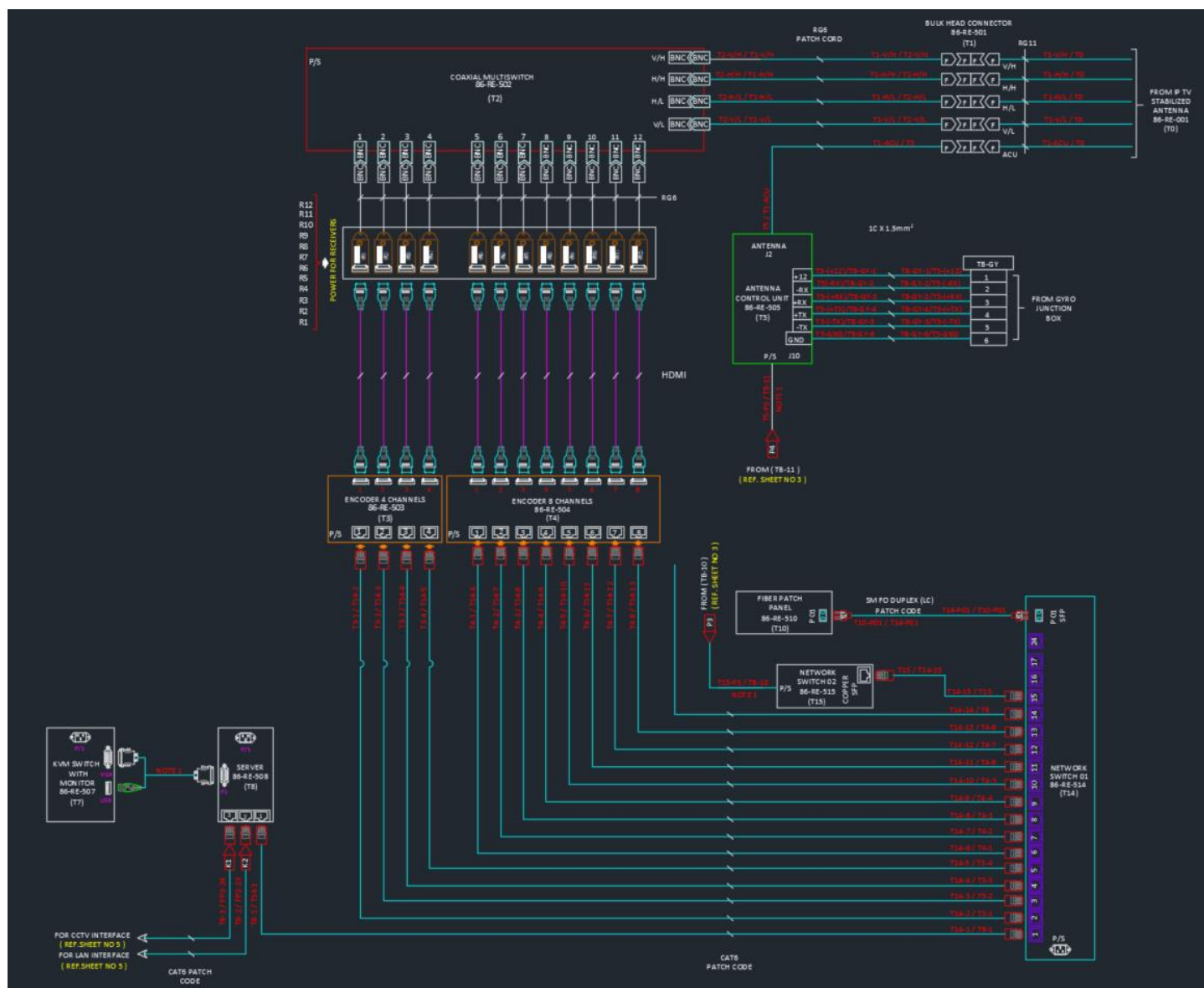
- Locate the rear rail standoff on each side of the system and lower them into the rear J-slots on the slide assemblies.
- Rotate the system downward until all the rail standoffs are seated in the J-slots.
- Push the system inward until the lock levers click into place.



- Pull the blue slide release lock tabs forward on both rails and slide the system into the rack until the system is in the rack.



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Following RF distribution by the coaxial Mult switch, the satellite IF outputs are routed to multiple satellite receiver modules associated with the IPTV encoding chain. Each receiver converts the selected DVB-S/S2 RF feed into baseband audio/video signals. These baseband outputs are then connected via HDMI cables to the multi-channel hardware encoders, including the 4-channel and 8-channel models (CLEAR-H3118AV and CLEAR-H311AV). The encoders perform real-time compression of the incoming audio/video signals using H.264/H.265 encoding and convert them into IP-based transport streams suitable for IPTV distribution.

Each encoder provides multiple Ethernet outputs carrying encoded IP streams. These outputs are terminated using Cat6 patch cords and connected to the primary aggregation switch, Switch-1 (24-port Huawei S5735-L24T4S-A-V). This switch acts as the central IPTV distribution node, handling multicast/unicast streaming traffic and integrating the IPTV subsystem into the vessel's wider IP infrastructure.

From Switch-1, the encoded IPTV streams are distributed across the shipboard LAN. The switch is also interconnected with the KVM console system (ATEN CL5708), allowing centralized monitoring and control of the IPTV server and associated hardware from a single rack-mounted console. The Dell PowerEdge R740 server is connected to the same switch and functions as the IPTV middleware and content server, providing channel management, VOD services, and

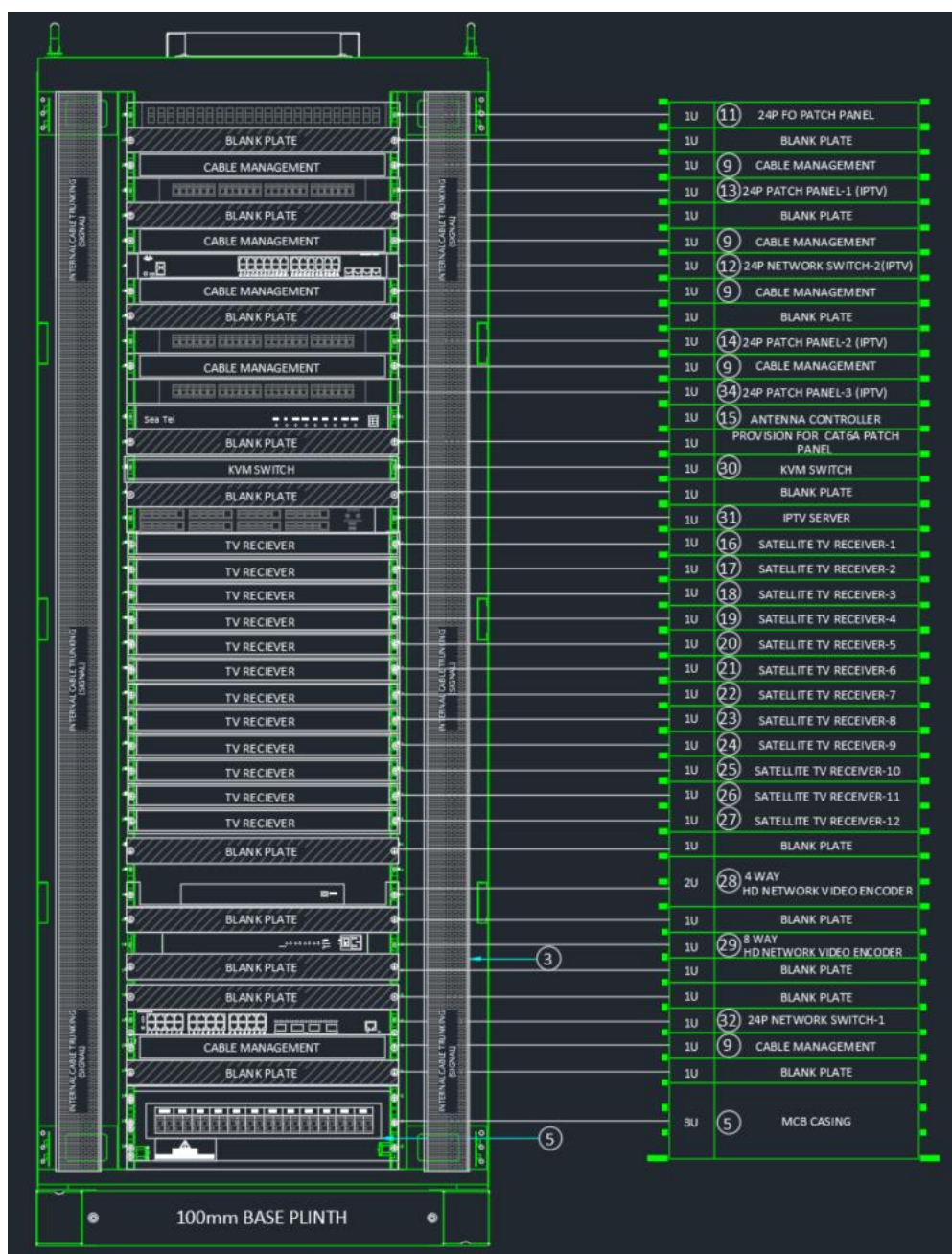
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system administration. The server also interfaces with other IP-based services such as CCTV and general LAN connectivity, enabling unified network management over the same backbone.

For network expansion and redundancy, Switch-1 is up linked to a secondary distribution switch, Switch-2 (Cisco C1300-24FP-4X), via high-speed Ethernet uplink ports. This inter-switch link extends IPTV and data services to network segments, supporting wider coverage across accommodation and operational areas. The Cisco switch also provides Power-over-Ethernet and additional port density for Smart TVs, IPTV set-top boxes, and auxiliary IP devices.

Overall, the signal flow progresses from RF satellite reception (Mult switch) to HDMI-based encoding, then into IP streaming via the Huawei aggregation switch. The network core interconnects with servers, KVM management, CCTV systems, and downstream access switches, forming a fully integrated IPTV and IP distribution architecture suitable for marine deployment.

3.1.10 INTERNAL CABINET LAYOUT AND EQUIPMENT MOUNTING



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The illustrated rack layout shows the IPTV headend equipment installed inside a floor-standing 19-inch cabinet designed for centralized onboard media distribution. The cabinet integrates all active and passive components required for satellite TV reception, IPTV processing, and network distribution within a structured and serviceable rack arrangement.

At the top of the cabinet, fibre optic and copper patch panels are installed together with horizontal cable management units to maintain structured routing and segregation of signal paths. Dedicated IPTV patch panels and network switches are arranged in the upper sections to provide organized interconnection between encoders, server, and vessel LAN infrastructure. Blank panels are used to maintain airflow control and allow space for future expansion.

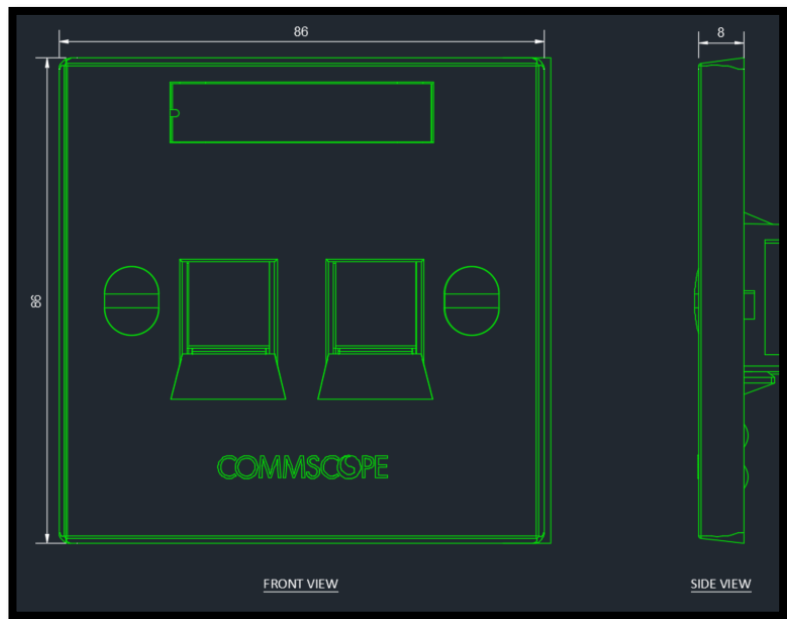
The mid-section of the rack houses the SeaTel antenna control interface, KVM switch, and IPTV server platform, enabling centralized monitoring and control of the headend system. Below this, multiple satellite TV receivers are installed in stacked configuration, providing multi-channel satellite signal demodulation for redistribution across the IPTV system.

The lower portion of the cabinet contains video encoding equipment, including multi-channel HD encoders that convert receiver outputs into IP streams. Additional network switches are installed to aggregate encoded streams and distribute them across the onboard network. Structured cable management panels are placed between active devices to ensure proper routing, airflow, and maintenance accessibility.

Power distribution and protection equipment are installed at the bottom section within an integrated MCB enclosure, providing controlled AC supply to all rack-mounted devices. The cabinet is mounted on a raised base plinth for mechanical stability and cable entry clearance.

3.1.11 TV OUTLET SOCKET

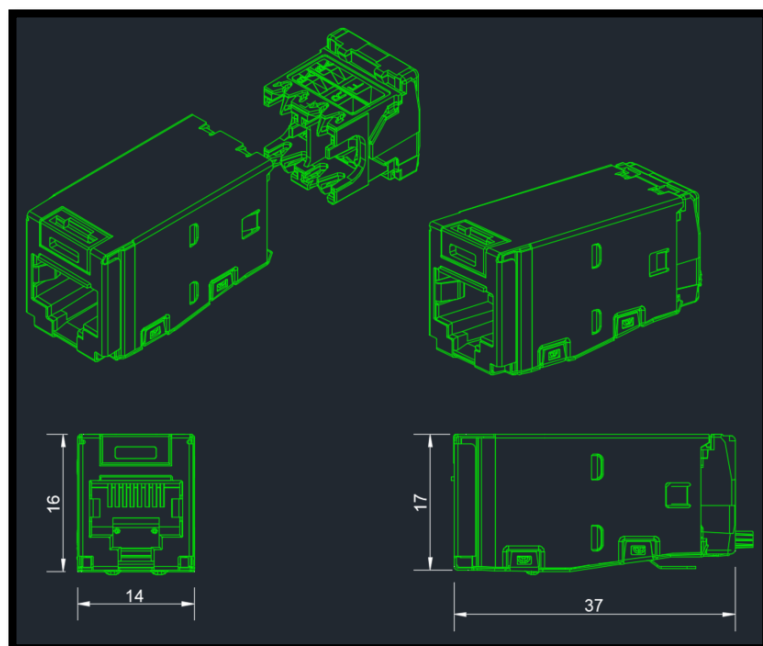
- COMMSCOPE 760249251



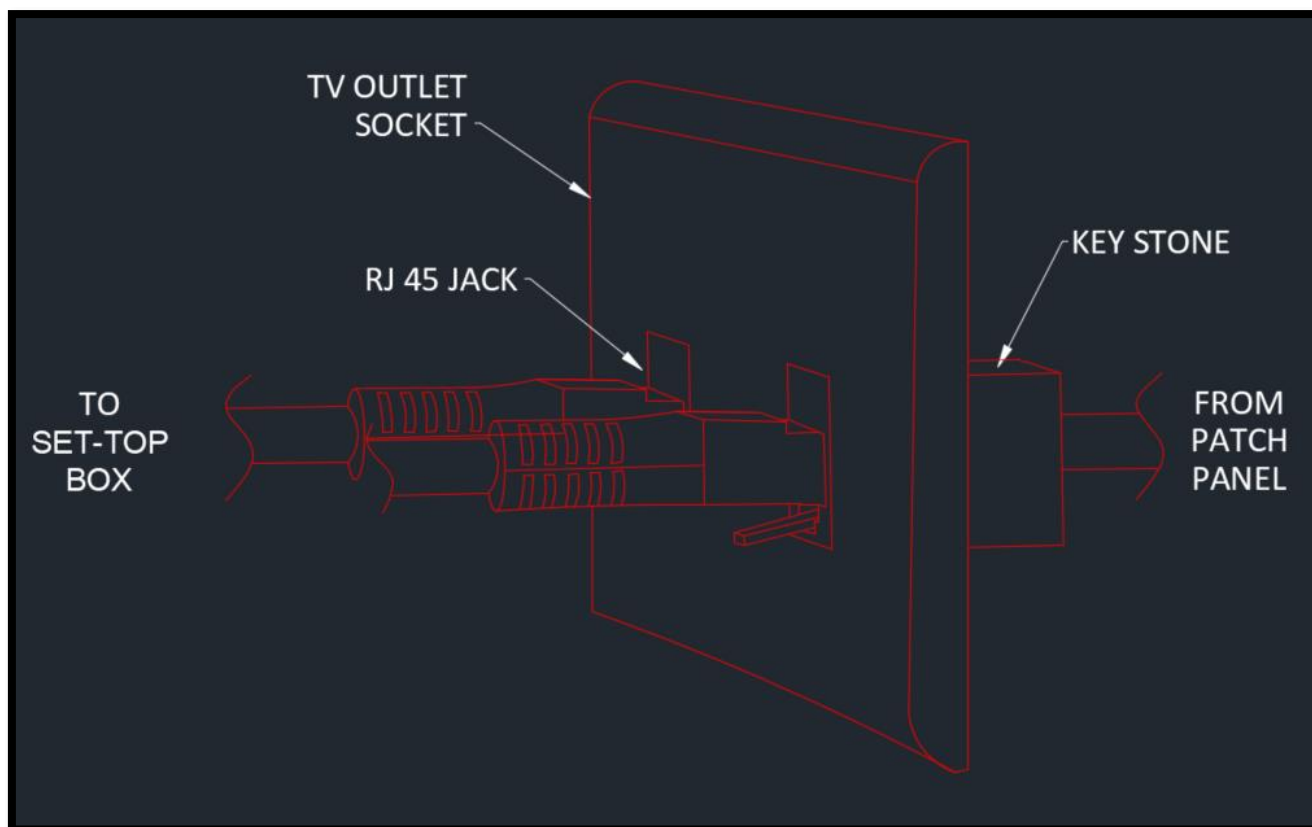
TECHNICAL SPECIFICATIONS:	
MANUFACTURER	: COMMSCOPE
MODEL	: 760249251
GENERAL SPECIFICATIONS	
COLOUR	: PRO WHITE
GANGS, QUANTITY	: 1
INTEGRATED DUST COVER TYPE	: SLIDING
OUTLET ORIENTATION	: FLAT
PORT CAPACITY, MAXIMUM	: 2
PORT MARKING TYPE	: LABEL
TOTAL PORTS, QUANTITY	: 2
PRODUCT TYPE	: FACEPLATE KIT
DIMENSIONS	
HEIGHT	: 85.98 mm 3.36 in
LENGTH	: 85.98 mm 3.36 in
DEPTH	: 8.00 mm 0.32 in
MATERIAL SPECIFICATIONS	
MATERIAL TYPE	: HIGH-IMPACT, FLAME RETARDANT, THERMOPLASTIC, UL LISTED MATERIAL
PACKING AND WEIGHTS	
WEIGHT, GROSS	: 0.036Kg 0.08 lb

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- COMMSCOPE 2153449-4



TECHNICAL SPECIFICATIONS:	
MANUFACTURER	: COMMSCOPE
MODEL	: 2153449-4
PRODUCT CLASSIFICATION	
PRODUCT TYPE	: MODULAR JACK
PRODUCT SERIES	: SLX SERIES
GENERAL SPECIFICATION	
NSI/TIA CATEGORY	: 6A
CABLE TYPE	: SHIELDED
COLOR	: GRAY
CONDUCTOR TYPE	: SOLID STRANDED
INTEGRATED DUST COVER TYPE	: NONE
MOUNTING	: SL SERIES FACEPLATES AND PANELS
OUTLET TYPE	: STANDARD
TERMINATION TOOL	: JTT TERMINATION TOOL
TERMINATION TYPE	: IDC
MECHANICAL	
PANEL CUTOUT	: 14.78 x 20.07mm (0.58 x 0.79in)
PANEL THICKNESS	: 1.6mm 0.063 in
HEIGHT	: 16.13 mm 0.635 in
WIDTH	: 14.44 mm 0.569 in



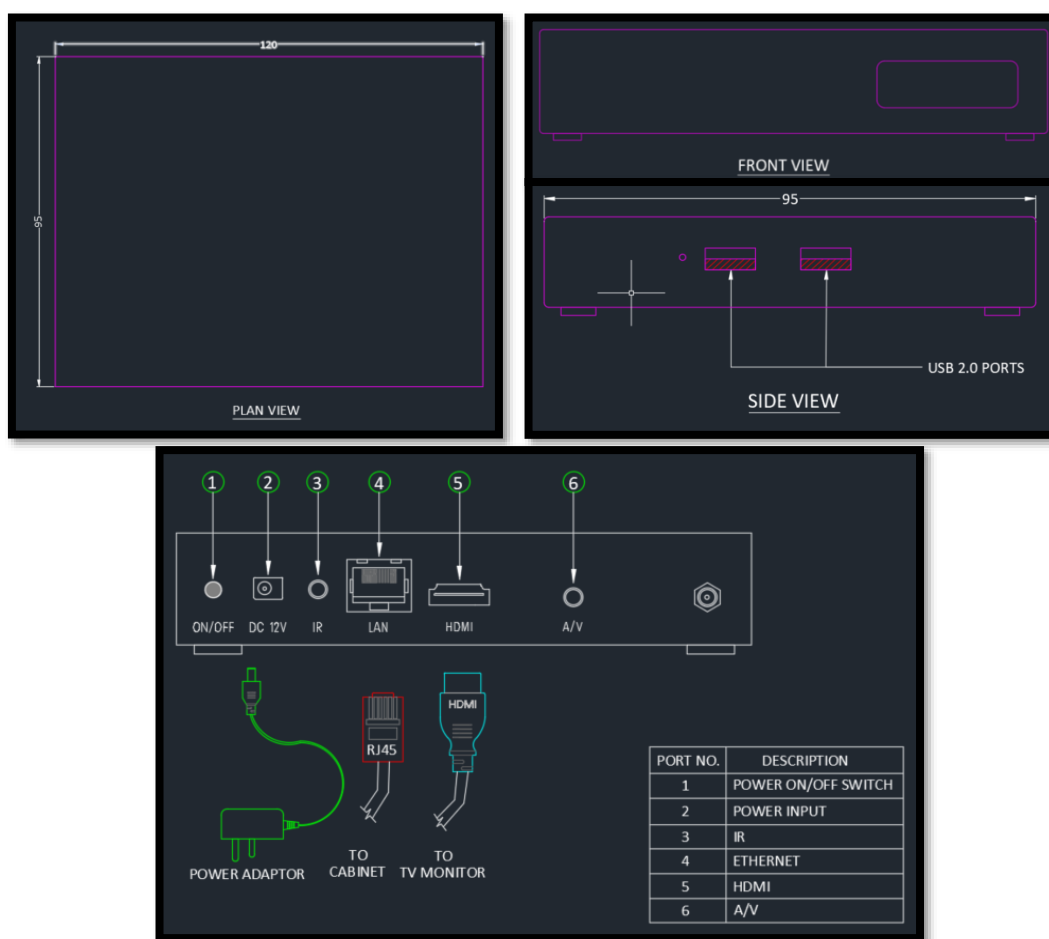
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The CommScope 760249251 faceplate is installed at the wall outlet location to provide a structured termination point for IPTV and TV signal connections. It accommodates modular keystone inserts, including RJ45 data jacks and TV outlet modules, ensuring a clean and secure interface for end-user connectivity.

The CommScope 2153449-4 TV outlet module is fitted into the faceplate and connected internally to the coaxial cable routed from the patch panel. As shown in the installation diagram, the RJ45 keystone jack connects to the structured cabling system, while the coaxial outlet connects to the set-top box or television equipment.

Together, the faceplate and outlet module provide a professional wall-mounted termination solution, ensuring organized cabling, reliable signal transmission, and easy maintenance within the IPTV distribution system.

3.1.12 IPTV SET-TOP BOX CLEAR H98MUV3



Component	Description
Model	CLEAR H98MUV3
Main Chip	GK6323V100C
CPU	Quad-core 64-bit ARM Cortex A55
GPU	Multi Core Mali-450, OpenGL ES 2.0/1.1, Open VG 1.1
System	Android 9.0
EMMC	EMMC 8GB (16GB Optional)
SDRAM	LPDDR4 2GB

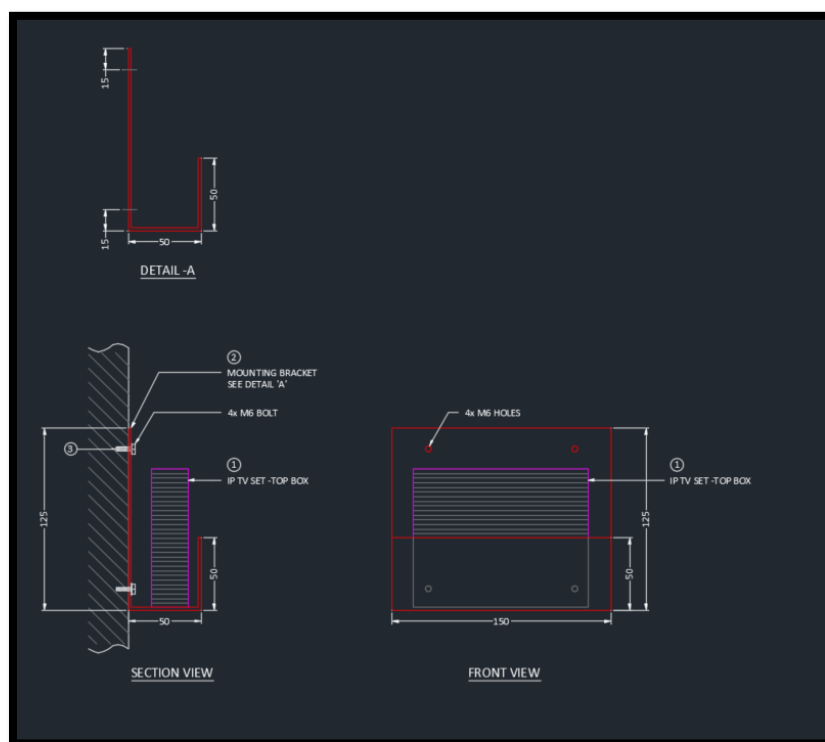
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WIFI	IEEE 802.11 a/b/g/n/ac, 2.4G/5G
Bluetooth	5.0
Video Decoding	▪ support for H.265 / AVS 2 video decoding -Maximum support of 4 Kp 60 ▪ Support for H.264 video decoding-Maximum support of 4 Kp 30 ▪ Support for MPEG / AVS decoding-Maximum support of 1080p60
Audio Decoding	▪ support for Dolby Digital / Dolby Digital Plus -Decoder-Converter ▪ Support for AAC-LC, HE AAC V1 / V2 decoding ▪ Support for APE / FLAC / Ogg / AMR-NB / WB decoding ▪ Supports G.711 (u / a) audio decoding ▪ Support for G.711 (u / a) / AMR-NB / AMR-WB / AAC-LC audio

The Clear H98MUV3 IPTV set-top box is a compact Android-based media receiver designed for IPTV and VOD service delivery in marine or enterprise environments. It functions as the end-user decoding device within the IPTV system, receiving multicast or unicast IP video streams from the IPTV server over the local network and converting them into HDMI output for display on televisions.

The unit supports high-definition video playback, including H.264/H.265 formats, ensuring smooth decoding of live TV channels and on-demand content. It connects to the network via an Ethernet interface and integrates easily with structured cabling systems using RJ45 outlets. The HDMI output provides direct connection to smart TVs or standard displays, while USB ports allow firmware updates or local media playback if required.

In an FPSO or offshore installation, the H98MUV3 operates as the subscriber endpoint of the IPTV distribution system, enabling crew access to live satellite TV, internal broadcast channels, and VOD services with stable performance and simple plug-and-play operation



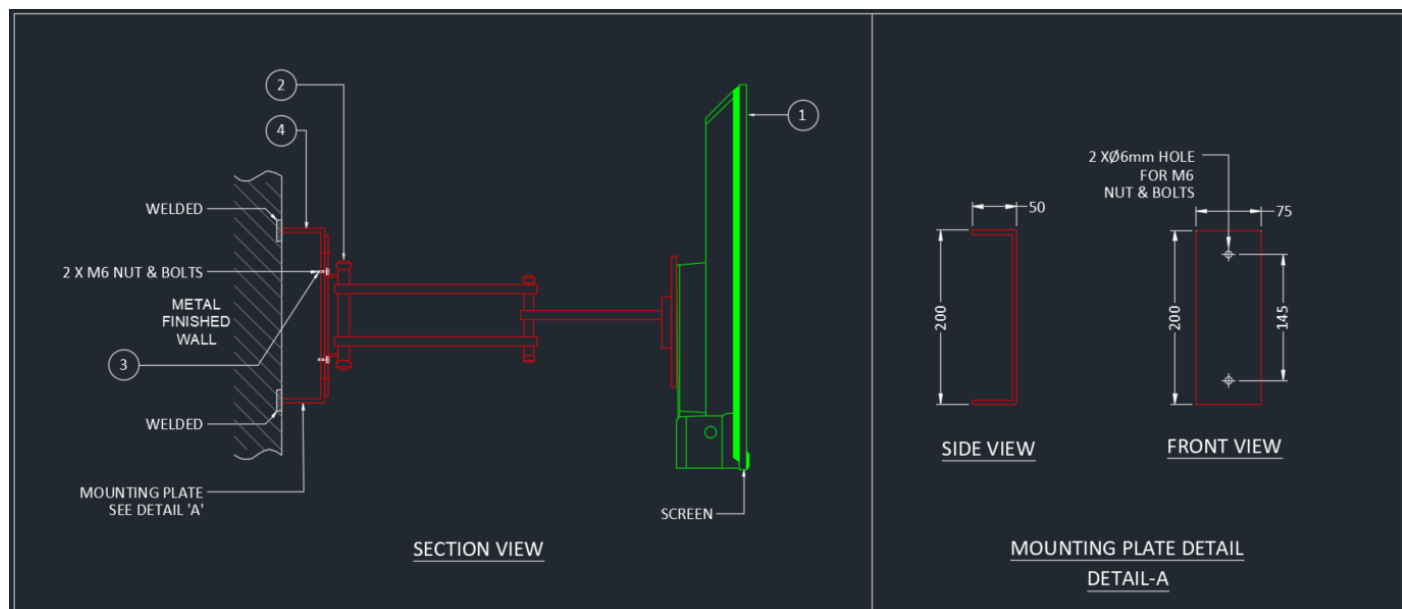
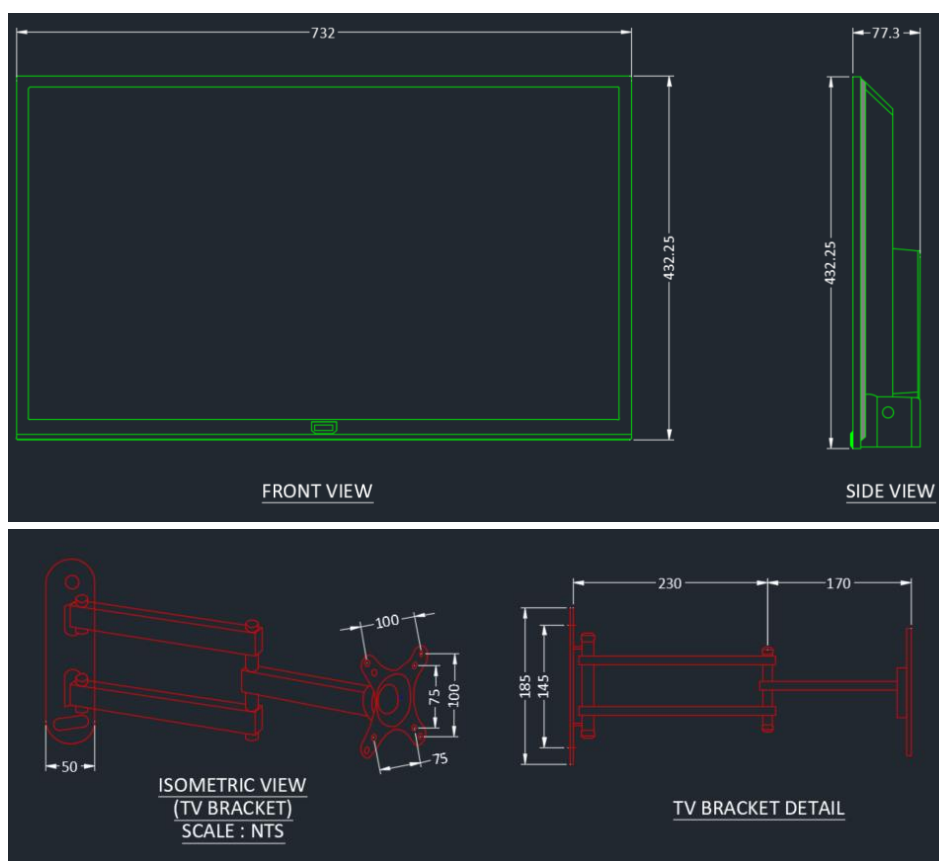
The IPTV set-top box is wall-mounted using a dedicated steel mounting bracket as shown in the installation detail. The bracket is fabricated in an L-shape profile and secured to the bulkhead using four M6 bolts, ensuring firm mechanical

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support in marine environments. The set-top box is then fixed onto the bracket with appropriate fasteners, maintaining a minimum clearance for ventilation and cable routing.

The installation allows easy access to rear connections, including power, HDMI output, and Ethernet interface, while keeping the unit securely fixed against vibration and movement. This mounting method ensures stable operation of the IPTV receiver and facilitates maintenance or replacement when required.

3.1.13 SMART TV PHILIPS 32HFL3014

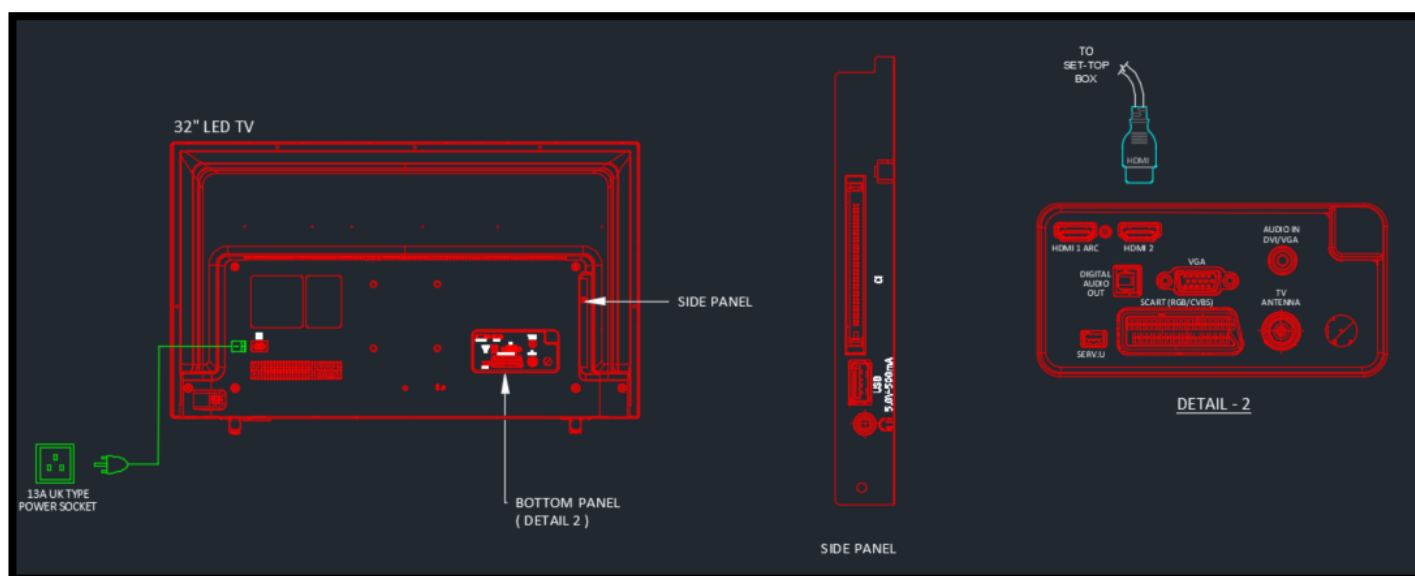


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The SMART TV is installed using a heavy-duty wall-mounted articulating bracket suitable for marine environments. A steel mounting plate is first welded or securely fixed to the metal bulkhead, ensuring a rigid and vibration-resistant foundation. The wall plate of the TV bracket is then aligned with the mounting plate and secured using two M6 bolts and nuts through the pre-drilled holes, as indicated in Detail-A.

Once the bracket base is firmly fixed, the articulated arm assembly is attached and tightened to eliminate unwanted movement. The TV mounting interface is fixed to the rear of the SMART TV using manufacturer-supplied screws, ensuring proper load distribution. The television is then hooked or bolted onto the bracket arm and mechanically locked in position.

After mechanical installation, HDMI and power cables are routed with adequate slack to allow arm movement while preventing strain on connectors. The articulating arm allows viewing angle adjustment and facilitates maintenance access. All fasteners are tightened and checked to ensure the installation remains secure under vibration and vessel motion conditions.



The SMART TV wiring is completed by connecting the HDMI output from the IPTV set-top box to the HDMI input located on the TV rear interface panel. Power is supplied via a 13A AC plug connected to the nearest utility socket. If required, network or service connections can be accessed through the side panel ports.

All cables are routed neatly along the bracket and secured to avoid strain during TV movement. Adequate slack is maintained to allow articulation while ensuring connectors remain firmly seated. Wiring is dressed and tied to prevent interference, mechanical damage, or loosening due to vessel vibration.

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			2024-VIVO-EMEM-TS-IM-012		
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3.1.14 IP TV DISTRIBUTION



The IPTV distribution system delivers TV content from the central IPTV system cabinet in the Telecom Room to multiple TVs located across different decks and rooms. Video streams are transmitted over the IP network using structured CAT6 cabling via network switches, ensuring stable and high-quality signal delivery. Each TV point is terminated at a local data outlet and connected through HDMI to the display, allowing centralized content management, monitoring, and easy expansion of the IPTV services throughout the vessel.

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3.2 OPERATION – IP TV SYSTEM

3.2.1 OPERATION MANUAL AND CONFIGURATION FOR ANTENNA CONTROLLER SEATEL LMXP

The first thing you need to do is to configure your computer so that it will display the MXP screens. Follow these instructions to accomplish that.

- Initial Network Connection
 - Connect an Ethernet (LAN) cable to the network port of the computer. If connecting through the vessel LAN instead of directly to a single PC, ensure the LAN router/switch is linked to the LMXP.
 - Connect the other end of the LAN cable to the Ethernet port on the rear of the LMXP.
 - Power ON the LMXP and allow the unit to complete start-up.
 - From the computer desktop, open the Control Panel.
 - Select “View network status and tasks.”
- Network Adapter Settings
 - Click “Change adapter settings.”
 - Select “Local Area Connection.”
 - Click “Properties.”
 - Double-click “Internet Protocol Version 4 (IPv4).”
 - Select “Use the following IP address.”
- IP Configuration
 - In the IP address field, enter 10.1.1.102 for the computer.
 - Note: Alternative addresses such as 10.1.1.101, 10.1.1.103, etc. may be used, provided they do not match the LMXP default address (10.1.1.100).
 - Enter the Subnet Mask as 255.255.255.0.
 - Click “OK” to save the IPv4 configuration.
 - In the Local Area Connection Properties window, click “OK.”
 - Click “Close.”
 - Close the Control Panel window.
- Accessing the LMXP Interface
 - Launch a web browser and enter the following address in the URL bar: <http://10.1.1.100>
 - At the login screen, enter the appropriate username (Dealer, Sys Admin, or User). The password is issued by Cobham/Sea Tel service support.

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- After successful login, the System Status page will be displayed.

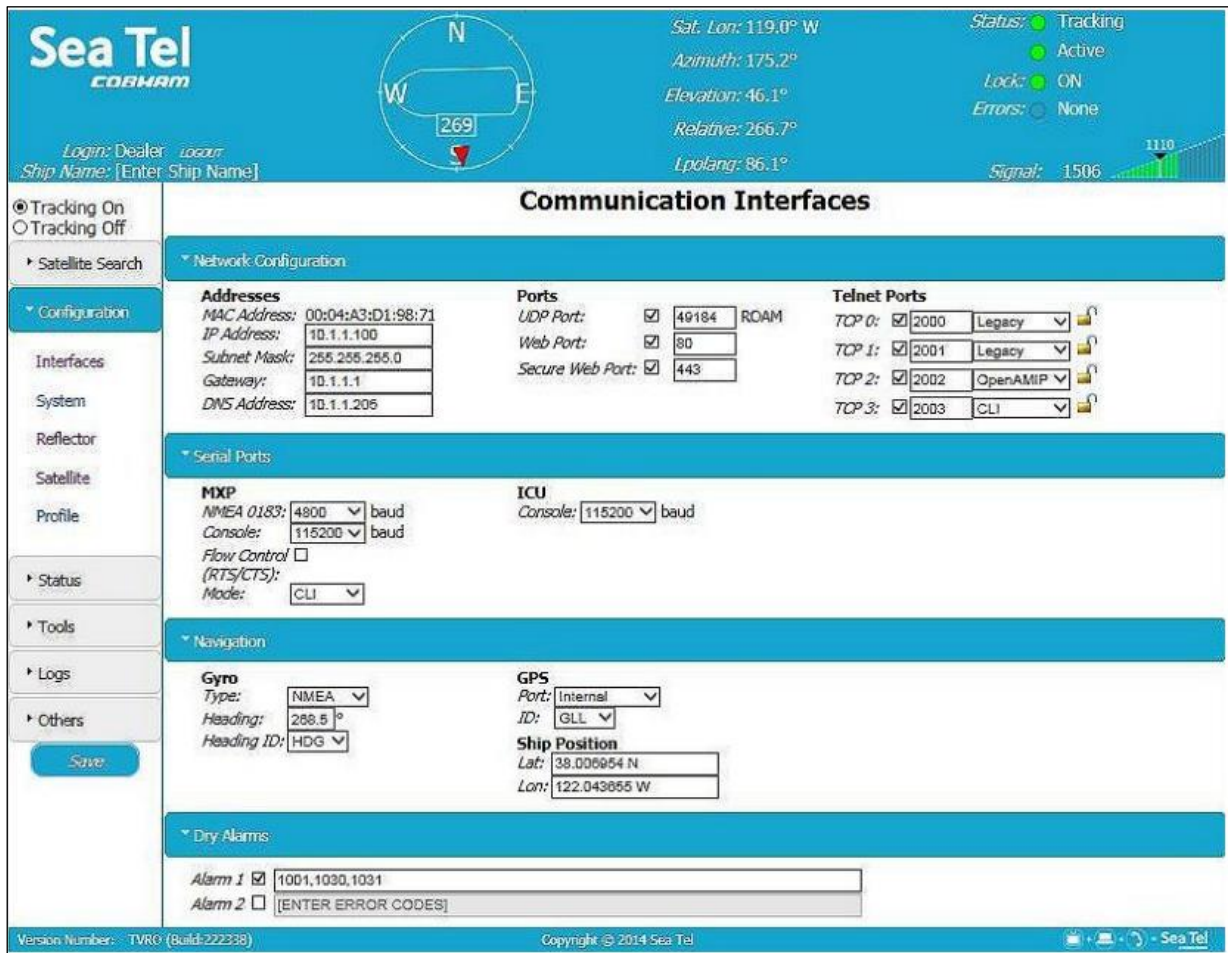


- The top banner on every page display:
 - Sea Tel/Cobham branding
 - User login level and a Logout option
 - Ship name
 - A compass graphic showing heading and antenna pointing
 - Satellite info (longitude, azimuth, elevation, polarization)
 - Signal bar graph with threshold indicator
- The menu on the left includes navigation tabs:
 - Satellite Search (Auto)
 - Configuration
 - Status

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- Tools
- Logs
- Help
- Save button

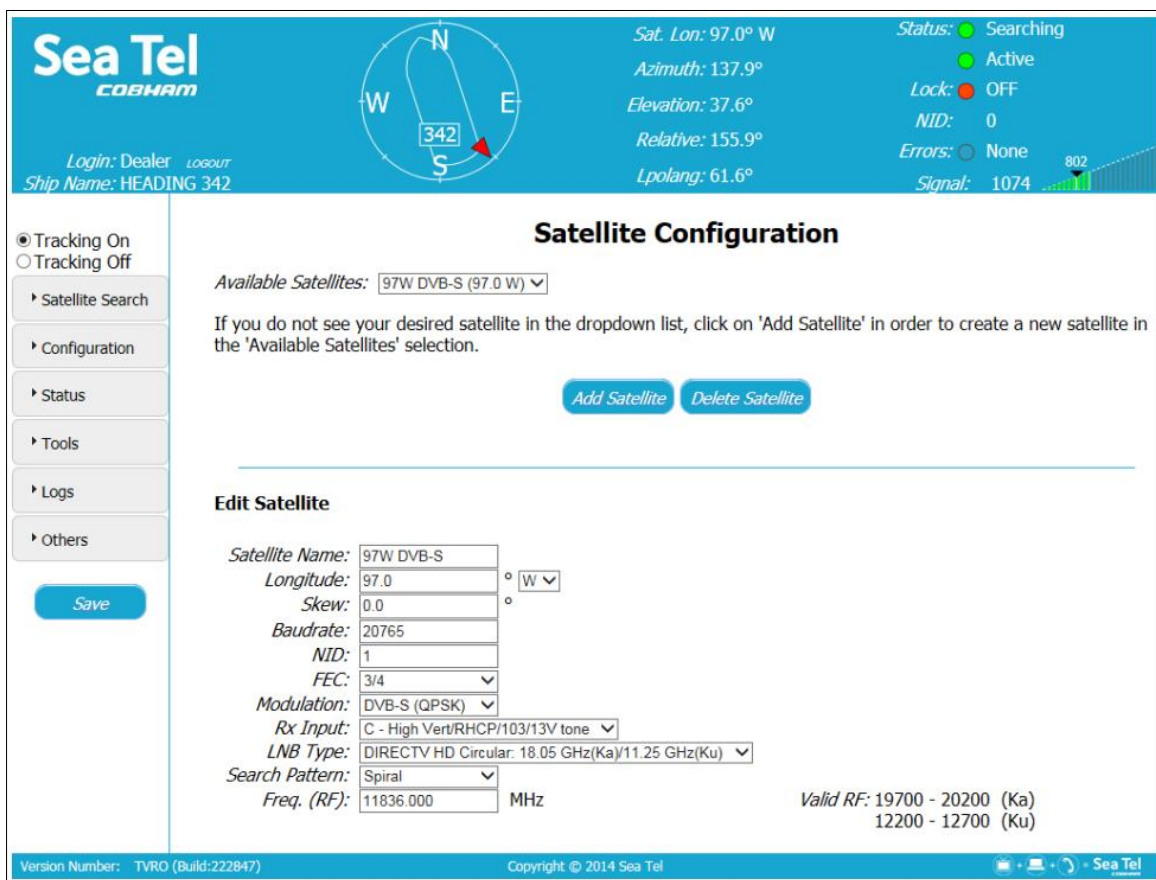
- Satellite Search – Auto
 - Displays current ship location (from GPS).
 - Contains a dropdown list of saved satellites (“favourites”)
 - Select the desired satellite and the system will target and track it
- Network Configuration
 - IP addresses, ports, and telnet settings for LMXP and network
 - Default settings are suitable for simple single-PC access; adapt for LAN use as needed



- Configures serial communication ports
 - NMEA 0183 Gyro input (default 4800 baud)
 - Console port (default 115200 baud)
 - Flow control and mode settings
 - Most leave factory defaults unless external devices are present

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- Defines how LMXP uses heading and position:
 - Choose gyro type: No Gyro/Fixed heading/NMEA 0183 (preferred for dynamic heading)
 - Internal GPS supplies latitude/longitude automatically
 - Manual entry only needed if GPS fails
- Dry Alarms
 - Assigns alarm contact outputs for selected warning or error conditions
 - When configured and connected, these dry contacts report system statuses externally
- Satellite Selection
 - Use a dropdown to select a saved satellite preset.
 - Once chosen, the antenna will target and attempt to acquire it
- To create a new satellite preset:



Sea Tel COBHAM

Login: Dealer Logout Ship Name: HEADING 342

Sat. Lon: 97.0° W Azimuth: 137.9° Elevation: 37.6° Relative: 155.9° Lpolang: 61.6°

Status: Searching Active Lock: OFF NID: 0 Errors: None Signal: 1074

Satellite Configuration

Available Satellites: 97W DVB-S (97.0 W)

If you do not see your desired satellite in the dropdown list, click on 'Add Satellite' in order to create a new satellite in the 'Available Satellites' selection.

Add Satellite Delete Satellite

Edit Satellite

Satellite Name: 97W DVB-S Longitude: 97.0 W Skew: 0.0 Baudrate: 20765 NID: 1 FEC: 3/4 Modulation: DVB-S (QPSK) Rx Input: C - High Vert/RHCP/103/13V tone LNB Type: DIRECTV HD Circular: 18.05 GHz(Ka)/11.25 GHz(Ku) Search Pattern: Spiral Freq. (RF): 11836.000 MHz

Valid RF: 19700 - 20200 (Ka) 12200 - 12700 (Ku)

Version Number: TVRO (Build:222847) Copyright © 2014 Sea Tel

- Click **Add Satellite**
 - Enter:
 - Satellite name
 - Longitude (with East/West selection)
 - RF frequency (MHz)

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- Skew (degrees, if required)
 - Baud rate and network ID (NID)
 - Save to the favorites list
 - To delete Click, Delete Satellite to remove it from the favorites list.
- System status displays read-only info such as:
 - GPS position values
 - Antenna pointing angles (azimuth, cross-level, elevation)
 - Current machine states and relevant telemetry

Quick Start Operation

Provided the system configuration is correct and the vessel has not moved since the last operation, the antenna will automatically perform satellite acquisition following a cold start. Upon successful acquisition, the modem will achieve lock, enabling normal service operation.

- Turn Power ON
 - Switch ON the LMXP and the antenna pedestal by setting the front-panel power switch on the LMXP to the **ON** position.



- Power ON all other below-decks equipment, including satellite receivers, distribution units, and audio/video devices.
- The antenna system will start up, complete its initialization sequence, and then automatically retarget the last satellite that was previously in use.

If satellite signal is found AND network lock is achieved:

- The system will automatically enter tracking mode (the Tracking LED on the front panel will light up) and the antenna will optimize its pointing to achieve the strongest possible received signal from the selected satellite.



- When the received signal level is above the configured threshold and the modem successfully establishes network lock, the antenna will continue to follow the satellite without user intervention.

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- The configured satellite name (if available), tracking status, modem lock status, and signal strength (both numeric readout and bar indicator) will be displayed on the top banner of the LMXP graphical interface.

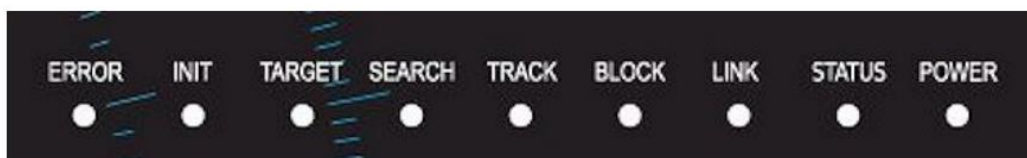
Once these conditions are satisfied, the system will remain in normal automatic operation until any of the following events occur:

- System AC power is switched off or lost, or
- The satellite line-of-sight becomes obstructed, or
- The vessel enters a coverage area with insufficient satellite signal strength.

To Target a different satellite:

- Log in to the LMXP graphical user interface (GUI).
- To switch to another satellite, open the Satellite Search – Auto page and choose the required satellite from the drop-down menu.
- After selecting the satellite, a temporary status message will appear: “Acquiring satellite signal... Please wait.”
- Once acquisition is successful, the status message will update to indicate: “Satellite signal detected. Lock: ON.”

Front Panel Status LEDs



The basic description of the front panel LED states are:

ERROR - ON [Solid or Flashing Red OR Solid or Flashing Amber] indicates that one, or more, discrete system errors have occurred. Refer to Status – Error Code information menu to determine which error(s) have occurred.

OFF indicates that no errors have occurred

INIT (Initializing) - (Solid Green) indicates that the Antenna is initializing. Initialization of the antenna will take approximately two minutes. (Flashing Amber) indicates that a software update in progress, the system (ADE-BDE) is synchronizing or in service/out of service testing is in progress.

TARGET -ON (Solid Green) indicates that the antenna is TARGETING (driving) to the specified Azimuth and/or Elevation position(s).

SEARCH (Searching) - ON (Solid Green) indicates that LMXP is actively searching for your satellite signal. OFF indicates that SEARCH is OFF.

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TRACK (Tracking) - ON (Solid Green) indicates that the LMXP has identified and is actively tracking the desired satellite to optimize the signal level (AGC). Blinking indicates that the LMXP is in search delay or that the system is analyzing a satellite signal. OFF indicates that Tracking is OFF.

BLOCK - ON (Solid Red) indicates that the antenna is within a defined blockage zone (therefore you should not have satellite signal until the antenna is no longer within the blockage zone). OFF indicates that the antenna is not within a defined blockage zone). If no blockage zones have been entered, the antenna could be blocked causing loss of signal.

LINK - (Green LED) ON (Solid Green) indicates that the LMXP has good communications with the antenna (therefore good antenna control). This is normal state for system operation. ON (Solid Red) indicates LOSS of communications with the antenna (therefore, loss of antenna control). Check the coax connections on the LMXP and in the base of the radome.

STATUS - (Green LED) ON (Solid Red) indicates that the signal is below threshold (satellite signal is low or lost). OFF indicates that status is normal.

POWER - (Green LED) OFF indicates that LMXP Power switch is OFF (No power to the LMXP or the antenna). ON (Flashing Green) indicates that the LMXP is booting up. ON (Solid Green) indicates that the LMXP boot sequence is complete and the system power is ON (normal).

3.2.2 BASIC OPERATION – IPTV SYSTEM

- System Power-Up Sequence
 - Ensure the LMXP and antenna system are powered ON and tracking the selected satellite with stable signal.
 - Power ON the coaxial Mult switch and verify satellite IF inputs are active.
 - Power ON the IPTV encoders (CLEAR-H3118AV / CLEAR-H311AV).
 - Power ON the IP network switches (Huawei S5735-L24T4S-A-V and Cisco C1300-24FP-4X).
 - Power ON the IPTV server (Dell PowerEdge R740) and confirm the IPTV application/services are running.
 - Power ON IPTV Set-Top Boxes (STBs) and/or Smart TVs at user locations.
- Signal Flow Check
 - Satellite RF from antenna → Coaxial Multi switch
 - Multi switch IF outputs → CLEAR Encoders (HDMI/AV inputs)
 - Encoders IP output → Huawei Switch (Core) → Cisco Switch (Access)
 - Network → IPTV Server → Smart TVs

Make sure Encoder channel LEDs/status show streaming active. Network switch ports show link/activity. IPTV server shows channels online.

- IPTV Channel Availability

On the IPTV server, verify that all encoder inputs are correctly mapped to the corresponding channels and that channel names and numbers are properly published. From the Smart TV, refresh or reload the channel list. Select a channel and confirm that video is displayed correctly, audio is clear, and there is no excessive buffering or freezing.

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- Normal Operation

Once the satellite link is stable, the IPTV system will operate continuously. Users can switch channels from the STB remote or TV IPTV menu.

The IPTV server manages:

- Channel distribution
- Multicast/unicast streaming
- User access

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3.3 MAINTENANCE AND TROUBLESHOOTING

The IPTV system requires minimal routine maintenance during normal operation. Periodic visual inspections shall be carried out to ensure that all equipment, including the satellite antenna system, coaxial multi switch, IPTV encoders, network switches, IPTV server, and end-user devices (Smart TVs), are powered ON, free from physical damage, and operating within normal status indications. Network connectivity and streaming performance should be verified at regular intervals by checking selected channels for correct video and audio quality without excessive buffering or interruptions. Cooling fans, ventilation paths, and rack-mounted equipment shall be kept clean and unobstructed to prevent overheating. Firmware and software updates for the LMXP, encoders, IPTV server, and network switches should be performed as per manufacturer recommendations and vessel IT change management procedures. Any abnormal alarms, signal degradation, or service interruptions shall be recorded and investigated promptly, and corrective actions shall be carried out by qualified technical personnel.

3.3.1 MAINTENANCE OF IPTV SYSTEM

Routine preventive maintenance shall be carried out to ensure continuous and reliable operation of the IPTV and TVRO antenna system. Shipboard personnel may perform basic inspections and functional checks; however, internal servicing and calibration of the antenna pedestal and LMXP controller shall be performed only by authorized service personnel.

TVRO Antenna – Sea Tel 120TV

- Perform periodic visual inspection of the radome to check for cracks, physical damage, or water ingress.
- Clean the radome surface regularly to remove salt deposits and oil residues.
- Inspect the antenna mounting base, pedestal, and fasteners to ensure they remain secure and free from abnormal vibration.
- Check coaxial cables, cable glands, and grounding connections for tightness, corrosion, and mechanical damage.
- Monitor antenna tracking performance, signal level, and stabilization status via the LMXP GUI to confirm stable satellite lock during normal operation.

Antenna Controller – Sea Tel LMXP

- Verify power status and front-panel LED indications during routine checks.
- Ensure adequate ventilation around the LMXP and keep air inlets/outlets free from obstruction to prevent overheating.
- Inspect Ethernet, coaxial, and power connections at the rear panel for secure termination and proper strain relief.
- Back up configuration files after commissioning and following any configuration changes.
- Perform firmware/software updates only in accordance with manufacturer recommendations and vessel IT change management procedures.

IPTV System

- Check operational status of IPTV encoders, IPTV server, and network switches, and confirm normal link/activity indications on switch ports.

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- Periodically verify selected TV channels to ensure correct video and clear audio without freezing, pixelation, or excessive buffering.
- Inspect rack-mounted equipment, cooling fans, and power distribution units, and keep ventilation paths clean and unobstructed.
- Monitor network performance to ensure stable streaming across the vessel.

3.3.2 TROUBLESHOOTING GUIDE

LMXP (No Satellite Acquisition After Cold Start)

Step	Observation / Check	Expected Indication	Corrective Action
1	Front panel LEDs during startup	Tracking LED flashes briefly, followed by Search LED ON	This indicates the system has entered satellite search mode. Allow the search cycle to continue.
2	Antenna search behaviour	Antenna moves in a programmed search pattern	Observe the signal bar graph on the LMXP GUI. Red bars indicate signal level is below the threshold.
3	Signal bar graph status	Bar graph remains red (signal below threshold)	This confirms that no valid satellite signal has been detected during the current search sweep.
4	Repeated search cycles	Antenna retargets and repeats the search sequence	Allow the system to continue searching until the satellite is acquired or operator intervention is required.
5	Operator access	Log in to the LMXP GUI	Proceed with manual verification and configuration checks.
6	Communication interface settings	Navigate to Configuration → Communication Interfaces	Access position and heading parameters for verification.
7	Vessel position data	Latitude, Longitude, and Heading values displayed	If values are correct, proceed to Step 10.
8	Position correction	Incorrect Latitude/Longitude	Enter the correct vessel Latitude and Longitude values and apply the changes.
9	Heading correction	Incorrect Heading value	Enter the correct heading value and apply the changes.
10	Satellite selection page	Open Satellite Search – Auto page	Prepare to verify the selected satellite.
11	Satellite selection	Correct satellite selected	If correct, proceed to Step 14.
12	Satellite re-selection	Incorrect satellite selected	Select the required satellite from the drop-down list.
13	Save configuration	Configuration changes applied	Click Save to store the updated settings.
14	Line-of-sight check	Possible physical blockage	Inspect for obstructions (masts, cranes, structures). Blockage is the most common cause of acquisition failure.
15	Cable inspection	Loose or disconnected cables	Verify all RF, control, and power cable connections between ADE and BDE are secure.
16	Escalation	System still not acquiring satellite	If the antenna is unobstructed and all connections are confirmed, contact the authorized dealer or service support.

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4
PROJECT SPECIFIC DRAWINGS

TELECOM CABLE BLOCK DIAGRAM - IPTV SYSTEM

REVISIONS

Z	23.12.2025	ISSUED FOR AS-BUILT	CK	KK	TS	
0	23.04.2025	ISSUED FOR CONSTRUCTION	CK	KK	TS	
B	16.10.2024	ISSUED FOR APPROVAL	CK	KK	TN	
A	04.10.2024	ISSUED FOR REVIEW	CK	KK	TN	
REV.	DATE	DESCRIPTION	DRWN	CHKD	APPD	CLIENT

COMPANY:



Oriental
OKWOK



ADDAX
PETROLEUM

CONTRACTOR:



WORLD
CARRIER
CORPORATION



HBA Offshore

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PROJECT:

OKWOK-FPSO
TOPSIDES PRODUCTION FACILITIES

LOCATION:

OKWOK FIELD OFFSHORE NIGERIA-OML 67

TITLE:

TELECOM CABLE BLOCK DIAGRAM – IPTV SYSTEM

SCALE:

DRAWING NO.
2024-VVO-EMEM-TS-BD-012

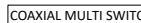
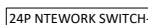
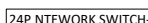
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SHEET:

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SHEET NO.	REVISION				DESCRIPTION	REMARKS
	A	B	O	Z		
1	04.10.2024	16.10.2024	23.04.2025	23.12.2025	COVER PAGE	
2	04.10.2024	16.10.2024	23.04.2025	23.12.2025	INDEX PAGE	
3	04.10.2024	16.10.2024	23.04.2025	23.12.2025	IPTV RACK- ANTENNA CONNECTION AND INTERFACE DIAGRAM	
4	04.10.2024	16.10.2024	23.04.2025	23.12.2025	IPTV SYSTEM CABLE BLOCK DIAGRAM	
5	04.10.2024	16.10.2024	23.04.2025	23.12.2025	IPTV SYSTEM CABLE BLOCK DIAGRAM	
6	04.10.2024	16.10.2024	23.04.2025	23.12.2025	IPTV SYSTEM CABLE BLOCK DIAGRAM	

BILL OF MATERIAL			
S/N	SYMBOL	DESCRIPTION	QTY
1		4 WAY ENCODER MODEL : H3114AV (4-WAY) MANUFACTURER : CLEAR	01
2		8 WAY ENCODER MODEL : H3118AV (8-WAY) MANUFACTURER : CLEAR	01
3		IPTV SERVER WITH VOD MODEL : CLEAR SKY MANUFACTURER : CLEAR	01
4		COAXIAL MULTI SWITCH MODEL : TMP5x24 MANUFACTURER : TRIAX	01
5		24P NTEWORK SWITCH MODEL : S5735-L24T4S-A-V MANUFACTURER : HUAWEI	01
5		24P NTEWORK SWITCH MODEL : C1300-24FP-4X MANUFACTURER : CISCO	01
6		KVM MODEL : CL5708/2L-5201U MANUFACTURER : ATEN	01
7		SMART TV MODEL : 32HFL3014 MANUFACTURER : PHILIPS	17
8	 NOTE-1	TV OUTLET SOCKET MODEL : 2153449-4 MANUFACTURER : COMMSCOPE	57
9		IPTV SET-TOP BOX MODEL : CLEAR H98MUV3 MANUFACTURER : CLEARTV	17
10		13A UK TYPE POWER SOCKET OUTLET (SUPPLY BY OTHERS)	34
11		IPTV / TVRO ANTENNA MODEL : 120TV MANUFACTURER : SEATEL	01
12		ANTENNA CONTROLLER MODEL : LMXP MANUFACTURER : SEATEL	01
13		SATELLITE TV RECEIVER (BY OTHERS) MODEL : TBA MANUFACTURER : TBA	12

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0	23.04.2025	ISSUED FOR CONSTRUCTION	CK	KK	TS	
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 E-mail : contract@vivoasia.com

PROJECT:

OKWOK-FPSO
TOPSIDES PRODUCTION FACILITIES

LOCATION:

OKWOK FIELD OFFSHORE NIGERIA-OML 67

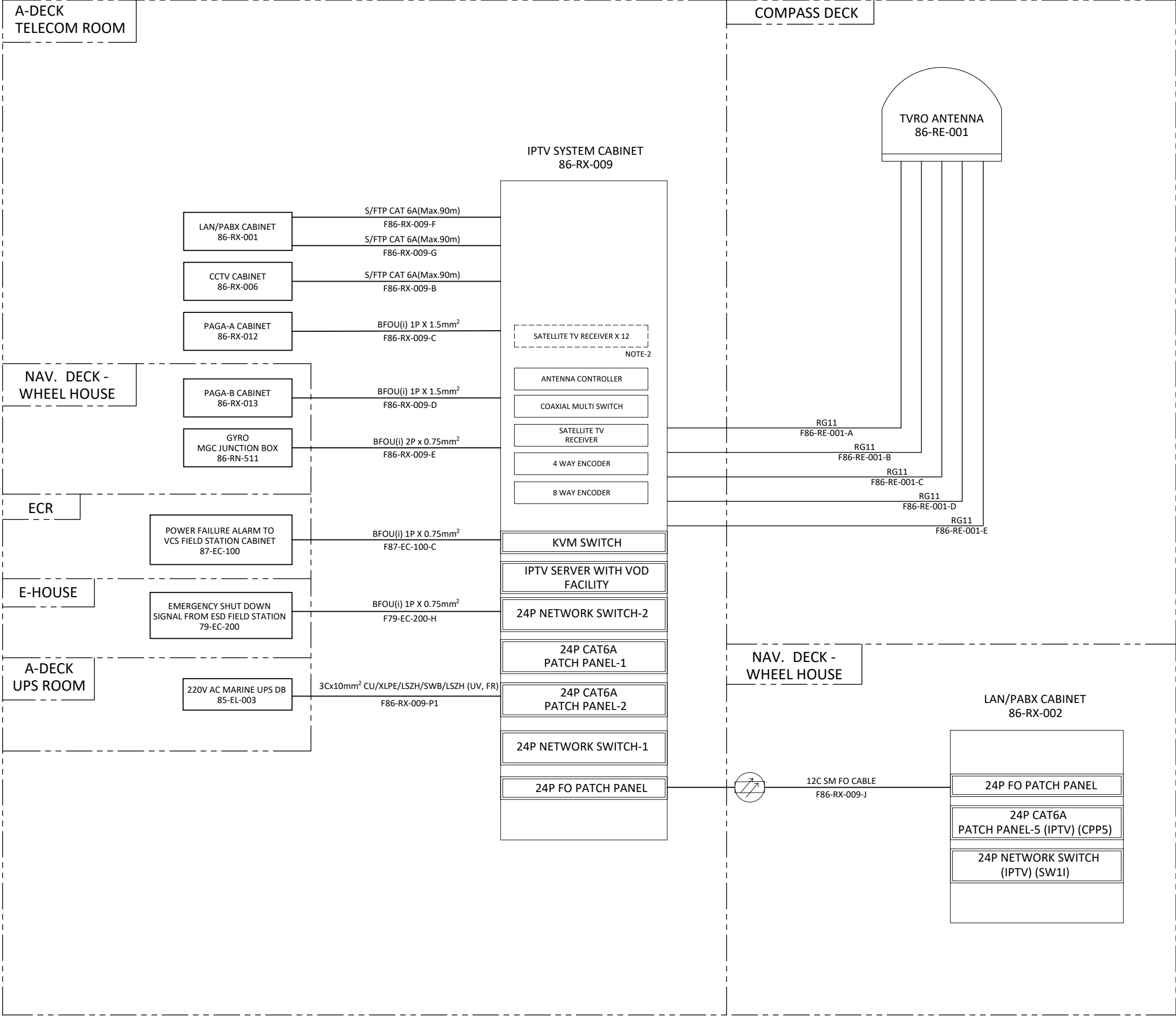
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TELECOM CABLE BLOCK DIAGRAM – IPTV SYSTEM

SCALE:	DRAWING NO. 2024-VIVO-EMEM-TS-BD-012	WCP DRAWING NO. WCPQKWTP-450-TE-(VAE)BF-010	REV.
	SHEET: 02 of 06		0

NOTE :

IPTV RACK- ANTENNA CONNECTION AND INTERFACE DIAGRAM



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A	04.10.2024	ISSUED FOR REVIEW	CK	KK	TN		
REV.	DATE	DESCRIPTION	DRWN	CHKD	APPD	CLIENT	
COMPANY:							
		ORIENTAL OIL	 ADDAX PETROLEUM				
CONTRACTOR:							
		WORLD CARRIER CORPORATION					
VENDOR:							
 VIVO ASIA ENGINEERING & TRADING		TP : +65 6826 2525 Head Office : 21 Toh Guan Rd E, #06-06A, Singapore 608609 Web : vivoasia.com E-mail : contract@vivoasia.com					
PROJECT:							
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TITLE:							
TELECOM CABLE BLOCK DIAGRAM – IPTV SYSTEM							
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IPTV SYSTEM CABLE BLOCK DIAGRAM

The diagram illustrates the IPTV system cable block diagram, showing connections between various rooms and decks. The rooms are categorized into four main sections: A-Deck - Telecom Room, A-Deck, Raised Deck, and C-Deck. The connections are detailed in the table below.

Room	Room ID	TV	HDMI Cable	Patch Cable	Connection
MEETING ROOM	86-RE-301	TV	HDMI CABLE	86-RE-201	F86-RE-101 S/FTP CAT 6A(Max.90m)
MESS ROOM	86-RE-302	TV	HDMI CABLE	86-RE-202	F86-RE-102 S/FTP CAT 6A(Max.90m)
RECREATION ROOM	86-RE-303	TV	HDMI CABLE	86-RE-203	F86-RE-103 S/FTP CAT 6A(Max.90m)
DOCTOR'S OFFICE	86-RE-304	TV	HDMI CABLE	86-RE-204	F86-RE-104 S/FTP CAT 6A(Max.90m)
GYM ROOM	86-RE-305	TV	HDMI CABLE	86-RE-205	F86-RE-105 S/FTP CAT 6A(Max.90m)
HOSPITAL	86-RE-306	TV	HDMI CABLE	86-RE-206	F86-RE-106 S/FTP CAT 6A(Max.90m)
MAINTENANCE SUPERINTENDENT OFFICE	86-RE-307	TV	HDMI CABLE	86-RE-207	F86-RE-107 S/FTP CAT 6A(Max.90m)
2M-BP1	86-RE-108	2M-BP1			F86-RE-108 S/FTP CAT 6A(Max.90m)
2M-BP2	86-RE-109	2M-BP2			F86-RE-109 S/FTP CAT 6A(Max.90m)
2M-BP3	86-RE-110	2M-BP3			F86-RE-110 S/FTP CAT 6A(Max.90m)
2M-BP4	86-RE-111	2M-BP4			F86-RE-111 S/FTP CAT 6A(Max.90m)
2M-BP5	86-RE-112	2M-BP5			F86-RE-112 S/FTP CAT 6A(Max.90m)
4M-BP6	86-RE-113	4M-BP6			F86-RE-113 S/FTP CAT 6A(Max.90m)
4M-BP7	86-RE-114	4M-BP7			F86-RE-114 S/FTP CAT 6A(Max.90m)
4M-BP8	86-RE-115	4M-BP8			F86-RE-115 S/FTP CAT 6A(Max.90m)
2M-BP9	86-RE-116	2M-BP9			F86-RE-116 S/FTP CAT 6A(Max.90m)
2M-BP10	86-RE-117	2M-BP10			F86-RE-117 S/FTP CAT 6A(Max.90m)
2M-BP11	86-RE-118	2M-BP11			F86-RE-118 S/FTP CAT 6A(Max.90m)
86-RE-120	86-RE-120	2M-BP13			F86-RE-120 S/FTP CAT 6A(Max.90m)
86-RE-119	86-RE-119	2M-BP12			F86-RE-119 S/FTP CAT 6A(Max.90m)

REVISIONS

REV.	DATE	DESCRIPTION	DRWN	CHKD	APPD	CLIENT
Z	23.12.2025	ISSUED FOR AS-BUILT	CK	KK	TS	
O	23.04.2025	ISSUED FOR CONSTRUCTION	CK	KK	TS	
B	16.10.2024	ISSUED FOR APPROVAL	CK	KK	TN	
A	04.10.2024	ISSUED FOR REVIEW	CK	KK	TN	

COMPANY:

CONTRACTOR:

VENDOR:

PROJECT:

LOCATION:

TITLE:

SCALE:

DRAWING NO.:

WCP DRAWING NO.:

SHEET:

REV

REVISIONS						
Z	23.12.2025	ISSUED FOR AS-BUILT	CK	KK	TS	
0	23.04.2025	ISSUED FOR CONSTRUCTION	CK	KK	TS	
B	16.10.2024	ISSUED FOR APPROVAL	CK	KK	TN	
A	04.10.2024	ISSUED FOR REVIEW	CK	KK	TN	
REV.	DATE	DESCRIPTION	DRWN	CHKD	APPD	CLIENT

COMPANY:



ORIENTAL
OIL



CONTRACTOR:




VENDOR:



TP : +65 6826 2525

Head Office : 21 Toh Guan Rd E, #06-06A, Singapore 608609

Web : vivoasia.com

E-mail : contract@vivoasia.com

PROJECT:

OKWOK –FPSO
TOPSIDES PRODUCTION FACILITIES

LOCATION:

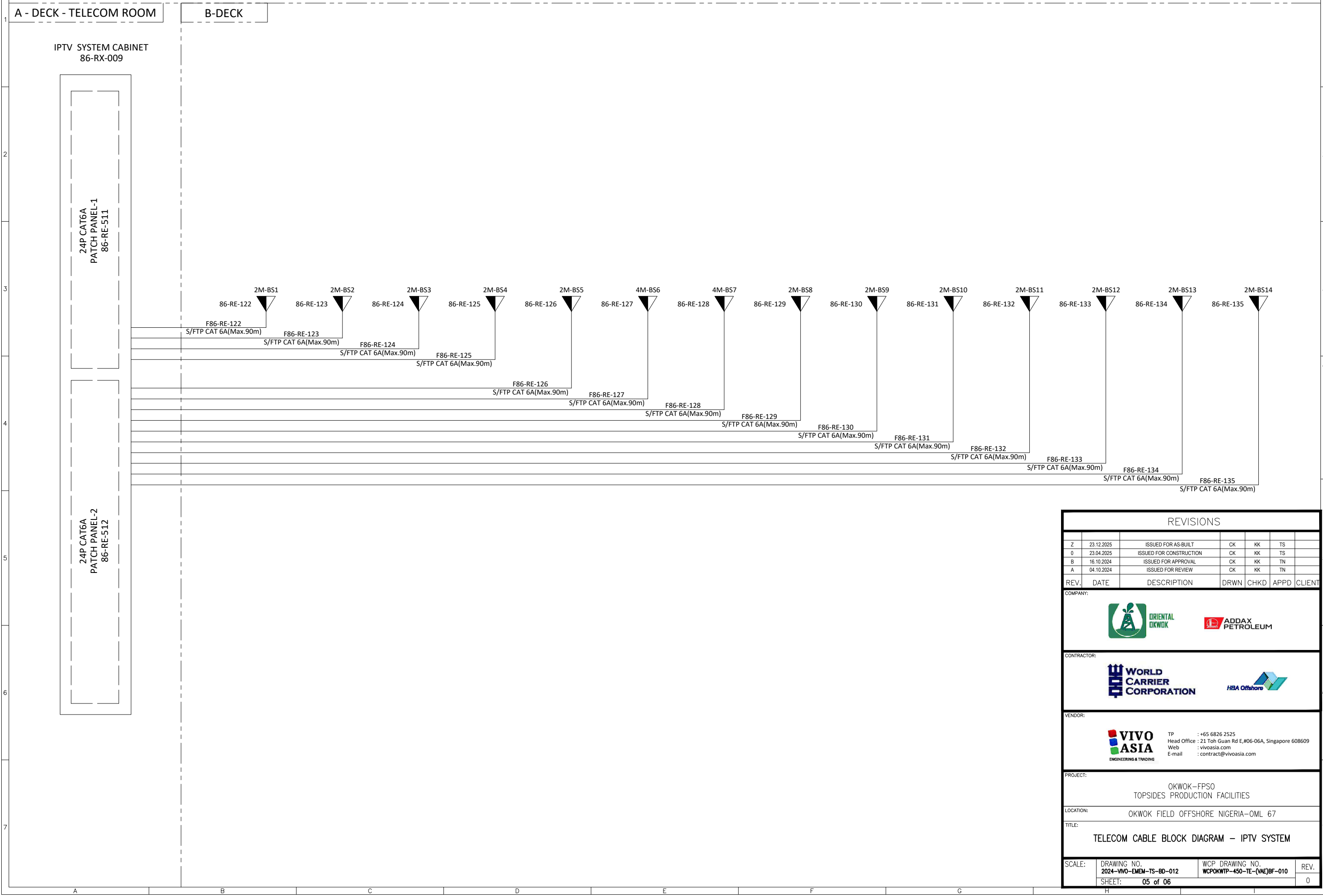
OKWOK FIELD OFFSHORE NIGERIA –OML 67

TITLE:

TELECOM CABLE BLOCK DIAGRAM – IPTV SYSTEM

SCALE:	DRAWING NO. 2024-VIVO-EMEM-TS-BD-012	WCP DRAWING NO. WCP0KWTP-450-TE-(VAE)BF-010	REV. 0
	SHEET: 04 of 06		

IPTV SYSTEM CABLE BLOCK DIAGRAM



IPTV SYSTEM CABLE BLOCK DIAGRAM

The diagram illustrates the IPTV system cable block diagram, showing connections between various rooms and a central patch panel. The rooms are organized into three main sections: Navigation Deck Wheel House, C-Deck, and Navigation Deck. The central patch panel is labeled 24P CAT6A PATCH PANEL-5 (IPTV) 86-RE-537.

Navigation Deck Wheel House:

- LAN/PABX CABINET 86-RX-002

C-Deck:

- MEETING ROOM-2 86-RE-341
- MARINE SUPERINTENDENT BED ROOM 86-RE-307
- VIP-P1 86-RE-343
- NUPRC REP BED ROOM 86-RE-344
- GYM 86-RE-346
- HELI BRIEFING ROOM 86-RE-347
- VIP-S1 86-RE-355
- SAFETY OFFICER 86-RE-356
- OIM BED ROOM 86-RE-357
- JANITOR ROOM
- MARINE SUPERINTENDENT BED ROOM
- PRODUCTION SUPERVISOR BED ROOM 86-RE-358

Navigation Deck:

- 2M-CP1, 2M-CP2, 2M-CP3, 2M-CP4, 2M-CP5
- 2M-CS1, 2M-CS2, 2M-CS3, 2M-CS4, 2M-CS5

Connections:

- Rooms are connected to the patch panel via HDMI CABLE and PATCH CABLE.
- Rooms are connected to the patch panel via S/FTP CAT 6A (Max.90m).

Revisions:

REV.	DATE	DESCRIPTION	DRWN	CHKD	APPD	CLIENT
Z	23.12.2025	ISSUED FOR AS-BUILT	CK	KK	TS	
O	23.04.2025	ISSUED FOR CONSTRUCTION	CK	KK	TS	
B	16.10.2024	ISSUED FOR APPROVAL	CK	KK	TN	
A	04.10.2024	ISSUED FOR REVIEW	CK	KK	TN	

COMPANY:

- ORIENTAL OIL
- ADDAX PETROLEUM

CONTRACTOR:

- WORLD CARRIER CORPORATION
- HBA Offshore

VENDOR:

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- TP : +65 6826 2525
- Head Office : 21 Toh Guan Rd E, #06-06A, Singapore 608609
- Web : vivoasia.com
- E-mail : contract@vivoasia.com

PROJECT:

- OKWOK-FPSO
- TOPSIDES PRODUCTION FACILITIES

LOCATION:

- OKWOK FIELD OFFSHORE NIGERIA-OML 67

TITLE:

- TELECOM CABLE BLOCK DIAGRAM – IPTV SYSTEM

SCALE:

- DRAWING NO. 2024-VIVO-EMEM-TS-BD-012
- WCP DRAWING NO. WCPKWT-450-TE-(VAE)BF-010
- SHEET: 06 of 06

COMPANY:  ORIENTAL OKWOK  ADDAX PETROLEUM			
CONTRACTOR:  			
VENDOR:  VIVO ASIA ENGINEERING & TRADING TP : +65 6826 2525 Head Office : 21 Toh Guan Rd E, #06-06A, Singapore 608609 Web : vivoasia.com E-mail : contract@vivoasia.com			
PROJECT: OKWOK-FPSO TOPSIDES PRODUCTION FACILITIES			
LOCATION: OKWOK FIELD OFFSHORE NIGERIA-OML 67			
TITLE: TELECOM CABLE BLOCK DIAGRAM – IPTV SYSTEM			
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	SHEET: 06 of 06		0

IPTV SYSTEM LOCATION LAYOUT

NOTES

HOLD LIST

KEYPLAN

REFERENCES

DOCUMENT NO.	DESCRIPTION

REVISIONS

Z	05.02.2026	RE-ISSUED FOR AS-BUILT	CK	KK	TS	
1	21.04.2025	RE-ISSUED FOR CONSTRUCTION	CK	KK	TS	
0	13.02.2025	ISSUED FOR CONSTRUCTION	CK	KK	TS	
B	12.12.2024	ISSUED FOR APPROVAL	CK	KK	TN	
A	04.10.2024	ISSUED FOR REVIEW	CK	KK	TN	
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COMPANY:



**ORIENTAL
OKWOK**



**ADDAX
PETROLEUM**

CONTRACTOR:



**WORLD
CARRIER
CORPORATION**



HBA Offshore

VENDOR:



**VIVO
ASIA**
ENGINEERING & TRADING

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Web : vivoasia.com
E-mail : contract@vivoasia.com

PROJECT:

OKWOK-FPSO
TOPSIDES PRODUCTION FACILITIES

LOCATION:

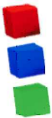
OKWOK FIELD OFFSHORE NIGERIA-OML 67

TITLE:

IPTV SYSTEM LOCATION LAYOUT

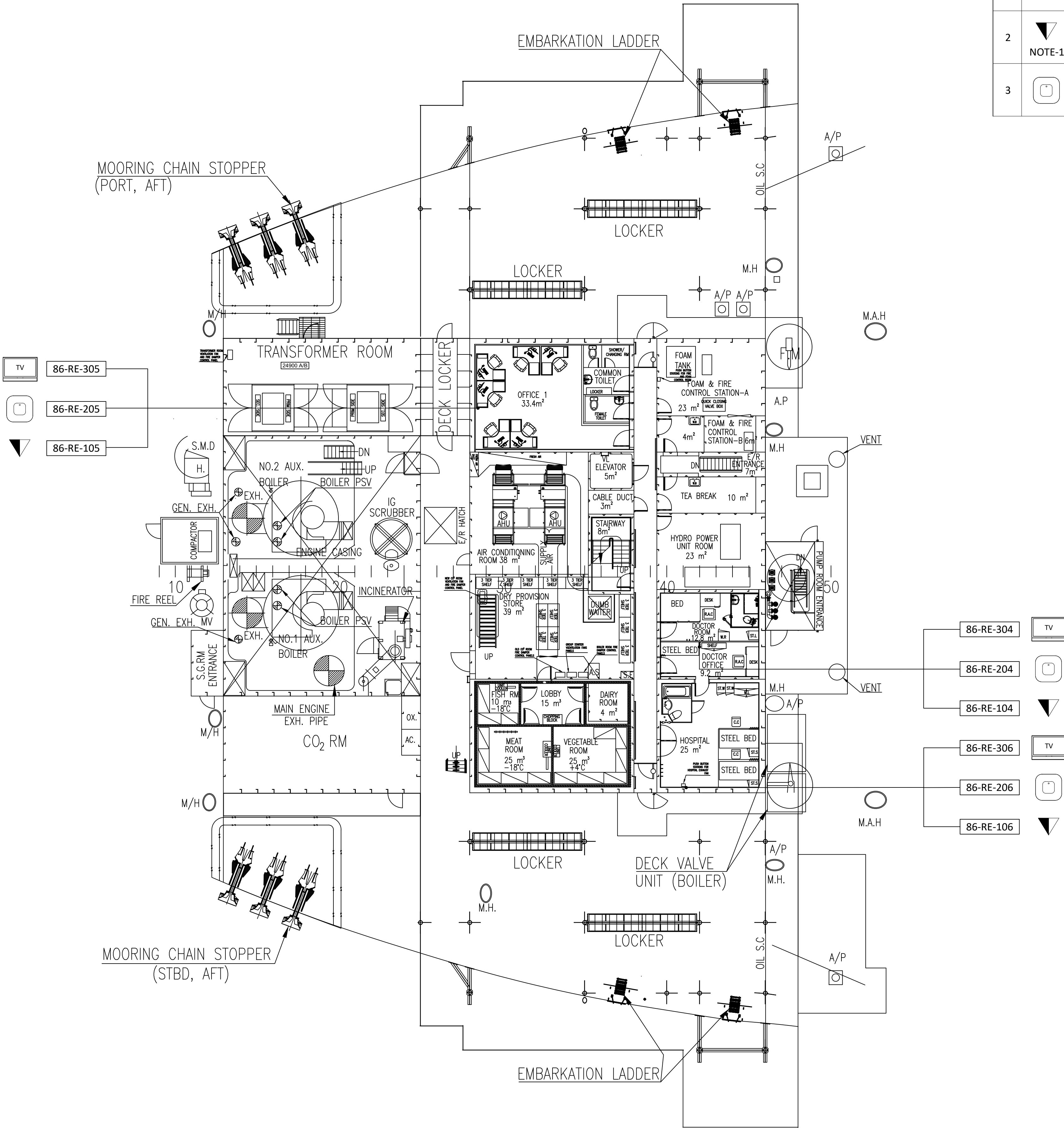
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SHEET NO.	REVISION					DESCRIPTION	REMARKS
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1	04.10.2024	12.12.2024	13.02.2025	21.04.2025	05.02.2026	COVER PAGE	
2	04.10.2024	12.12.2024	13.02.2025	21.04.2025	05.02.2026	INDEX PAGE	
3	04.10.2024	12.12.2024	13.02.2025	21.04.2025	05.02.2026	IPTV SYSTEM LOCATION LAYOUT RAISED DECK	
4	04.10.2024	12.12.2024	13.02.2025	21.04.2025	05.02.2026	IPTV SYSTEM LOCATION LAYOUT A DECK	
5	04.10.2024	12.12.2024	13.02.2025	21.04.2025	05.02.2026	IPTV SYSTEM LOCATION LAYOUT B DECK	
6	04.10.2024	12.12.2024	13.02.2025	21.04.2025	05.02.2026	IPTV SYSTEM LOCATION LAYOUT C DECK	
7	04.10.2024	12.12.2024	13.02.2025	21.04.2025	05.02.2026	IPTV SYSTEM LOCATION LAYOUT NAV. DECK	
8	04.10.2024	12.12.2024	13.02.2025	21.04.2025	05.02.2026	IPTV SYSTEM LOCATION LAYOUT COMPASS DECK	

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1	21.04.2025	RE-ISSUED FOR CONSTRUCTION	CK	KK	TS	
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B	12.12.2024	ISSUED FOR APPROVAL	CK	KK	TN	
A	04.10.2024	ISSUED FOR REVIEW	CK	KK	TN	
REV.	DATE	DESCRIPTION	DRWN	CHKD	APPD	CLIENT
COMPANY:						
 Oriental OKWOK  ADDAX PETROLEUM						
CONTRACTOR:						
 WORLD CARRIER CORPORATION 						
VENDOR:						
 VIVO ASIA ENGINEERING & TRADING TP : +65 6826 2525 Head Office : : 21 Toh Guan Rd E,#06-06A, Singapore 608609 Web : : vivoasia.com E-mail : : contract@vivoasia.com						
PROJECT:						
OKWOK-FPSO TOPSIDES PRODUCTION FACILITIES						
LOCATION:						
OKWOK FIELD OFFSHORE NIGERIA-OML 67						
TITLE:						
IPTV SYSTEM LOCATION LAYOUT						
SCALE:		DRAWING NO.		WCP DRAWING NO.		REV.
NTS		2024-VIVO-EMEM-TS-LD-012		WCP0KWTP-450-TE-(VAE)LY-011		Z
SHEET:		02 of 08				

REFERENCE			
NO	REVISION	CLIENT DOC NO.	TITLE
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IPTV SYSTEM LOCATION LAYOUT RAISED DECK

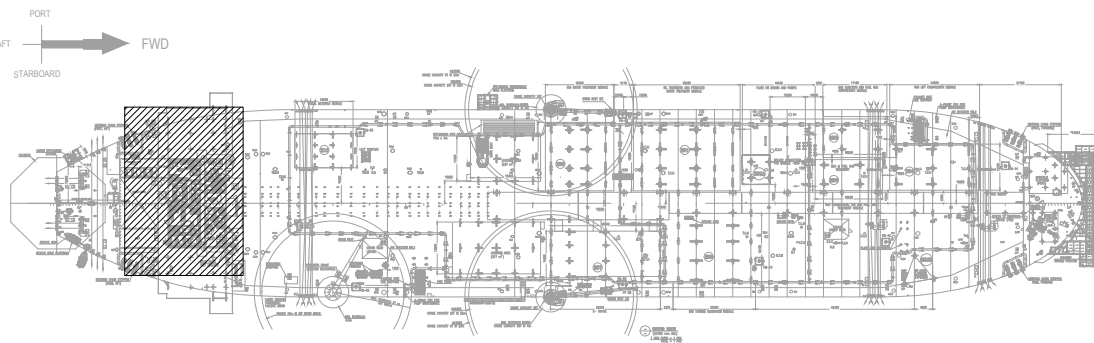


S/N	SYMBOL	DESCRIPTION	QTY
1		SMART TV MODEL : PHILIP 32HFL3014 MANUFACTURER : CLEAR	03
2	 NOTE-1	TV OUTLET SOCKET MODEL : 2153449-4 MANUFACTURER : COMMSCOPE	03
3		IPTV SET-TOP BOX MODEL : CLEARTV H98MUV3 MANUFACTURER : CLEARTV	03

NOTES

HOLD LIST

KEYPLAN



REFERENCES

DOCUMENT NO.	DESCRIPTION

REVISIONS

REV.	DATE	DESCRIPTION	DRWN	CHKD	APPD	CLIENT
Z	05.02.2026	RE-ISSUED FOR AS-BUILT	CK	KK	TS	
1	21.04.2025	RE-ISSUED FOR CONSTRUCTION	CK	KK	TS	
0	13.02.2025	ISSUED FOR CONSTRUCTION	CK	KK	TS	
B	12.12.2024	ISSUED FOR APPROVAL	CK	KK	TN	
A	04.10.2024	ISSUED FOR REVIEW	CK	KK	TN	

COMPANY:



CONTRACTOR:



VENDOR:



TP : +65 6826 2525
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Web : vivoasia.com
E-mail : contract@vivoasia.com

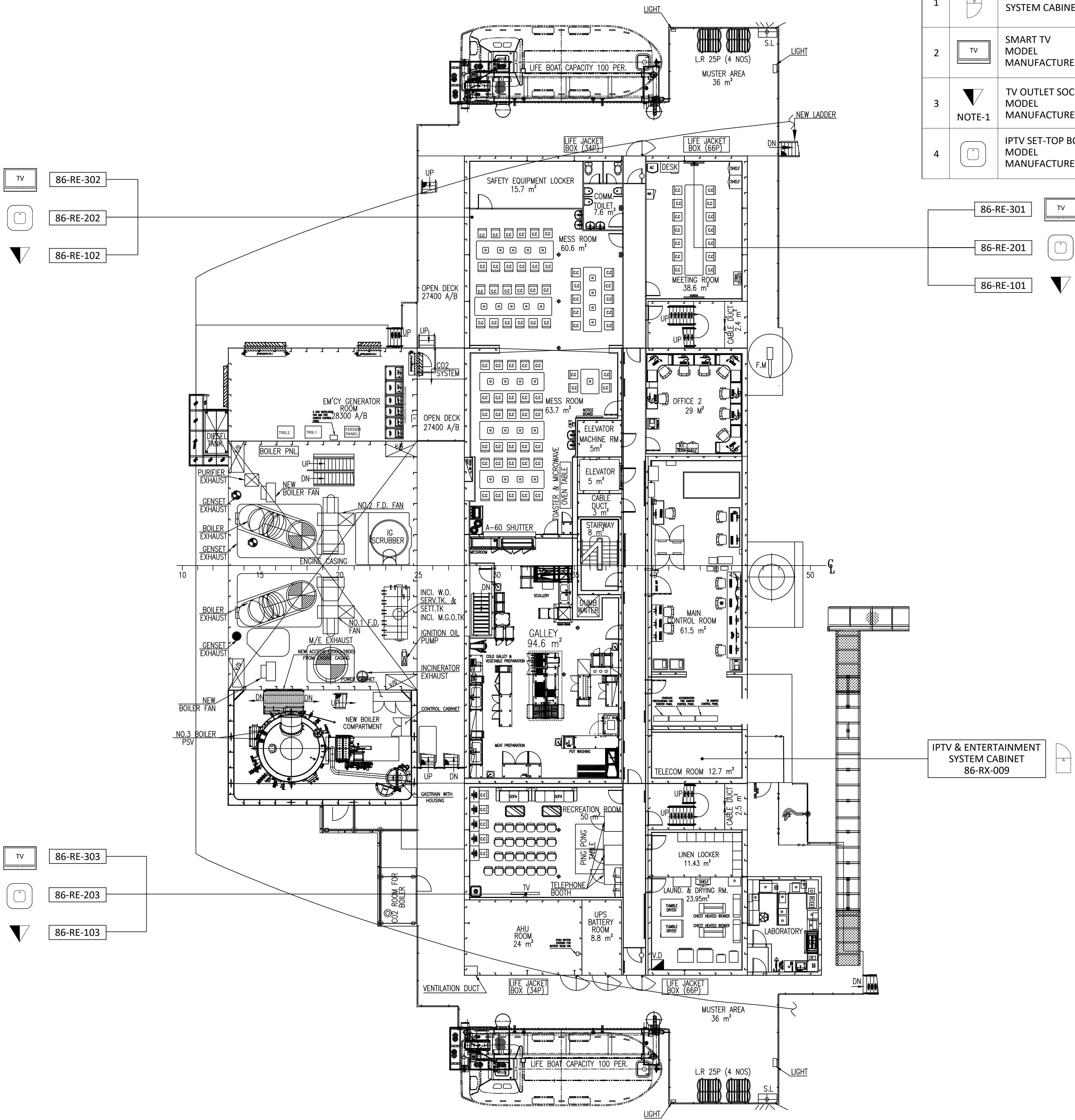
PROJECT: OKWOK-FPSO
TOPSIDES PRODUCTION FACILITIES

LOCATION: OKWOK FIELD OFFSHORE NIGERIA-OML 67

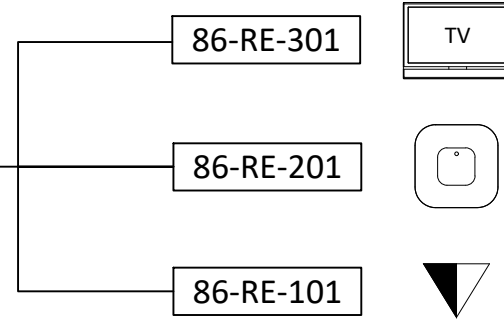
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IPTV SYSTEM LOCATION LAYOUT A DECK



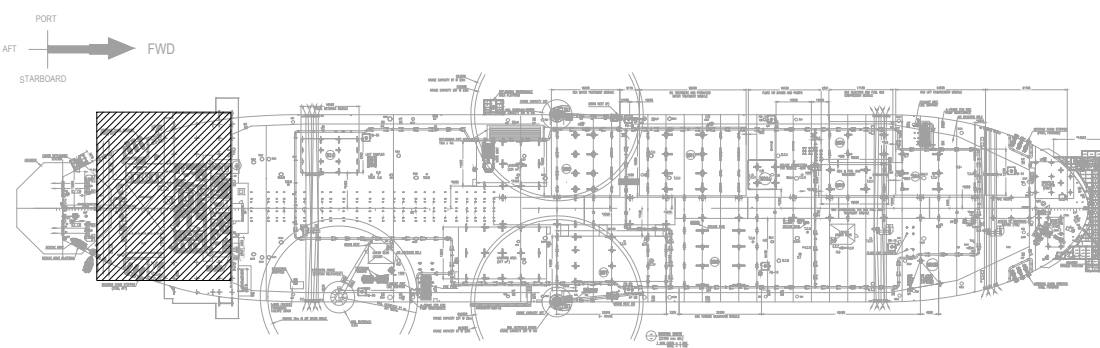
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1		IPTV & ENTERTAINMENT SYSTEM CABINET	01
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3		TV OUTLET SOCKET MODEL : 2153449-4 MANUFACTURER : COMMSCOPE	03
4		IPTV SET-TOP BOX MODEL : CLEARTV H98MUV3 MANUFACTURER : CLEARTV	03



NOTES

HOLD LIST

KEYPLAN



REFERENCES

DOCUMENT NO.	DESCRIPTION

REVISIONS

REV.	DATE	DESCRIPTION	DRWN	CHKD	APPD	CLIENT
Z	05.02.2026	RE-ISSUED FOR AS-BUILT	CK	KK	TS	
1	21.04.2025	RE-ISSUED FOR CONSTRUCTION	CK	KK	TS	
0	13.02.2025	ISSUED FOR CONSTRUCTION	CK	KK	TS	
B	12.12.2024	ISSUED FOR APPROVAL	CK	KK	TN	
A	04.10.2024	ISSUED FOR REVIEW	CK	KK	TN	

COMPANY:



CONTRACTOR:



VENDOR:



OKWOK-FPSO
TOPSIDES PRODUCTION FACILITIES

LOCATION: OKWOK FIELD OFFSHORE NIGERIA-OML 67

TITLE: IPTV SYSTEM LOCATION LAYOUT

SCALE: 1:125	DRAWING NO. 2024-VIVO-EMEM-TS-LD-012 SHEET: 04 of 08	WCP DRAWING NO. WCP0KWTP-450-TE-(VAE)LY-011	REV. Z
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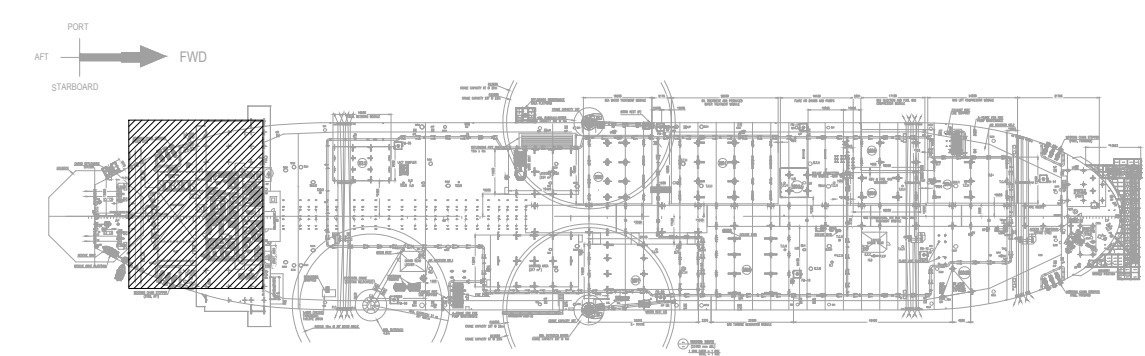
IPTV SYSTEM LOCATION LAYOUT B DECK

BILL OF MATERIAL			
S/N	SYMBOL	DESCRIPTION	QTY
1	 NOTE-1	TV OUTLET SOCKET MODEL : 2153449-4 MANUFACTURER : COMMSCOPE	27

NOTES

HOLD LIST

KEYPLAN



REFERENCES

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REVISIONS

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1	21.04.2025	RE-ISSUED FOR CONSTRUCTION	CK	KK	TS	
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A	04.10.2024	ISSUED FOR REVIEW	CK	KK	TN	
REV.	DATE	DESCRIPTION	DRWN	CHKD	APPD	CLIE

COMPANY:



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PROJECT:

OKWOK-FPSO
TOPSIDES PRODUCTION FACILITIES

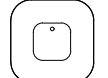
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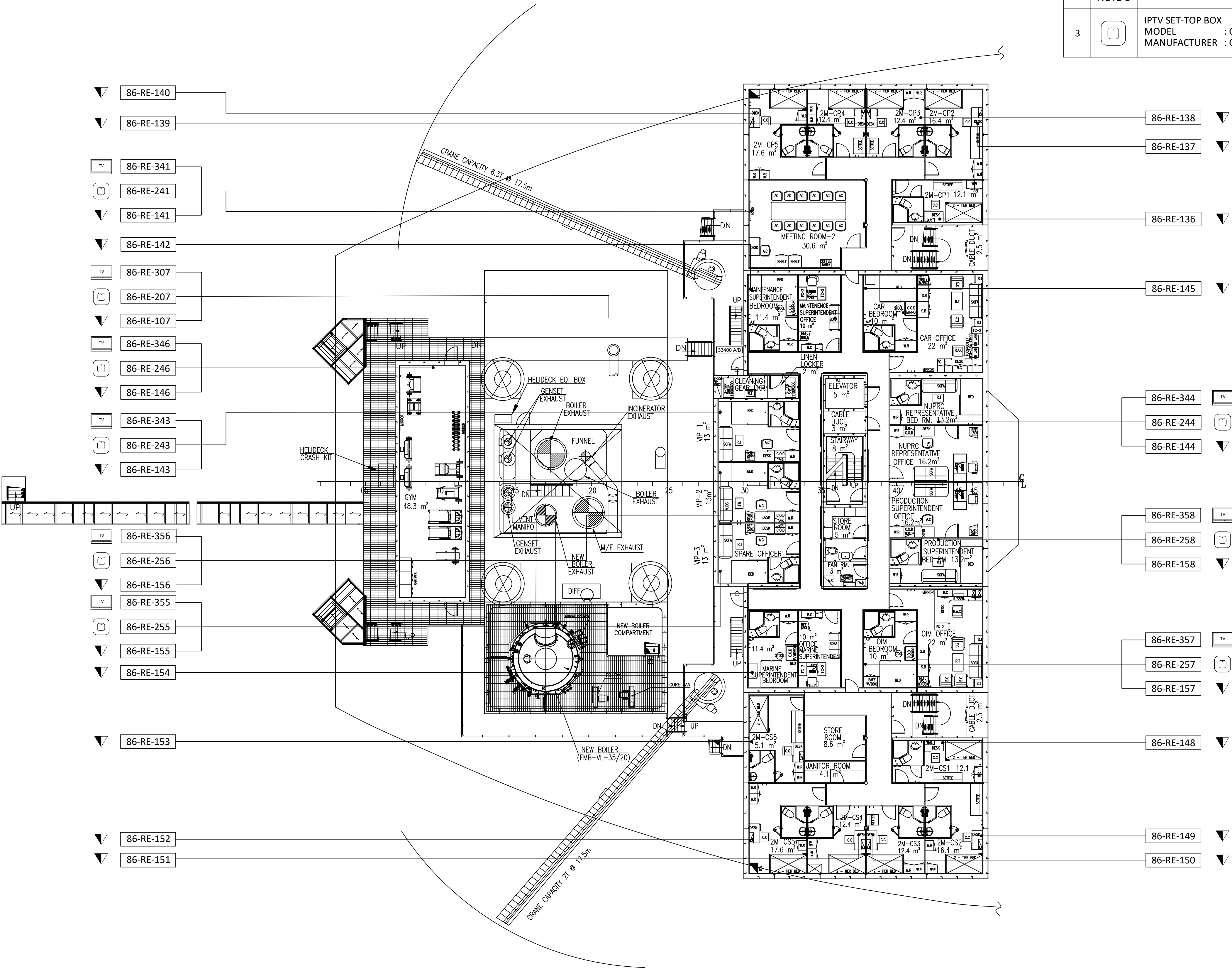
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IPTV SYSTEM LOCATION LAYOUT

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1:125	SHEET: 05 of 08		Z

IPTV SYSTEM LOCATION LAYOUT C DECK

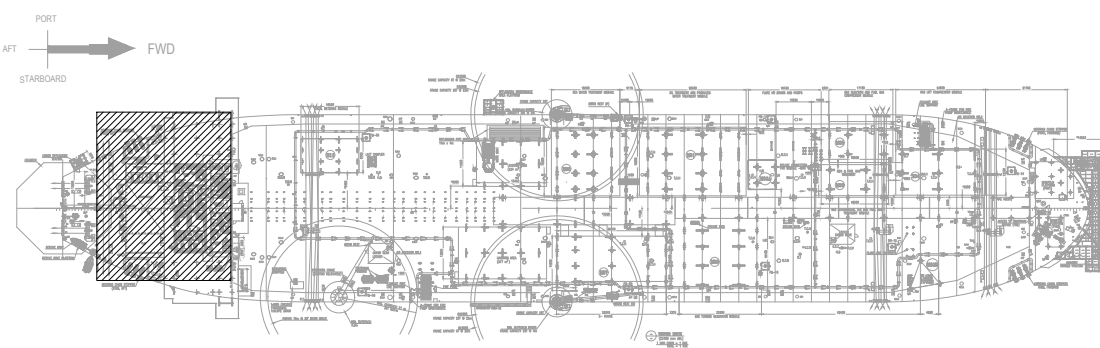
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2	 NOTE-1	TV OUTLET SOCKET MODEL : 2153449-4 MANUFACTURER : COMMSCOPE	23
3		IPTV SET-TOP BOX MODEL : CLEARTV H98MUV3 MANUFACTURER : CLEARTV	09



NOTES

HOLD LIST

KEYPLAN



REFERENCES

DOCUMENT NO.	DESCRIPTION

REVISIONS

REV.	DATE	DESCRIPTION	DRWN	CHKD	APPD	CLIENT
Z	05.02.2026	RE-ISSUED FOR AS-BUILT	CK	KK	TS	
1	21.04.2025	RE-ISSUED FOR CONSTRUCTION	CK	KK	TS	
0	13.02.2025	ISSUED FOR CONSTRUCTION	CK	KK	TS	
B	12.12.2024	ISSUED FOR APPROVAL	CK	KK	TN	
A	04.10.2024	ISSUED FOR REVIEW	CK	KK	TN	

COMPANY:



CONTRACTOR:



VENDOR:



TP : +65 6826 2525
Head Office : 21 Toh Guan Rd E, #06-06A, Singapore 608609
Web : vivoasia.com
E-mail : contract@vivoasia.com

PROJECT:

OKWOK-FPSO
TOPSIDES PRODUCTION FACILITIES

LOCATION:

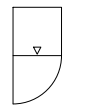
OKWOK FIELD OFFSHORE NIGERIA-OML 67

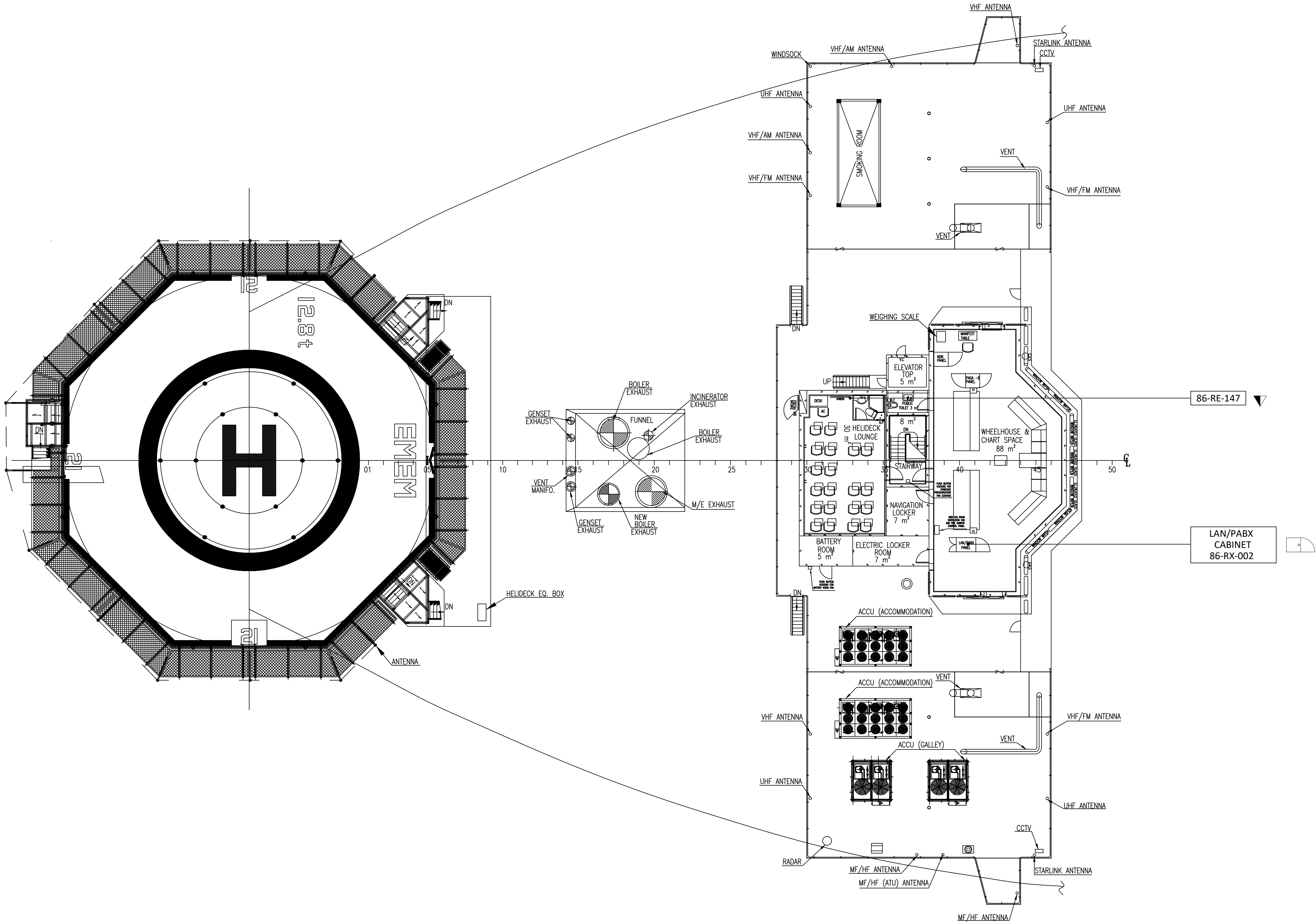
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IPTV SYSTEM LOCATION LAYOUT

SCALE:	DRAWING NO.	WCP DRAWING NO.	REV.
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SHEET:	06 of 08		

IPTV SYSTEM LOCATION LAYOUT NAV. DECK

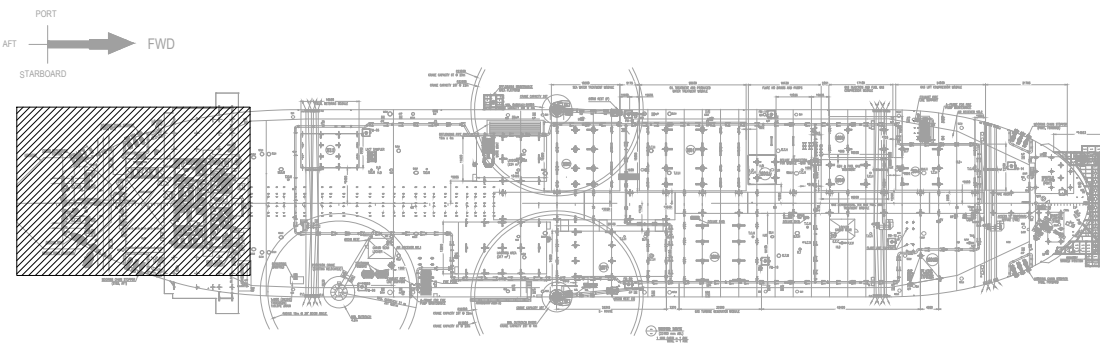
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1		IPTV & ENTERTAINMENT SYSTEM CABINET	01
2	 NOTE-1	TV OUTLET SOCKET MODEL : 2153449-4 MANUFACTURER : COMMSCOPE	01



NOTES

HOLD LIST

KEYPLAN



REFERENCES

DOCUMENT NO.	DESCRIPTION

REVISIONS

REV.	DATE	DESCRIPTION	DRWN	CHKD	APPD	CLIENT
Z	05.02.2026	RE-ISSUED FOR AS-BUILT	CK	KK	TS	
1	21.04.2025	RE-ISSUED FOR CONSTRUCTION	CK	KK	TS	
0	13.02.2025	ISSUED FOR CONSTRUCTION	CK	KK	TS	
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A	04.10.2024	ISSUED FOR REVIEW	CK	KK	TN	

COMPANY:



CONTRACTOR:



VENDOR:



PROJECT:

OKWOK-FPSO
TOPSIDES PRODUCTION FACILITIES

LOCATION:

OKWOK FIELD OFFSHORE NIGERIA-OML 67

TITLE:

IPTV SYSTEM LOCATION LAYOUT

SCALE:	DRAWING NO.	WCP DRAWING NO.	REV.
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SHEET:		07 of 08	

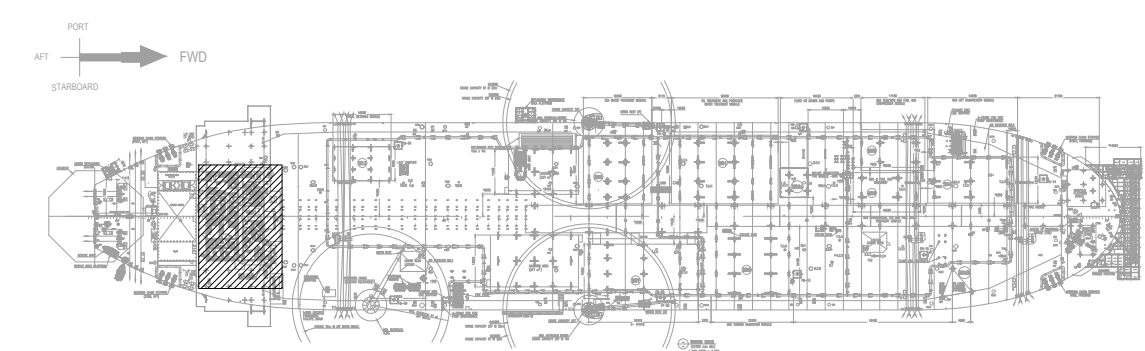
IPTV SYSTEM LOCATION LAYOUT COMPASS DECK

BILL OF MATERIAL			
S/N	SYMBOL	DESCRIPTION	QTY
1		IPTV ANTENNA MODEL : 120TV MANUFACTURER : SEATEL	01

NOTES

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KEYPLAN



REFERENCES

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REVISIONS

Z	05.02.2026	RE-ISSUED FOR AS-BUILT	CK	KK	TS
1	21.04.2025	RE-ISSUED FOR CONSTRUCTION	CK	KK	TS
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B	12.12.2024	ISSUED FOR APPROVAL	CK	KK	TN
A	04.10.2024	ISSUED FOR REVIEW	CK	KK	TN

COMPANY:



CONTRACTOR:



VENDOR:



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Web : vivoasia.com
E-mail : contract@vivoasia.com

PROJECT:

OKWOK-FPSO
TOPSIDES PRODUCTION FACILITIES

LOCATION:

OKWOK FIELD OFFSHORE NIGERIA—OML 67

TITLE:

IPTV SYSTEM LOCATION LAYOUT

SCALE: 1:50	DRAWING NO. 2024-VIVO-EMEM-TS-LD-012	WCP DRAWING NO. WCPOKWTP-450-TE-(VAE)LY-011	REV
	SHEET: 08 of 08		Z

TELECOM TERMINATION DRAWING

IPTV SYSTEM

REVISIONS

REV.	DATE	DESCRIPTION	DRWN	CHKD	APPD	CLIENT
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B	12.12.2024	ISSUED FOR APPROVAL	DD	HS	TN	
A	21.11.2024	ISSUED FOR REVIEW	DD	HS	TN	

COMPANY:



ORIENTAL
OKWOK



ADDAX
PETROLEUM

CONTRACTOR:



WORLD
CARRIER
CORPORATION



HBA Offshore

VENDOR:



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ASIA
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Web : vivoasia.com

E-mail : contract@vivoasia.com

PROJECT:

OKWOK-FPSO
TOPSIDES PRODUCTION FACILITIES

LOCATION:

OKWOK FIELD OFFSHORE NIGERIA-OML 67

TITLE:

TELECOM TERMINATION DRAWING – IPTV SYSTEM

SCALE:	DRAWING NO. 2024-VVO-EMEM-TS-TD-012	WCP DRAWING NO. WCP0KWTP-450-TE-(VAE)DG-032	REV.
NTS	SHEET: 01 of 07		0

LETTER SYMBOLS FOR COLORS	
ABBREVIATION	DESCRIPTION
BK	BLACK
BR	BROWN
RD	RED
OR	ORANGE
YE	YELLOW
GR	GREEN
BL	BLUE
PU	PURPLE
GY	GRAY
WH	WHITE
PK	PINK
LB	LIGHT BLUE
TQ	TURQUOISE

ABBREVIATIONS	
ABBREVIATION	DESCRIPTION
C	CORE
P	PORT
LAN	LOCAL AREA NETWORK
IPTV	INTERNET PROTOCOL TELEVISION
PSU	POWER SUPPLY UNIT
UPS	UNINTERRUPTIBLE POWER SUPPLY
DB	DISTRIBUTION BOARD
V	VOLTAGE
MDF	MAIN DISTRIBUTION FRAME
FO	FIBER OPTIC
HDMI	HIGH-DEFINITION MULTIMEDIA INTERFACE
TV	TELEVISION
PR	PAIR

BILL OF MATERIAL		
SYMBOL	DESCRIPTION	QTY.
	BULKHEAD CONNECTOR MODEL : FNSN6U MANUFACTURER : BELDEN	02
	SMART TV MODEL : 32HFL3014 MANUFACTURER : PHILIPS	15
 NOTE-1	TV SOCKET OUTLETS MODEL : 2153449-4 MANUFACTURER : COMMSCOPE	57
	IPTV SET-TOP BOX MODEL : CLEAR H98MUV3 MANUFACTURER : CLEARTV	15
	13A UK TYPE POWER SOCKET OUTLET (SUPPLY BY OTHERS)	30
	IPTV / TVRO ANTENNA MODEL : 120TV MANUFACTURER : SEATEL	01

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2	21.11.2024	12.12.2024	14.05.2025	INDEX PAGE	
3	21.11.2024	12.12.2024	14.05.2025	TELECOM TERMINATION DRAWING (ANTENNA CONNECTION AND INTERFACE) - IPTV SYSTEM	
4	21.11.2024	12.12.2024	14.05.2025	TELECOM TERMINATION DRAWING - IPTV SYSTEM	
5	21.11.2024	12.12.2024	14.05.2025	TELECOM TERMINATION DRAWING - IPTV SYSTEM	
6	21.11.2024	12.12.2024	14.05.2025	TELECOM TERMINATION DRAWING - IPTV SYSTEM	
7	21.11.2024	12.12.2024	14.05.2025	TELECOM TERMINATION DETAIL DRAWING - IPTV SYSTEM	

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A	21.11.2024	ISSUED FOR REVIEW	DD	HS	TN		
REV.	DATE	DESCRIPTION	DRWN	CHKD	APPD	CLIENT	

COMPANY:	
 ORIENTAL OKWOK	 ADDAX PETROLEUM

CONTRACTOR:	
 WORLD CARRIER CORPORATION	 HSA Offshore

VENDOR:	
 VIVO ASIA ENGINEERING & TRADING	TP : +65 6826 2525 Head Office : 21 Toh Guan Rd E, #06-06A, Singapore 608609 Web : vivoasia.com E-mail : contract@vivoasia.com

PROJECT:	
OKWOK-FPSO TOPSIDES PRODUCTION FACILITIES	
LOCATION:	
OKWOK FIELD OFFSHORE NIGERIA-OML 67	
TITLE:	
TELECOM TERMINATION DRAWING – IPTV SYSTEM	

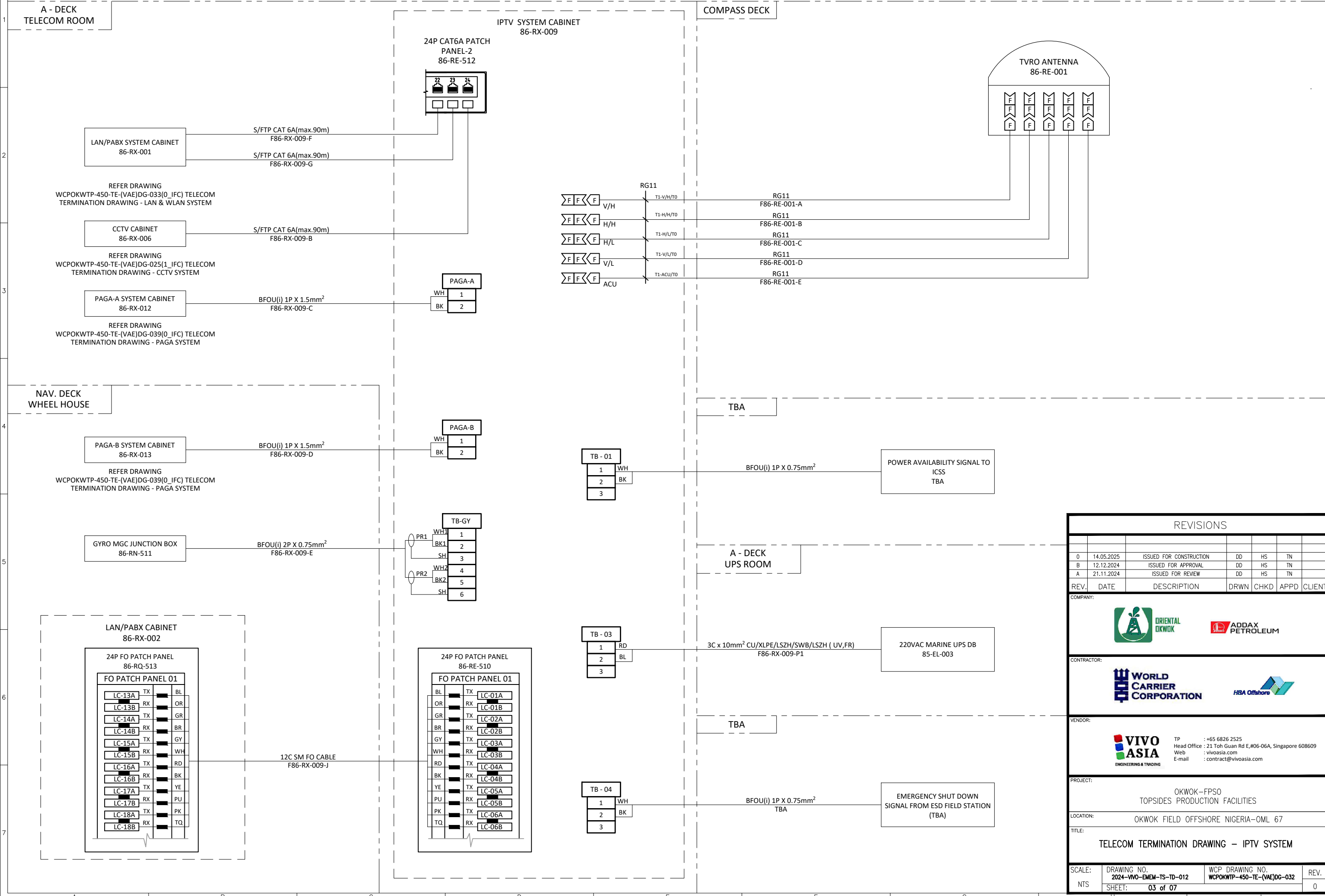
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NTS	2024-VVO-EMEM-TS-TD-012	WCPOKWTP-450-TE-(VAE)DG-032	
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REFERENCE			
NO	REVISION	CLIENT DOC NO.	TITLE
1	0	WCPOKWTP-450-TE-(VAE)BF-010	Telecom Cable Block Diagram - IPTV System
2	C	WCPOKWTP-450-TE-(VAE)DG-004	Telecom Internal Wiring Drawing- IPTV System
3	A	WCPOKWTP-450-TE-(VAE)DS-010	Telecom System Data Sheet- IPTV System

NOTE :

1. OUTLET TO BE MOUNTED ON THE WALL.

TELECOM TERMINATION DRAWING (ANTENNA CONNECTION AND INTERFACE) - IPTV SYSTEM



TELECOM TERMINATION DRAWING - IPTV SYSTEM

A - DECK - TELECOM ROOM

A - DECK

MAIN DECK

C - DECK

IPTV SYSTEM CABINET
86-RX-009

24P CAT6A (S/FTP)
PATCH PANEL-1
86-RE-511

MEETING ROOM
86-RE-301

MESS ROOM
86-RE-302

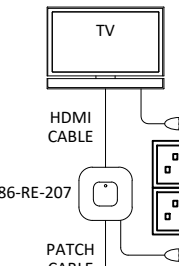
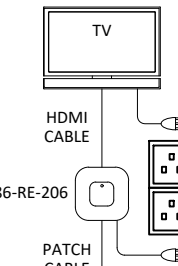
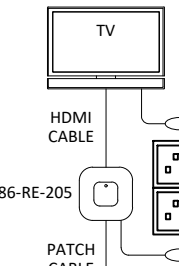
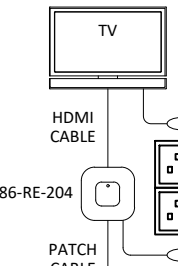
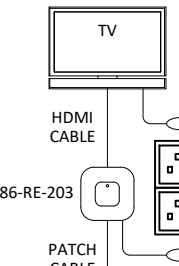
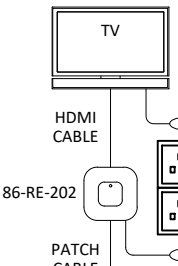
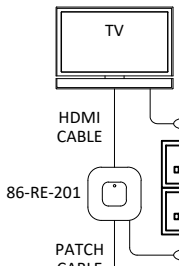
RECREATION ROOM
86-RE-303

DOCTOR'S OFFICE
86-RE-304

GYM ROOM
86-RE-305

HOSPITAL
86-RE-306

MAINTENANCE
SUPERINTENDENT
OFFICE
86-RE-307



B - DECK

2M-BP1
86-RE-108

2M-BP2
86-RE-109

2M-BP3
86-RE-110

2M-BP4
86-RE-111

2M-BP5
86-RE-112

4M-BP6
86-RE-113

4M-BP7
86-RE-114

4M-BP8
86-RE-115

2M-BP9
86-RE-116

2M-BP10
86-RE-117

F86-RE-108
S/FTP CAT 6A(max.90m)

F86-RE-109
S/FTP CAT 6A(max.90m)

F86-RE-110
S/FTP CAT 6A(max.90m)

F86-RE-111
S/FTP CAT 6A(max.90m)

F86-RE-112
S/FTP CAT 6A(max.90m)

F86-RE-113
S/FTP CAT 6A(max.90m)

F86-RE-114
S/FTP CAT 6A(max.90m)

F86-RE-115
S/FTP CAT 6A(max.90m)

F86-RE-116
S/FTP CAT 6A(max.90m)

F86-RE-117
S/FTP CAT 6A(max.90m)

2M-BP11
86-RE-118

2M-BP12
86-RE-119

2M-BP13
86-RE-120

2M-BS1
86-RE-122

2M-BS2
86-RE-123

2M-BS3
86-RE-124

2M-BS4
86-RE-125

S/FTP CAT 6A(max.90m)
F86-RE-118

F86-RE-119
S/FTP CAT 6A(max.90m)

F86-RE-120
S/FTP CAT 6A(max.90m)

F86-RE-122
S/FTP CAT 6A(max.90m)

F86-RE-123
S/FTP CAT 6A(max.90m)

F86-RE-124
S/FTP CAT 6A(max.90m)

F86-RE-125
S/FTP CAT 6A(max.90m)

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CONTRACTOR:



VENDOR:



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E-mail : contract@vivoasia.com

PROJECT:

OKWOK-FPSO
TOPSIDES PRODUCTION FACILITIES

LOCATION:

OKWOK FIELD OFFSHORE NIGERIA-OML 67

TITLE:

TELECOM TERMINATION DRAWING - IPTV SYSTEM

SCALE:

DRAWING NO.

2024-VIVO-EMEM-TS-TD-012

WCP DRAWING NO.

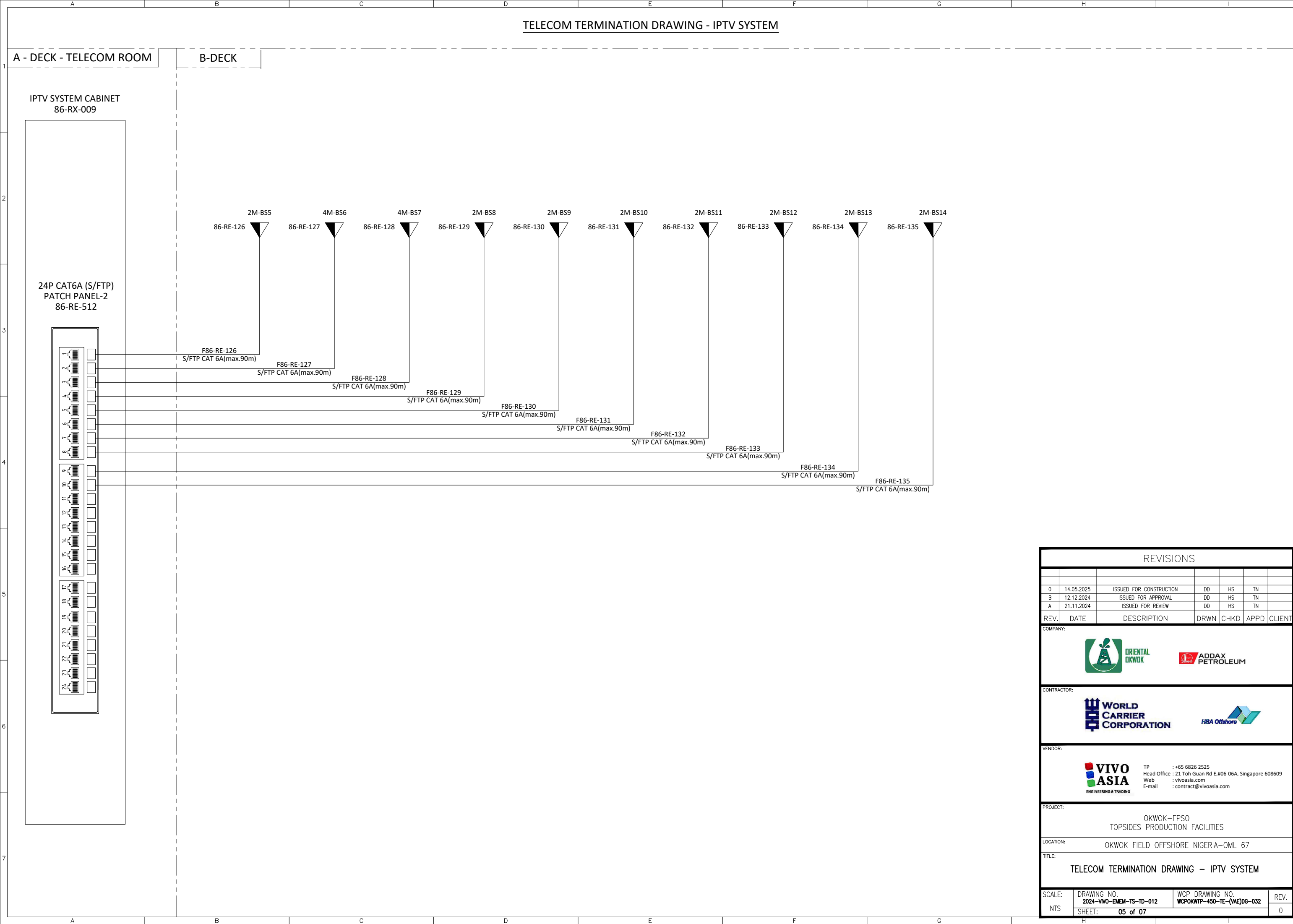
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ORIENTAL
OKWOK

ADDAX
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WCP0KWTP-450-TE-(VAE)DG-032

REV.
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TELECOM TERMINATION DRAWING - IPTV SYSTEM

NAVIGATION DECK ELECTRICAL EQUIPMENT ROOM

LAN/PABX CABINET 86-RX-002

24P CAT6A (S/FTP) PATCH PANEL-5 86-RE-537

NAVIGATION DECK

ROOMS AND TERMINATION POINTS:

- MEETING ROOM-2 86-RE-341: TV, HDMI CABLE, PATCH CABLE, 86-RE-241, 86-RE-141, 86-RE-142, 86-RE-143, 86-RE-144, 86-RE-145, 86-RE-146, 86-RE-148, 86-RE-149
- VIP-P1 86-RE-343: TV, HDMI CABLE, PATCH CABLE, 86-RE-243, 86-RE-143, 86-RE-156, 86-RE-157, 86-RE-158
- NUPRC REP BED ROOM 86-RE-344: TV, HDMI CABLE, PATCH CABLE, 86-RE-244, 86-RE-144, 86-RE-157, 86-RE-158
- GYM 86-RE-346: TV, HDMI CABLE, PATCH CABLE, 86-RE-246, 86-RE-146, 86-RE-158
- CFR BED ROOM: 86-RE-142, 86-RE-156
- VIP-S1 86-RE-355: TV, HDMI CABLE, PATCH CABLE, 86-RE-255, 86-RE-155, 86-RE-156, 86-RE-157, 86-RE-158
- SAFETY OFFICER 86-RE-356: TV, HDMI CABLE, PATCH CABLE, 86-RE-256, 86-RE-156, 86-RE-157, 86-RE-158
- OIM BED ROOM 86-RE-357: TV, HDMI CABLE, PATCH CABLE, 86-RE-257, 86-RE-157, 86-RE-158
- PRODUCTION SUPERVISOR BED ROOM 86-RE-358: TV, HDMI CABLE, PATCH CABLE, 86-RE-258, 86-RE-158
- 2 MEN ROOM: 86-RE-147
- JANITOR ROOM: 86-RE-153
- MARINE SUPERINTENDENT BED ROOM: 86-RE-154
- 2M-CP1 to 2M-CP5: 86-RE-136 to 86-RE-140
- 2M-CS1 to 2M-CS5: 86-RE-150 to 86-RE-154

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COMPANY:

ORIENTAL OKWOK

ADDAX PETROLEUM

CONTRACTOR:

WORLD CARRIER CORPORATION

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PROJECT:

OKWOK-FPSO
TOPSIDES PRODUCTION FACILITIES

LOCATION:

OKWOK FIELD OFFSHORE NIGERIA-OML 67

TITLE:

TELECOM TERMINATION DRAWING - IPTV SYSTEM

SCALE:

NTS

DRAWING NO.

2024-VIVO-EMEM-TS-TD-012

WCP DRAWING NO.

WCP0KWTP-450-TE-(VAE)06-032

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COMPANY:





**ADDAX
PETROLEUM**

CONTRACTOR:




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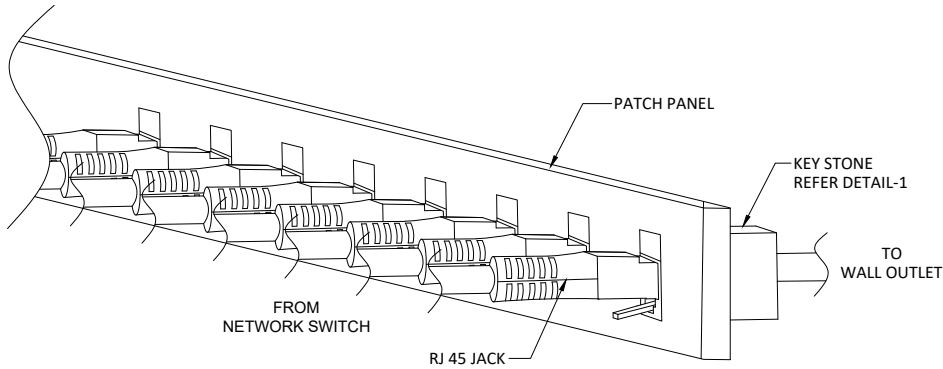
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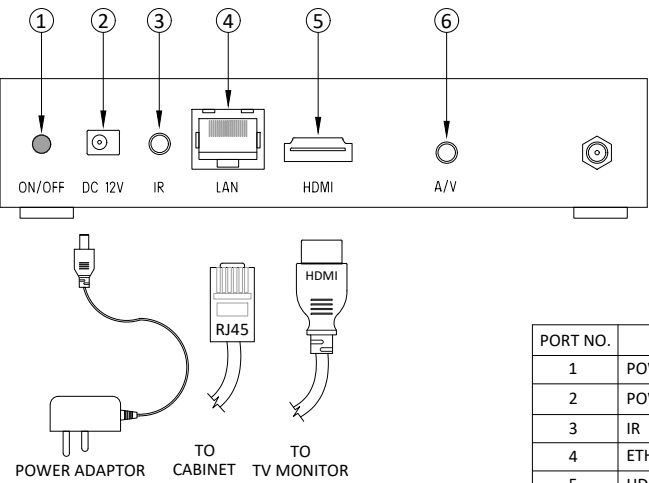
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	SHEET: 06 of 07		

TELECOM TERMINATION DETAIL DRAWING - IPTV SYSTEM

PATCH PANEL TERMINATION DETAIL



COMMON DETAIL - SET-TOP BOX CABLE TERMINATION

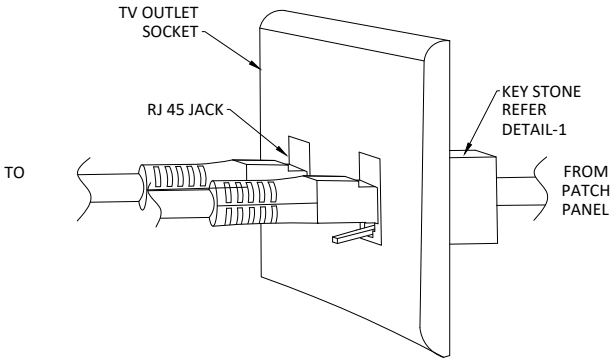


PORT NO.	DESCRIPTION
1	POWER ON/OFF SWITCH
2	POWER INPUT
3	IR
4	ETHERNET
5	HDMI
6	A/V

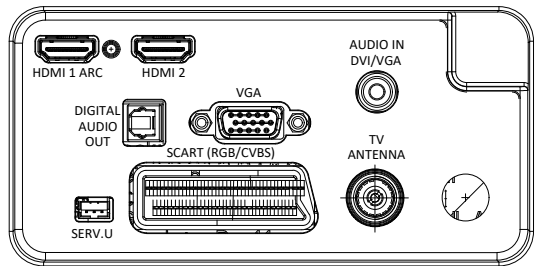
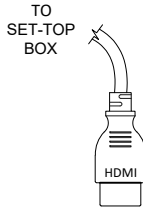
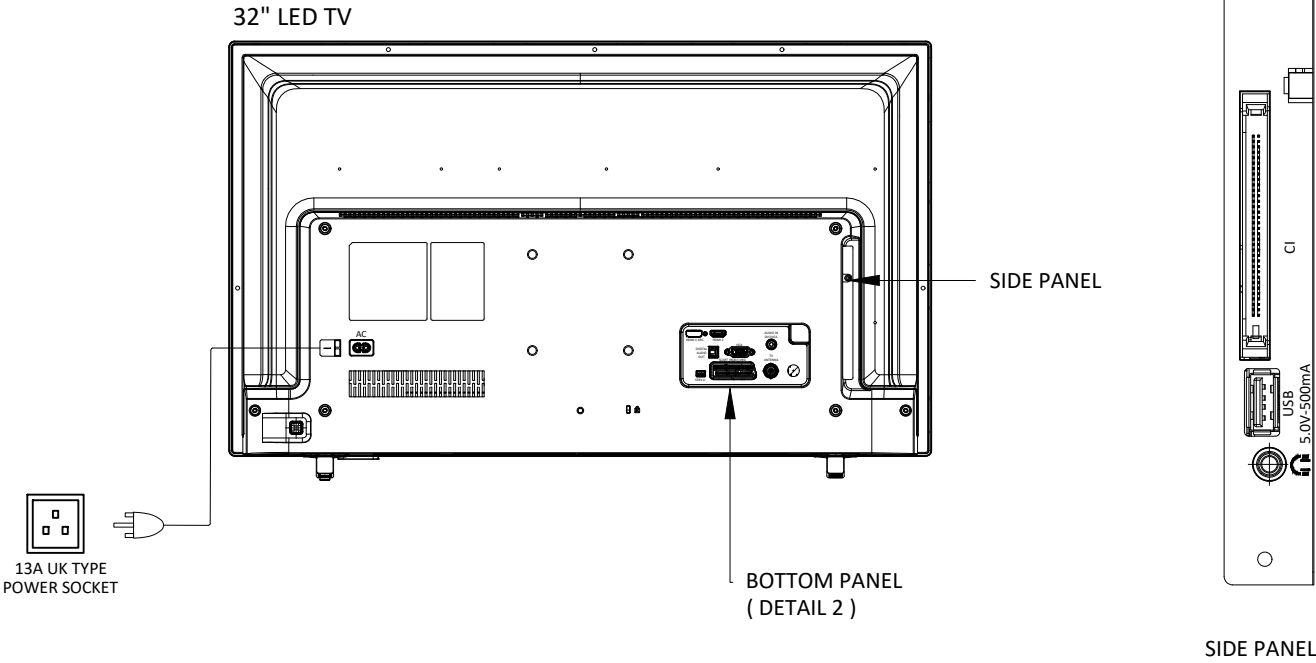
DETAIL - 1

WH/BU	5	3	WH/GN
BU	4	6	GN
WH/BR	7	1	WH/OG
BR	8	2	OG
RJ45 B.O.B			

COMMON DETAIL
CABLE TERMINATION METHOD FOR RJ45
KEYSTONE IN PATCH PANEL
(T - 568B STANDARD)



COMMON DETAIL - CABLE TERMINATION 32" LED TV



DETAIL - 2

REVISIONS

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PROJECT:

OKWOK-FPSO
TOPSIDES PRODUCTION FACILITIES

LOCATION:

OKWOK FIELD OFFSHORE NIGERIA-OML 67

TITLE:

TELECOM TERMINATION DRAWING - IPTV SYSTEM

SCALE:

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2024-VVO-EMEM-TS-TD-012
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TELECOM INTERNAL WIRING DRAWING- IPTV SYSTEM

REVISIONS

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C	11.12.2024	ISSUED FOR DESIGN	CK	HS	TN	
B	25.11.2024	ISSUED FOR APPROVAL	CK	HS	TN	
A	01.11.2024	ISSUED FOR REVIEW	CK	HS	TN	

COMPANY:



ORIENTAL
KWOK



ADDAX
PETROLEUM

CONTRACTOR:



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PROJECT:

OKWOK-FPSO
TOPSIDES PRODUCTION FACILITIES

LOCATION:

OKWOK FIELD OFFSHORE NIGERIA-OML 67

TITLE:

TELECOM INTERNAL WIRING DRAWING- IPTV SYSTEM

SCALE:	DRAWING NO. 2024-VIVO-EMEM-TS-IW-012	WCP DRAWING NO. WCP0KWTP-450-TE-(VAE)0G-004	REV.
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	A	B	C	D	E	F	G	H	I
1									
2									
3									
4									
5									
6									
7									
	A	B	C	D	E	F	G	H	I

LETTER SYMBOLS FOR COLORS

ABBREVIATION	DESCRIPTION
BL	BLUE
OR	ORANGE
GR	GREEN
BR	BROWN
SL	SLATE
WH	WHITE
RD	RED
BK	BLACK
G/Y	GREEN/YELLOW

ABBREVIATIONS

ABBREVIATION	DESCRIPTION
TB	TERMINAL BLOCKS
MCB	MINIATURE CIRCUIT BREAKER
CB	CIRCUIT BREAKER
PDU	POWER DISTRIBUTION UNIT
F	FRONT
R	REAR
L	LIVE
N	NEUTRAL
FO	FIBER OPTIC
HDMI	HIGH-DEFINITION MULTIMEDIA INTERFACE
SM	SINGLE MODE

LEGEND

SYMBOL	DESCRIPTION
	ETHERNET PORT
	CAT6A CONNECTOR
	VGA PORT
	VGA CONNECTOR
	USB PORT
	USB CONNECTOR
	POWER SUPPLY PORT
	FO LC TYPE CONNECTOR
	SFP CONNECTOR

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2	01.11.2024	25.11.2024	11.12.2024	INDEX PAGE	
3	01.11.2024	25.11.2024	11.12.2024	IPTV SYSTEM CABINET - INTERNAL WIRING DIAGRAM	
4	01.11.2024	25.11.2024	11.12.2024	IPTV SYSTEM CABINET - INTERNAL WIRING DIAGRAM	
5	01.11.2024	25.11.2024	11.12.2024	IPTV SYSTEM CABINET (86-RX-009) INTERNAL WIRING	
6	01.11.2024	25.11.2024	11.12.2024	IPTV SYSTEM CABINET (86-RX-002) INTERNAL WIRING	

NOTE :

1. THE CABLE SUPPLIED WITH THE EQUIPMENT.
WIRING COLOURS:
BR : 230VAC : L
BL : 230VAC : N
RD : +24V DC : +
BK : OV : -
YL/GN: PE/ IE

2. THIS DRAWING IS REVISED. (REFER SHEET 06)

REFERENCE			
NO	REVISION	CLIENT DOC NO.	TITLE
1	B	WCPOKWTP-450-TE-(VAE)BF-010	Telecom Cable Block Diagram - IPTV System
2	B	WCPOKWTP-450-TE-(VAE)DG-032	Telecom Termination Drawing - IPTV System
3	A	WCPOKWTP-450-TE-(VAE)DS-010	Telecom System Data Sheet- IPTV System

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LOCATION:

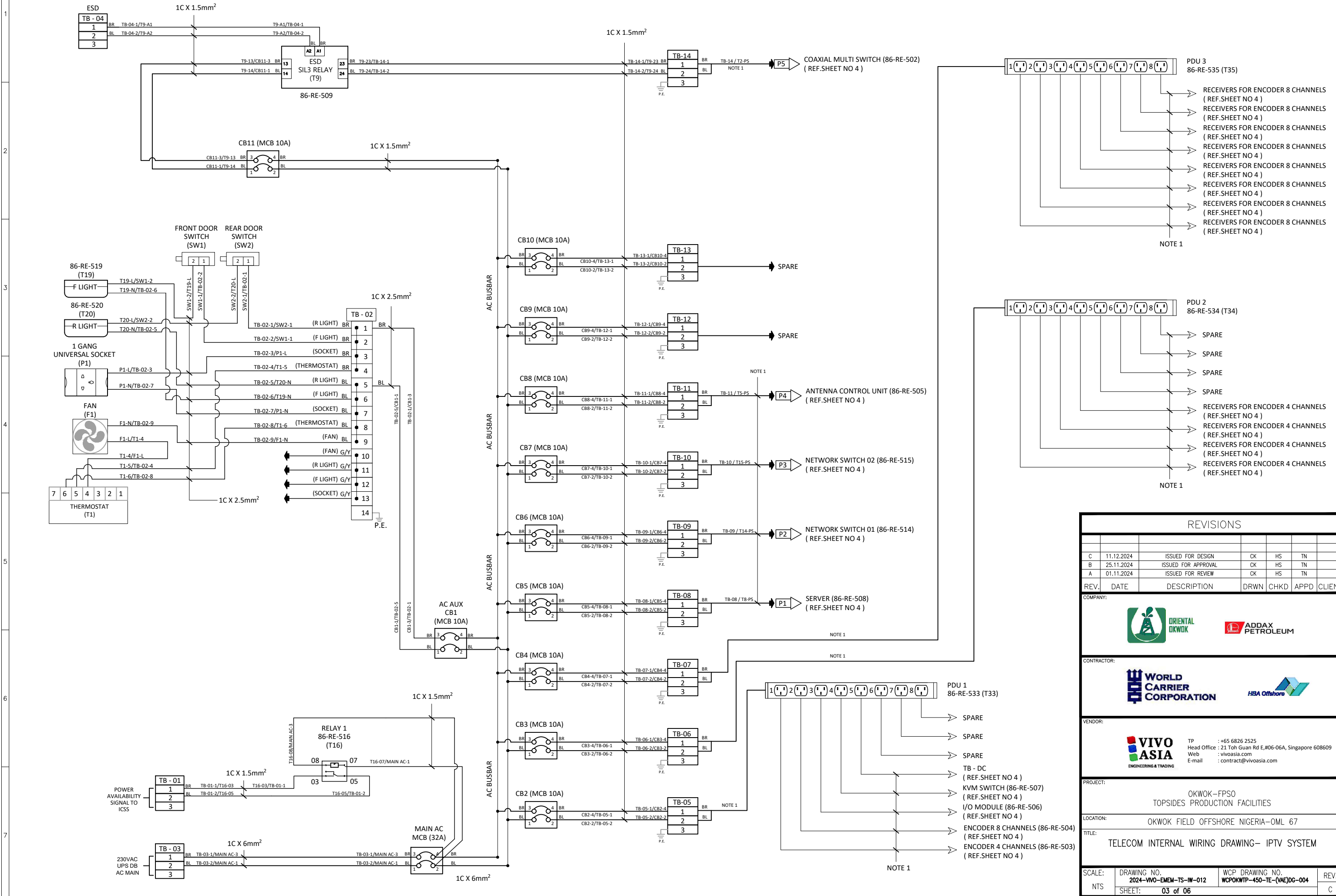
OKWOK FIELD OFFSHORE NIGERIA-OML 67

TITLE:

TELECOM INTERNAL WIRING DRAWING- IPTV SYSTEM

SCALE:	DRAWING NO.	WCP DRAWING NO.	REV.
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IPTV SYSTEM CABINET - INTERNAL WIRING DIAGRAM



REVISIONS

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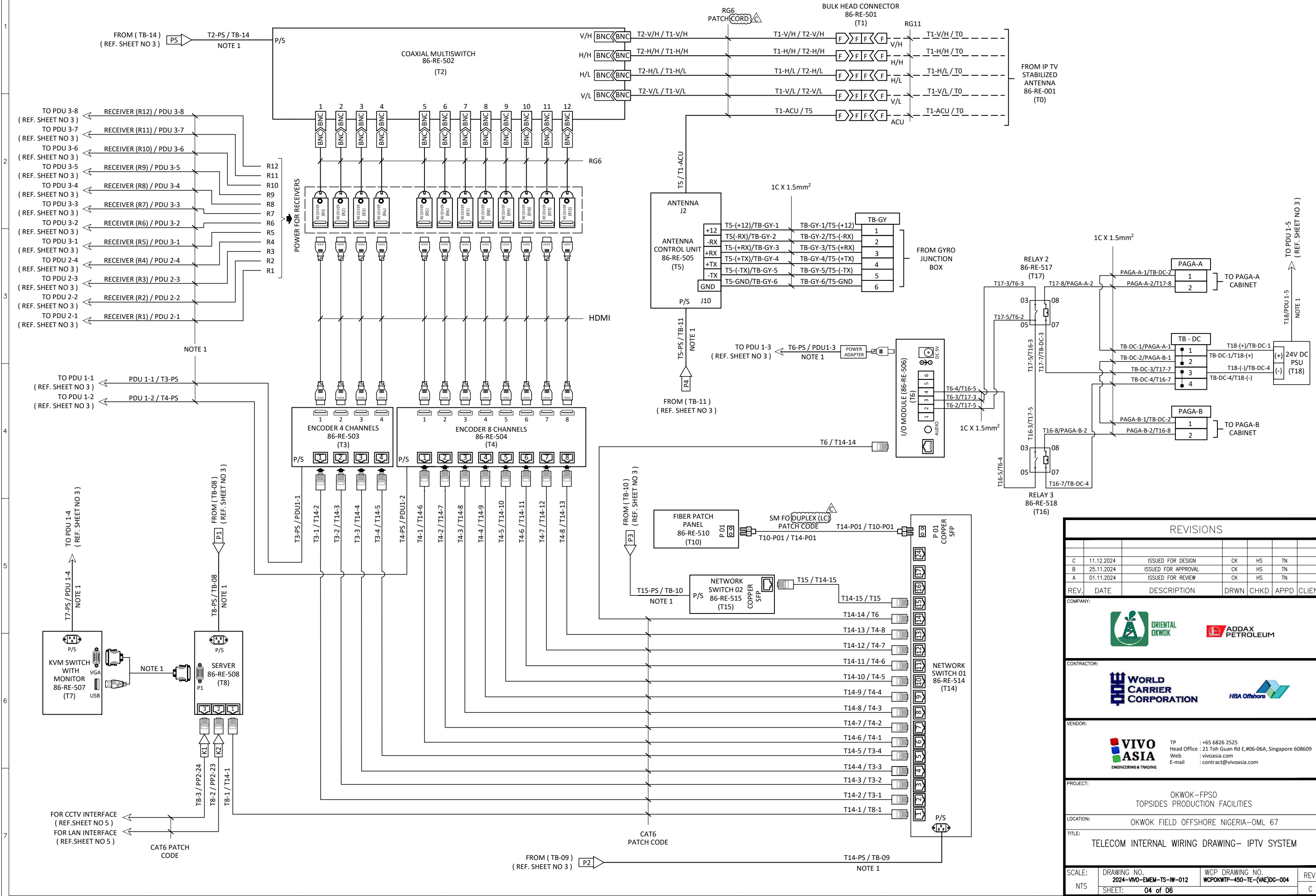
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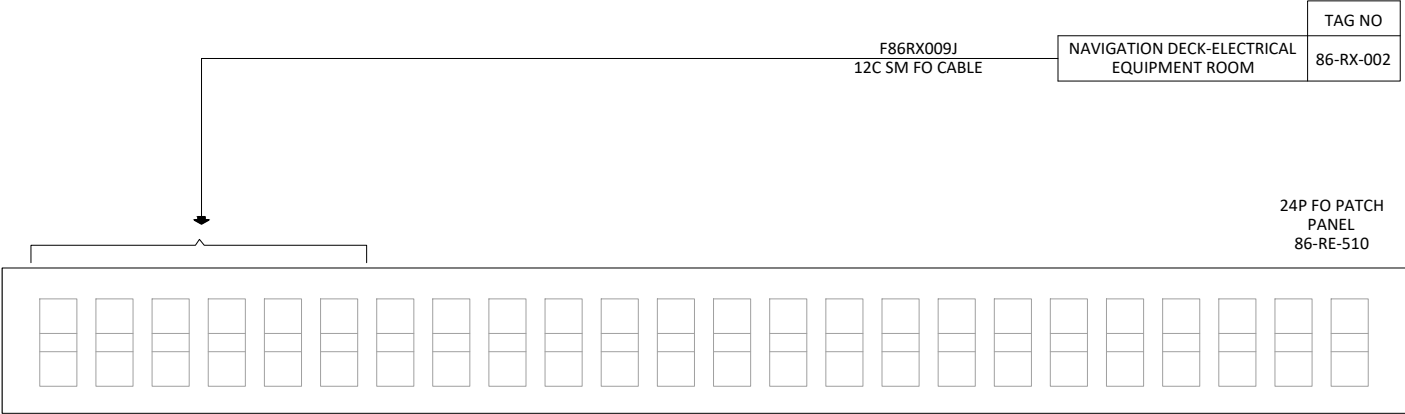
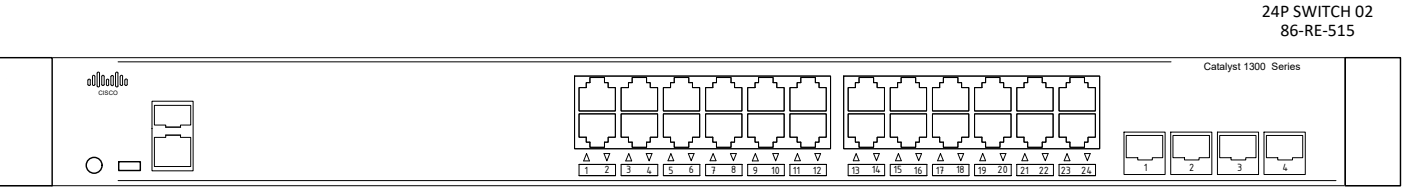
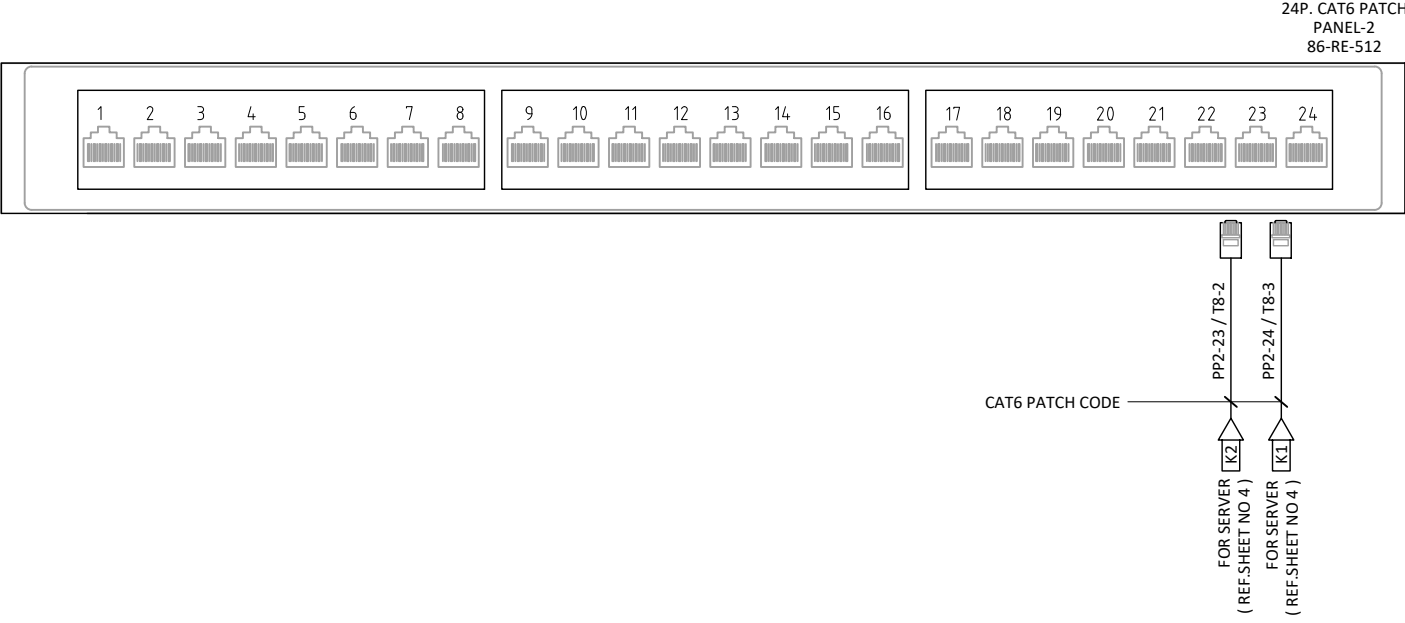
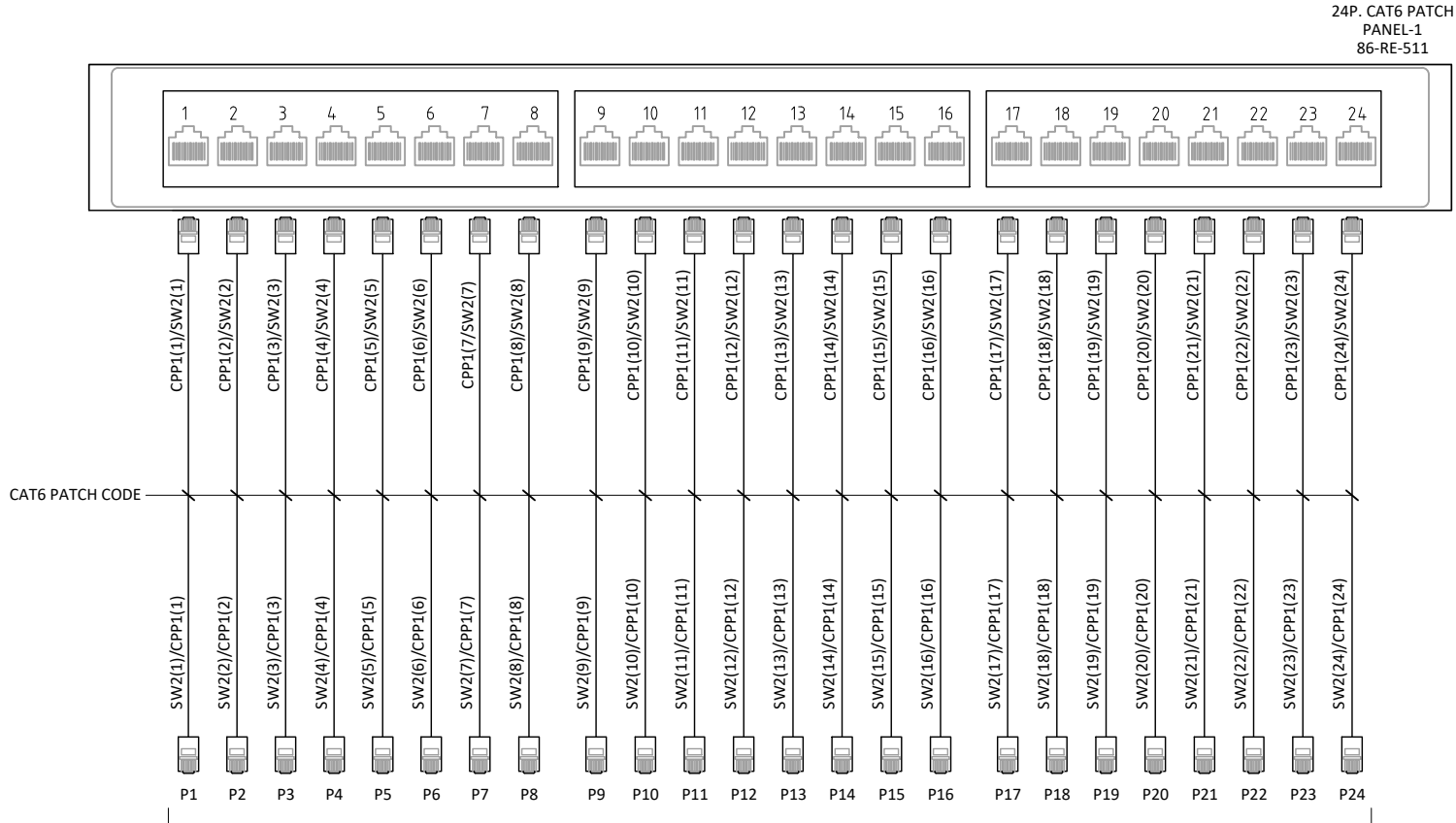
03 of 06

IPTV SYSTEM CABINET - INTERNAL WIRING DIAGRAM



REVISIONS						
C	11.12.2024	ISSUED FOR DESIGN	CK	HS	TN	
B	25.11.2024	ISSUED FOR APPROVAL	CK	HS	TN	
A	01.11.2024	ISSUED FOR REVIEW	CK	HS	TN	
REV.	DATE	DESCRIPTION	DRWN	CHKD	APPD	CLIENT
COMPANY:						
		ORIENTAL KWOK	 ADDAX PETROLEUM			
CONTRACTOR:						
		WORLD CARRIER CORPORATION	 HBA Offshore			
VENDOR:						
 VIVO ASIA ENGINEERING & TRADING		TP : +65 6826 2525 Head Office : 21 Toh Guan Rd E, #06-06A, Singapore 608609 Web : vivoasia.com E-mail : contract@vivoasia.com				
PROJECT:						
OKWOK-FPSO TOPSIDES PRODUCTION FACILITIES						
LOCATION:						
OKWOK FIELD OFFSHORE NIGERIA-OML 67						
TITLE:						
TELECOM INTERNAL WIRING DRAWING- IPTV SYSTEM						
SCALE:	DRAWING NO.		WCP DRAWING NO.		REV.	
NTS	2024-VIVO-EMEM-TS-W-012		WCP0KWTP-450-TE-(VAE)0G-004		C	
SHEET:		04 of 06				

IPTV SYSTEM CABINET (86-RX-009) INTERNAL WIRING



REVISIONS						
REV.	DATE	DESCRIPTION	DRWN	CHKD	APPD	CLIENT
C	11.12.2024	ISSUED FOR DESIGN	CK	HS	TN	
B	25.11.2024	ISSUED FOR APPROVAL	CK	HS	TN	
A	01.11.2024	ISSUED FOR REVIEW	CK	HS	TN	

COMPANY:



CONTRACTOR:



VENDOR:



TP : +65 6826 2525
Head Office : 21 Toh Guan Rd E, #06-06A, Singapore 608609
Web : vivoasia.com
E-mail : contract@vivoasia.com

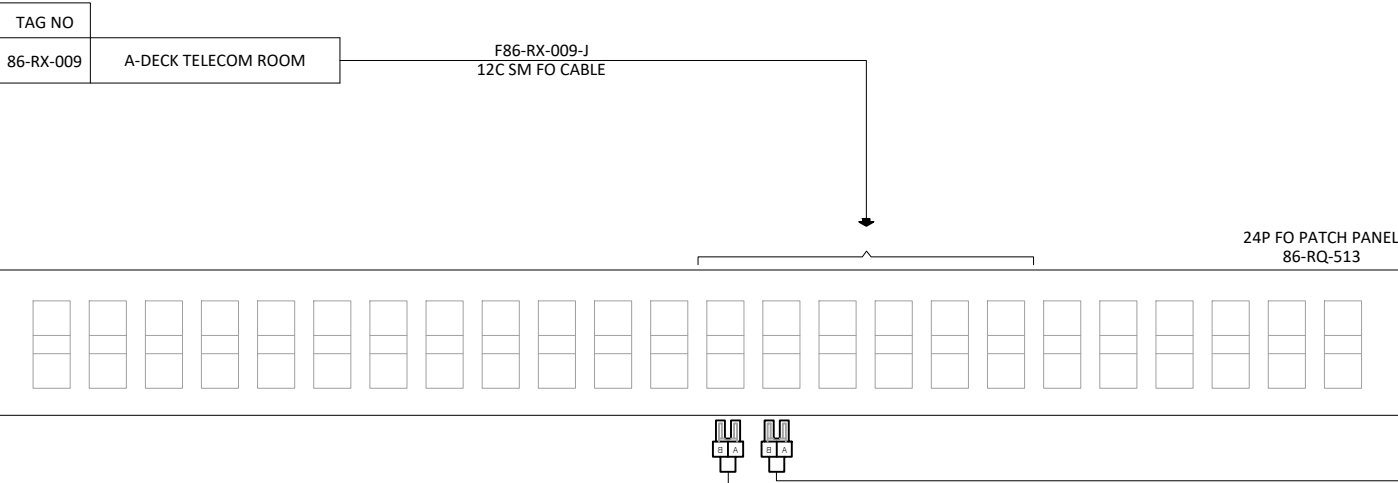
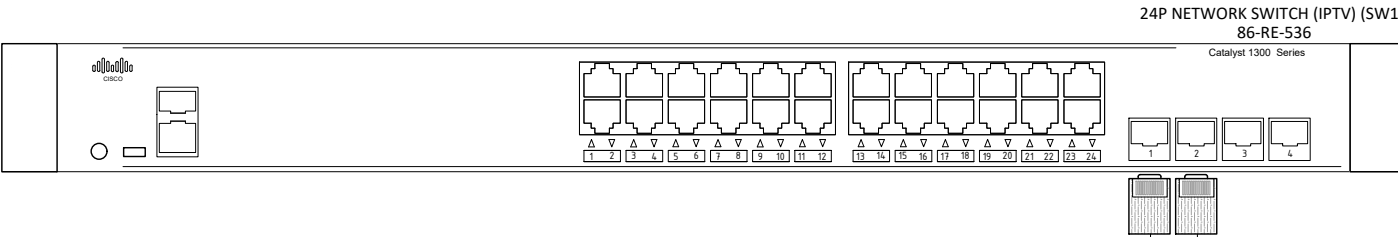
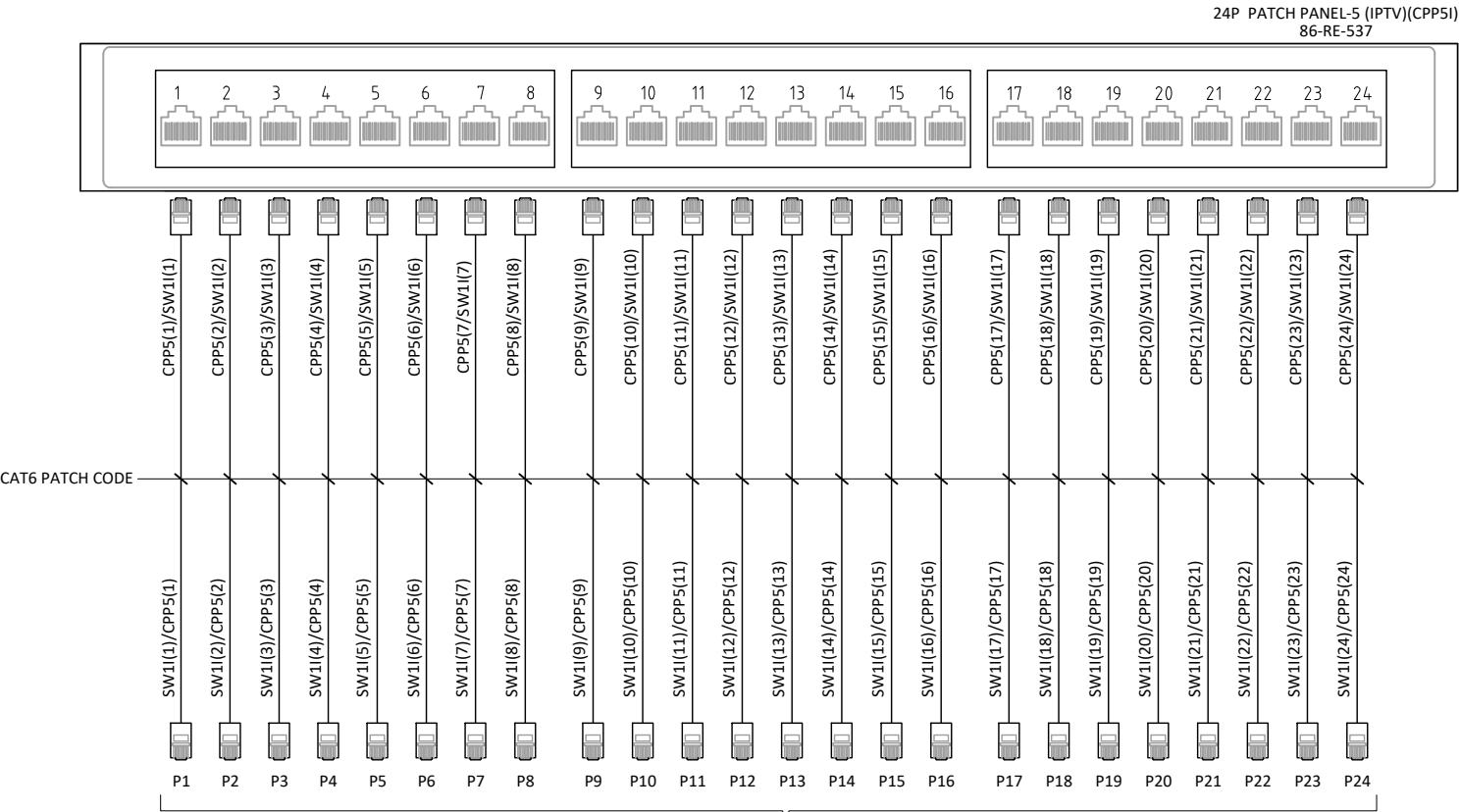
PROJECT:
OKWOK-FPSO
TOPSIDES PRODUCTION FACILITIES

LOCATION:
OKWOK FIELD OFFSHORE NIGERIA-OML 67

TITLE:
TELECOM INTERNAL WIRING DRAWING- IPTV SYSTEM

SCALE: NTS	DRAWING NO. 2024-VIVO-EMEM-TS-IW-012 SHEET: 05 of 06	WCP DRAWING NO. WCP0KWTP-450-TE-(VAE)06-004	REV. C
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IPTV SYSTEM CABINET (86-RX-002) INTERNAL WIRING



REVISIONS

REV.	DATE	DESCRIPTION	DRWN	CHKD	APPD	CLIENT
C	11.12.2024	ISSUED FOR DESIGN	CK	HS	TN	
B	25.11.2024	ISSUED FOR APPROVAL	CK	HS	TN	
A	01.11.2024	ISSUED FOR REVIEW	CK	HS	TN	

COMPANY:



CONTRACTOR:



VENDOR:



PROJECT:
OKWOK-FPSO
TOPSIDES PRODUCTION FACILITIES

LOCATION:
OKWOK FIELD OFFSHORE NIGERIA-OML 67

TITLE:
TELECOM INTERNAL WIRING DRAWING- IPTV SYSTEM

SCALE: NTS	DRAWING NO. 2024-VIVO-EMEM-TS-IW-012 SHEET: 06 of 06	WCP DRAWING NO. WCP0KWTP-450-TE-(VAE)06-004	REV. C
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TELECOM TYPICAL INSTALLATION DRAWING - IPTV SYSTEM

REVISIONS

REV.	DATE	DESCRIPTION	DRWN	CHKD	APPD	CLIENT
0	23.12.2024	ISSUED FOR CONSTRUCTION	VM	HS	TN	
A	29.10.2024	ISSUED FOR REVIEW	VM	HS	TN	

COMPANY:



Oriental
OKWOK



ADDAX
PETROLEUM

CONTRACTOR:



WORLD
CARRIER
CORPORATION



HBA Offshore

VENDOR:



VIVO
ASIA
ENGINEERING & TRADING

TP : +65 6826 2525

Head Office : 21 Toh Guan Rd E, #06-06A, Singapore 608609

Web : vivoasia.com

E-mail : contract@vivoasia.com

PROJECT:

OKWOK-FPSO
TOPSIDES PRODUCTION FACILITIES

LOCATION:

OKWOK FIELD OFFSHORE NIGERIA-OML 67

TITLE:

TELECOM TYPICAL INSTALLATION DRAWING – IPTV SYSTEM

SCALE:	DRAWING NO.	WCP DRAWING NO.	REV.
NTS	2024-VIVO-EMEM-TS-TY-012	WCP0KWTP-450-TE-(VAE)DG-014	0
	SHEET: 01 of 09		

INDEX PAGE

SHEET NO.	REVISION		DESCRIPTION	REMARKS
	A	O		
1	29.10.2024	23.12.2024	COVER PAGE	
2	29.10.2024	23.12.2024	INDEX PAGE	
3	29.10.2024	23.12.2024	IPTV SET-TOP BOX CT-2211 - DIMENSION & MOUNTING DETAILS DRAWING	
4	29.10.2024	23.12.2024	32" TV MONITOR 32HFL3014/12 - DIMENSION & MOUNTING DETAILS DRAWING	
5	29.10.2024	23.12.2024	SEATEL 120TV ANTENNA - DIMENSION & MOUNTING DETAILS DRAWING	
6	29.10.2024	23.12.2024	HIGH-PERFORMANCE SET-TOP BOX CT-2211 - BRACKET WALL MOUNTED (HORIZONTAL) TYPICAL INSTALLATION DRAWING	
7		23.12.2024	HIGH-PERFORMANCE SET-TOP BOX CT-2211 - BRACKET WALL MOUNTED (VERTICAL) TYPICAL INSTALLATION DRAWING	
8	29.10.2024	23.12.2024	32" TV MONITOR 32HFL3014/12 - WALL MOUNTED TYPICAL INSTALLATION DRAWING	
9	29.10.2024	23.12.2024	SEATEL 120TV ANTENNA - POLE MOUNTING TYPICAL INSTALLATION DRAWING	

NOTES:

1. ALL DIMENSIONS ARE IN MILLIMETER UNLESS OTHERWISE STATED
2. THE INSTALLATION MATERIALS INCLUDING MOUNTING PLATES, BOLTS, WASHER AND NUTS ARE PROVIDED BY OTHERS.
3. MECHANICAL ACCESSORIES FOR INSTALLATION LIKE CABLE TRAYS, CABLE GLANDS, SUPPORT CLAMPS SHALL BE PROVIDED BY YARD
4. THIS DRAWING IS REVISED. (REFER SHEET NO. 03,08)
5. THIS SHEET NEWLY ADDED. (REFER SHEET NO. 07)

REFERENCE				
NO	REVISION	CLIENT DOC NO.	TITLE	
1	A	WCPOKWTP-450-TE-(VAE)DS-010	Telecom System Data Sheet- IPTV System	
2	B	WCPOKWTP-450-TE-(VAE)LY-011	Telecom Device Location Layout Drawing- IPTV System	

REVISIONS

0	23.12.2024	ISSUED FOR CONSTRUCTION	VM	HS	TN	
A	29.10.2024	ISSUED FOR REVIEW	VM	HS	TN	
REV.	DATE	DESCRIPTION	DRWN	CHKD	APPD	CLIENT

COMPANY:

ORIENTAL
OKWOK

ADDAX
PETROLEUM

CONTRACTOR:

WORLD
CARRIER
CORPORATION

HBA Offshore

VENDOR:

VIVO
ASIA
ENGINEERING & TRADING

TP : +65 6826 2525
Head Office : 21 Toh Guan Rd E,#06-06A, Singapore 608609
Web : vivoasia.com
E-mail : contract@vivoasia.com

PROJECT:

OKWOK-FPSO
TOPSIDES PRODUCTION FACILITIES

LOCATION:

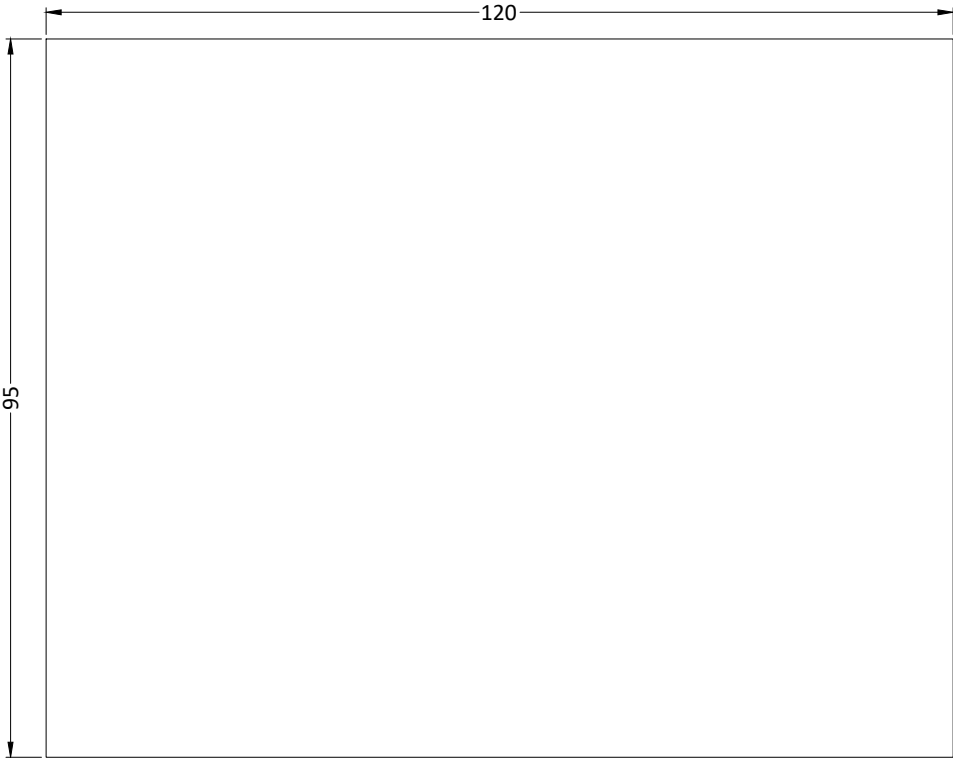
OKWOK FIELD OFFSHORE NIGERIA-OML 67

TITLE:

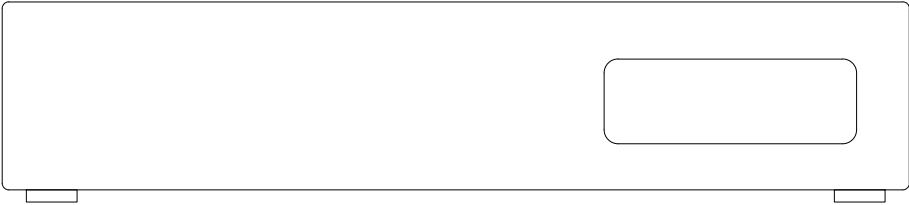
TELECOM TYPICAL INSTALLATION DRAWING – IPTV SYSTEM

SCALE:	DRAWING NO.	WCP DRAWING NO.	REV.
NTS	2024-VIVO-EMEM-TS-TY-012	WCPOKWTP-450-TE-(VAE)DG-014	0
	SHEET: 02 of 09		

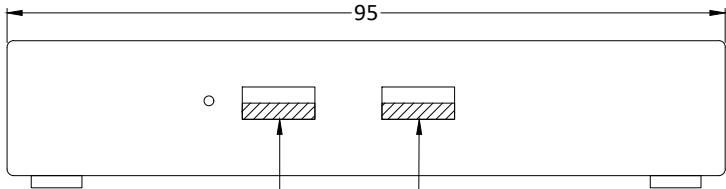
IPTV SET-TOP BOX CT-2211- DIMENSION & MOUNTING DETAILS DRAWING



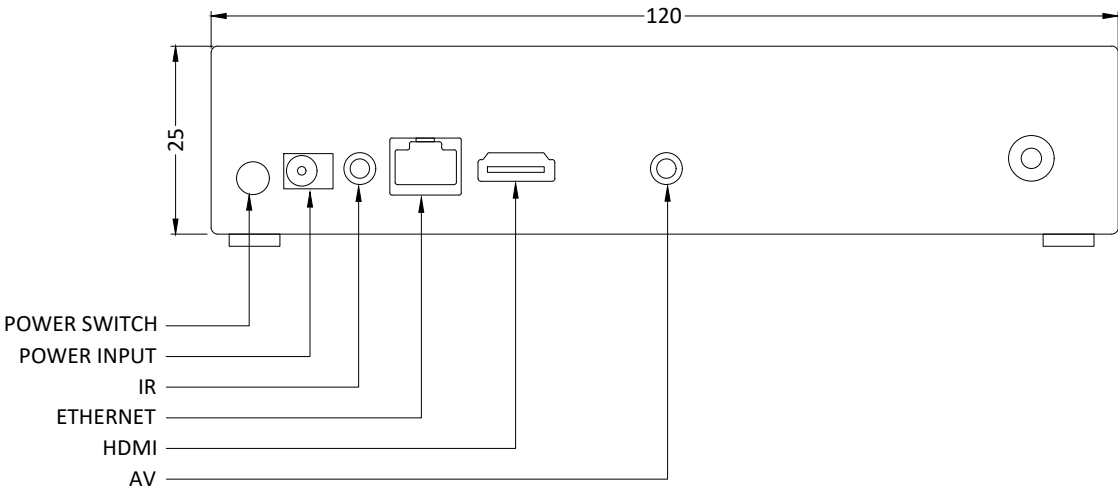
PLAN VIEW



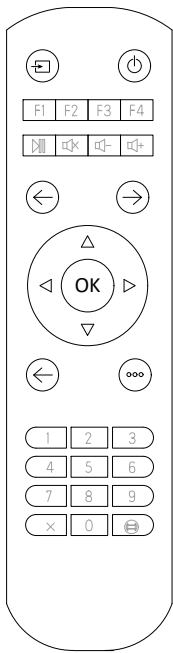
FRONT VIEW



SIDE VIEW



REAR VIEW

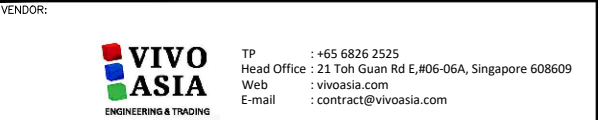


REMOTE CONTROL

TECHNICAL SPECIFICATIONS:	
MANUFACTURER	: CLEAR
MODEL	: H98MUV3
MAIN CHIP	: GK6323V100C
CPU	: QUAD-CORE 64 BIT ARM CORTEX A55
GPU	: MULTI CORE MALI-450 , OPENGL ES 2.0/1.1, OPEN VG 1.1
SYSTEM	: ANDROID 9.0
EMMC	: EMMC 8GB(16GB OPTIONAL)
SDRAM	: LPDDR4 2GB
WIFI	: IEEE 802.11 A/B/G/N/AC, 2.4G/5G
BLUETOOTH	: 5.0
IR	: 38KHZ
HDMI PORT	: SUPPORT HDCP2.2
ETHERNET PORT	: RJ45(10/100MBPS)
AV PORT	: 3.5MM AV JACK
USB PORT	: 2XUSB 2.0
POWER INTERFACE SUPPORTS	: 12V POWER INPUT
REMOTE CONTROL	
IR REMOTE CONTROL	
BATTERY TYPE	: 2 X 1.5V AAA
APPEARANCE SIZE	: 112X112X26mm

REVISIONS

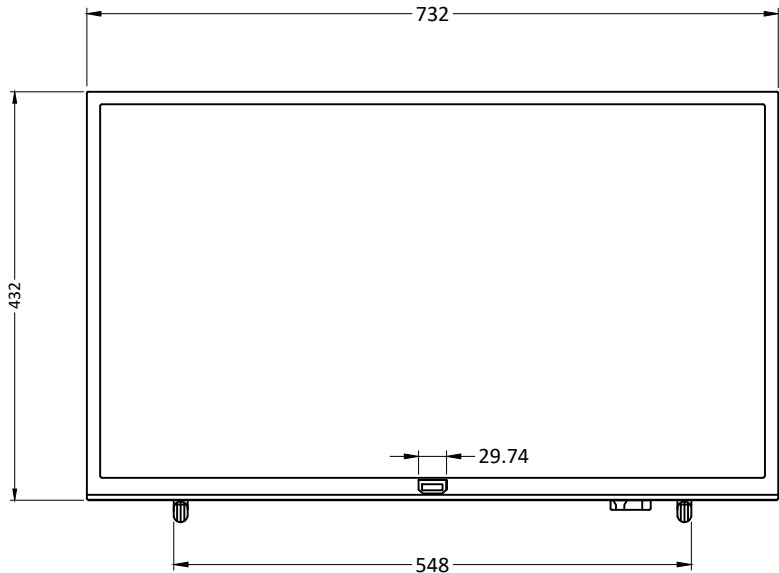
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A	29.10.2024	ISSUED FOR REVIEW	VM	HS	TN	



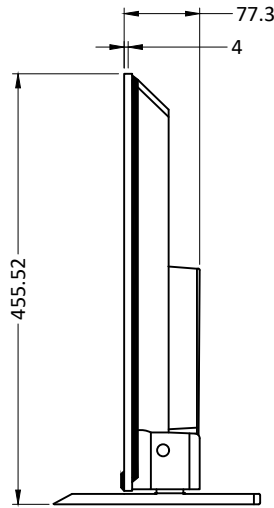
PROJECT:	OKWOK-FPSO TOPSIDES PRODUCTION FACILITIES
LOCATION:	OKWOK FIELD OFFSHORE NIGERIA-OML 67
TITLE:	TELECOM TYPICAL INSTALLATION DRAWING – IPTV SYSTEM

SCALE:	DRAWING NO.	WCP DRAWING NO.	REV.
1:1	2024-VIVO-EMEM-TS-TY-012	WCP0KWTP-450-TE-(VAE)DG-014	0
	SHEET:	03 of 09	

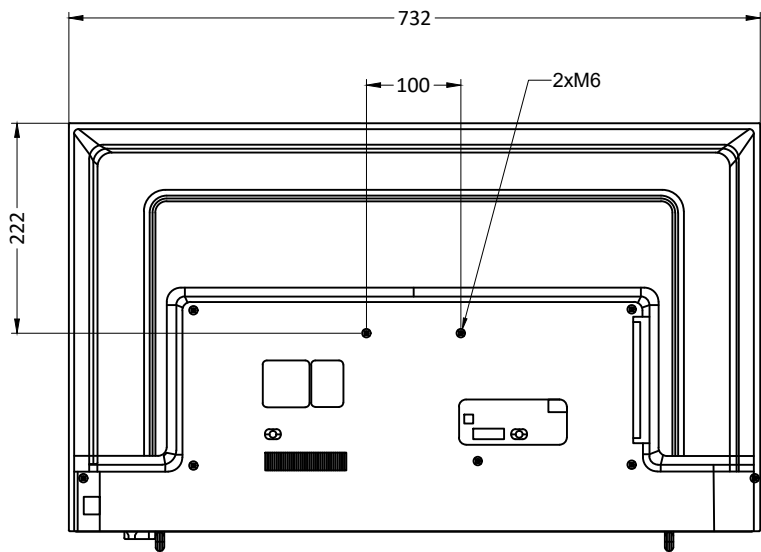
32" TV MONITOR 32HFL3014/12 - DIMENSION & MOUNTING DETAILS DRAWING



FRONT VIEW



SIDE VIEW



REAR VIEW

TECHNICAL SPECIFICATIONS:

MANUFACTURER	: PHILIPS
MODEL	: 32HFL3014/12
MECHANICAL	
DIMENSIONS (W X H X D) [mm]	: 732 X 432 X 77
WITHOUT STAND	: 4.6
WEIGHT [kg]	: 4.7 KG
PRODUCT WEIGHT (+STAND)	: 2XM6 (100mm)
VESA WALL MOUNT COMPATIBLE	
MULTIMEDIA	
MULTIMEDIA CONNECTIONS	: USB
VIDEO PLAYBACK SUPPORTED	: FORMATS: H.264/MPEG4 AVC, MPEG1, MPEG2, MPEG4, WMV9/VC1, MP4, MPG, TS, WMV, HEVC, CONTAINERS: AVI, MKV
SUBTITLE FORMATS SUPPORTED	: SRT, SMI, SSA SUB, TXT
MUSIC FORMATS SUPPORTED	: MP3, AAC, WAV, WMA (V2 UP TO V9.2)
PICTURE FORMATS SUPPORTED	: BMP, JPG, PNG, GIF
SUPP. VIDEO RESOLUTION ON USB	: UP TO 1920X1080P@60HZ
ELECTRICAL	
MAINS POWER	: AC 220-240V; 50-60HZ
ENERGY LABEL CLASS	: A+
EU ENERGY LABEL POWER	: 28 W
ANNUAL ENERGY CONSUMPTION	: 41 KW·H
STANDBY POWER CONSUMPTION	: <0.3W
POWER SAVING FEATURES	: ECO MODE, AUTO SWITCH-OFF TIMER
AMBIENT TEMPERATURE	: 0 °C TO 40 °C
DISPLAY	
DISPLAY	: LED HD TV, D-LED
DIAGONAL SCREEN SIZE	: 32 INCH / 80 CM
PANEL RESOLUTION	: 1366 X 768P
BRIGHTNESS	: 250 CD/M²
VIEWING ANGLE [°]	: 178º (H) / 178º (V)

REVISIONS

REV.	DATE	DESCRIPTION	DRWN	CHKD	APPD	CLIENT
0	23.12.2024	ISSUED FOR CONSTRUCTION	VM	HS	TN	
A	29.10.2024	ISSUED FOR REVIEW	VM	HS	TN	

COMPANY:



CONTRACTOR:



VENDOR:



PROJECT:

OKWOK-FPSO
TOPSIDES PRODUCTION FACILITIES

LOCATION:

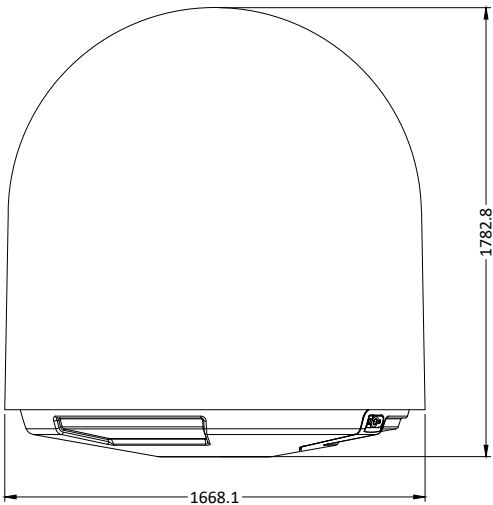
OKWOK FIELD OFFSHORE NIGERIA-OML 67

TITLE:

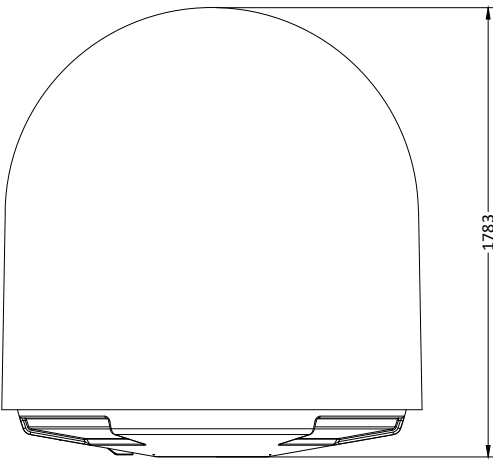
TELECOM TYPICAL INSTALLATION DRAWING – IPTV SYSTEM

SCALE:	DRAWING NO.	WCP DRAWING NO.	REV.
1:8	2024-VIVO-EMEM-TS-TY-012	WCP0KWTP-450-TE-(VAE)DG-014	0
	SHEET:	04 of 09	

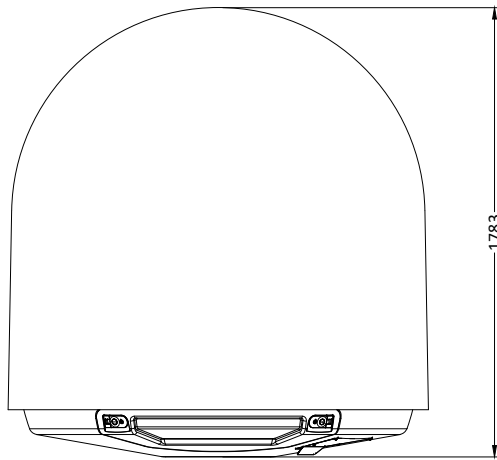
SEATEL 120TV ANTENNA - DIMENSION & MOUNTING DETAILS DRAWING



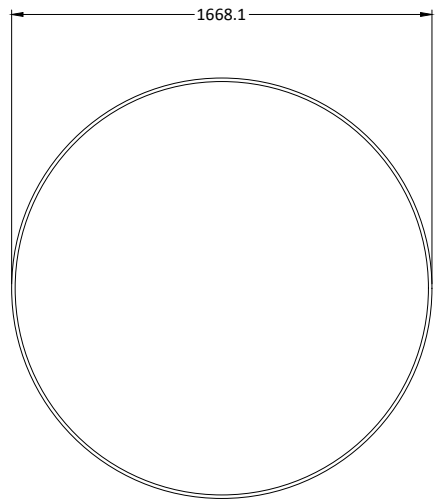
SIDE VIEW



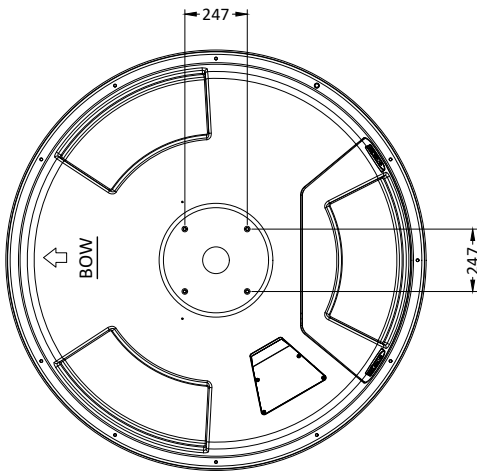
FRONT VIEW



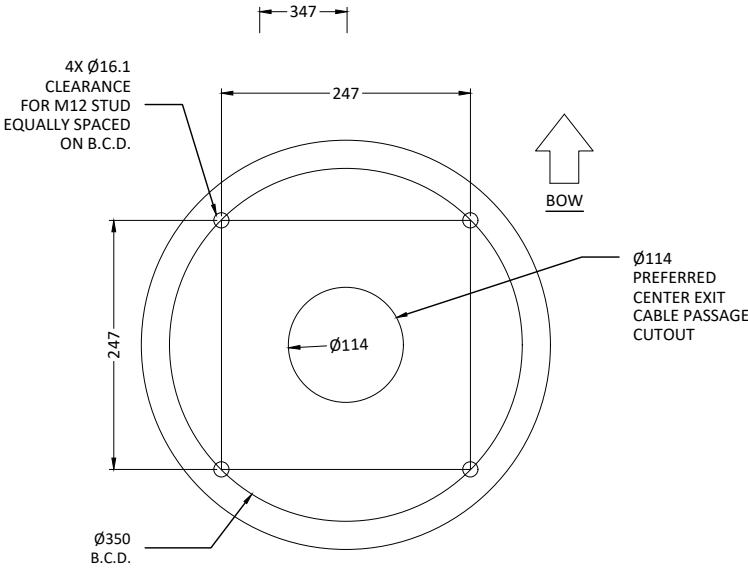
REAR VIEW



TOP VIEW



BOTTOM VIEW



MOUNTING SURFACE PATTERN

TECHNICAL SPECIFICATIONS:

MANUFACTURER	: SEATEL
MODEL	: 120TV
MIN. EIRP KU-BAND	: 40.5 DBW
DISH DIAMETER	: 124.46 CM (49 IN)
RADOME DIAMETER	: 166.88 CM (65.7 IN)
RADOME HEIGHT	: 178.31 CM (70.2 IN)
ADE WEIGHT	: 115.67 Kg (255 LBS.)
OPERATING TEMPERATURE RANGE	: -20°C TO 55°C
STORAGE TEMPERATURE RANGE	: -40°C TO 70°C
PEDESTAL TYPE	: 3 AXIS (AZIMUTH, ELEVATION, CROSS LEVEL)
SHIPS MOTION	: ROLL +/- 25°; PITCH +/- 15°
TURNING RATE	: 15°DEG /SEC @ 5°DEG /SEC2
AZIMUTH RANGE	: 680°
ELEVATION RANGE	: -15° TO 115°
POLARIZATION	: LINEAR/CIRCULAR SELECTABLE
STABILIZATION ACCURACY	: 0.2° PEAK ERROR UNDER SPECIFIED SHIP'S MOTION
SKEW CONTROL	: AUTOMATIC
LNB TYPE	: KU-BAND WORLDWIDE PROGRAMMABLE
FREQUENCY RANGE	: KU-BAND (10.7 - 12.75 GHZ)

REVISIONS

REV.	DATE	DESCRIPTION	DRWN	CHKD	APPD	CLIENT
0	23.12.2024	ISSUED FOR CONSTRUCTION	VM	HS	TN	
A	29.10.2024	ISSUED FOR REVIEW	VM	HS	TN	

COMPANY:



CONTRACTOR:



VENDOR:



TP : +65 6826 2525
Head Office : 21 Toh Guan Rd E, #06-06A, Singapore 608609
Web : vivoasia.com
E-mail : contract@vivoasia.com

PROJECT:

OKWOK-FPSO
TOPSIDES PRODUCTION FACILITIES

LOCATION:

OKWOK FIELD OFFSHORE NIGERIA-OML 67

TITLE:

TELECOM TYPICAL INSTALLATION DRAWING – IPTV SYSTEM

SCALE:

1:30

DRAWING NO.

2024-VIVO-EMEM-TS-TY-012

WCP DRAWING NO.

WCPQKWTP-450-TE-(VAE)DG-014

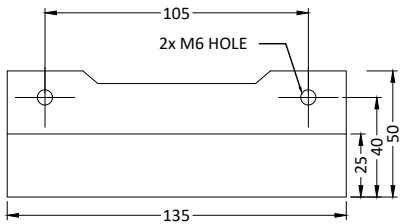
REV.

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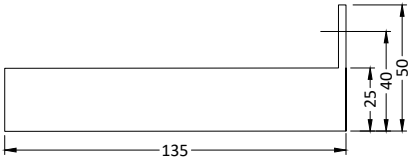
SHEET: 05 of 09

HIGH-PERFORMANCE SET-TOP BOX CT-2211 - BRACKET WALL MOUNTED (HORIZONTAL) TYPICAL INSTALLATION DRAWING

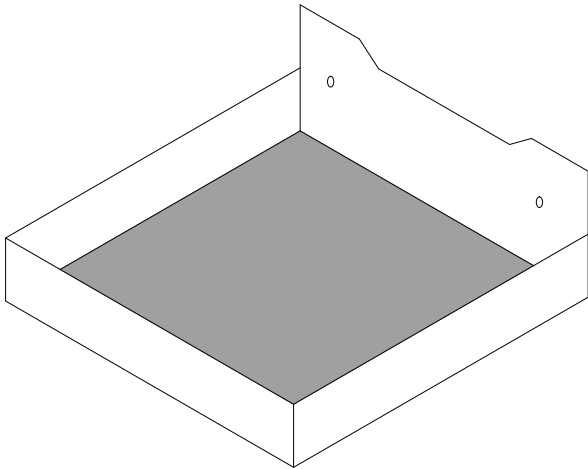
ITEM LIST				
ITEM NO.	DESCRIPTION	MANUFACTURER	SUPPLIED BY	QTY
1	IP TV SET -TOP BOX (CT-2211)	CLEAR	VIVO	01
2	SET TOP BOX MOUNTING BRACKET	BY YARD	BY YARD	01
3	M6 ANCHOR BOLT 2 PCS	BY YARD	BY YARD	01



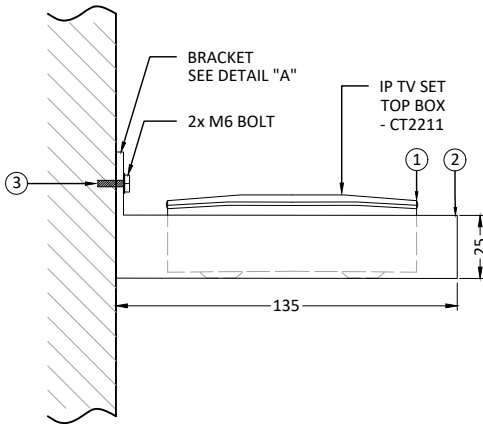
FRONT VIEW



SIDE VIEW



ISOMETRIC VIEW



SECTION VIEW

REVISIONS

REV.	DATE	DESCRIPTION	DRWN	CHKD	APPD	CLIENT
0	23.12.2024	ISSUED FOR CONSTRUCTION	VM	HS	TN	
A	29.10.2024	ISSUED FOR REVIEW	VM	HS	TN	

COMPANY:



ORIENTAL
OKWOK



ADDAX
PETROLEUM

CONTRACTOR:



WORLD
CARRIER
CORPORATION



HBA Offshore

VENDOR:



VIVO
ASIA
ENGINEERING & TRADING

TP : +65 6826 2525
Head Office : 21 Toh Guan Rd E, #06-06A, Singapore 608609
Web : vivoasia.com
E-mail : contract@vivoasia.com

PROJECT:

OKWOK-FPSO
TOPSIDES PRODUCTION FACILITIES

LOCATION:

OKWOK FIELD OFFSHORE NIGERIA-OML 67

TITLE:

TELECOM TYPICAL INSTALLATION DRAWING – IPTV SYSTEM

SCALE:

1:3

DRAWING NO.

2024-VIVO-EMEM-TS-TY-012

WCP DRAWING NO.

WCP0KWTP-450-TE-(VAE)DG-014

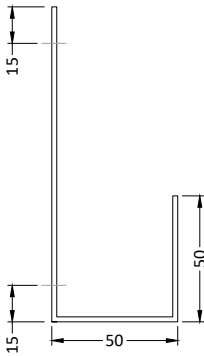
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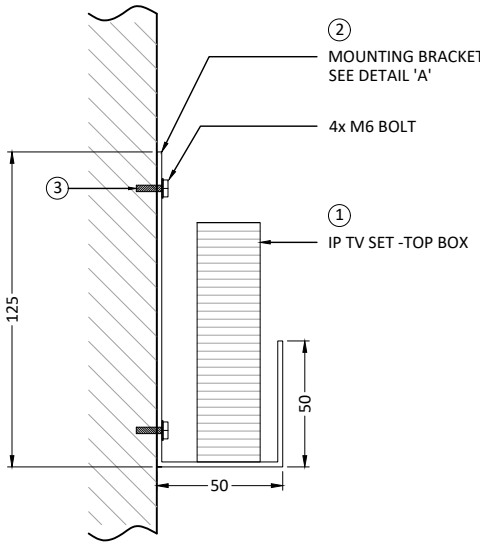
SHEET: 06 of 09

HIGH-PERFORMANCE SET-TOP BOX CT-2211 - BRACKET WALL MOUNTED (VERTICAL) TYPICAL INSTALLATION DRAWING

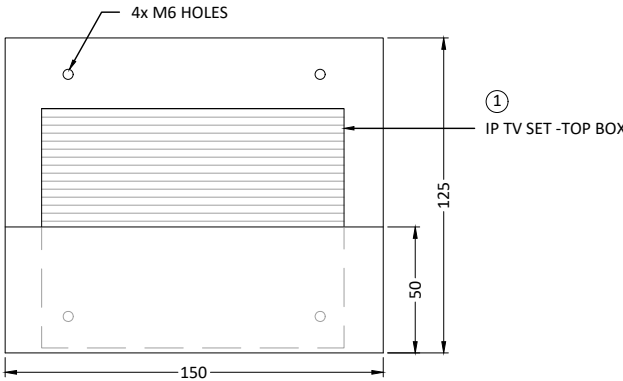
ITEM LIST				
ITEM NO.	DESCRIPTION	MANUFACTURER	SUPPLIED BY	QTY
1	IP TV SET -TOP BOX (CT-2211)	CLEAR	VIVO	01
2	SET TOP BOX MOUNTING BRACKET	BY YARD	BY YARD	01
3	M6 ANCHOR BOLT 4 PCS	BY YARD	BY YARD	01



DETAIL -A



SECTION VIEW



FRONT VIEW

REVISIONS

REV.	DATE	DESCRIPTION	DRWN	CHKD	APPD	CLIENT
0	23.12.2024	ISSUED FOR CONSTRUCTION	VM	HS	TN	
A	29.10.2024	ISSUED FOR REVIEW	VM	HS	TN	

COMPANY:



CONTRACTOR:



VENDOR:



TP : +65 6826 2525
Head Office : 21 Toh Guan Rd E, #06-06A, Singapore 608609
Web : vivoasia.com
E-mail : contract@vivoasia.com

PROJECT:

OKWOK-FPSO
TOPSIDES PRODUCTION FACILITIES

LOCATION:

OKWOK FIELD OFFSHORE NIGERIA-OML 67

TITLE:

TELECOM TYPICAL INSTALLATION DRAWING – IPTV SYSTEM

SCALE:

1:3

DRAWING NO.

2024-VIVO-EMEM-TS-TY-012

WCP DRAWING NO.

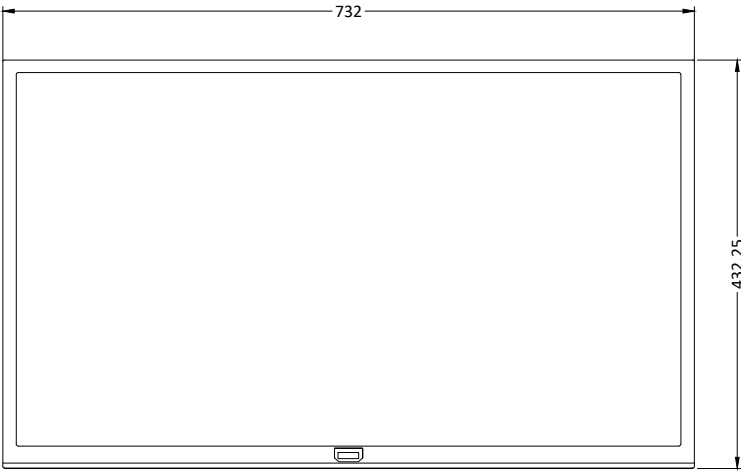
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REV.

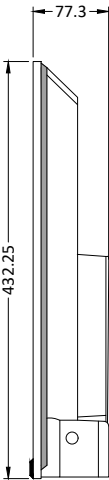
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SHEET: 07 of 09

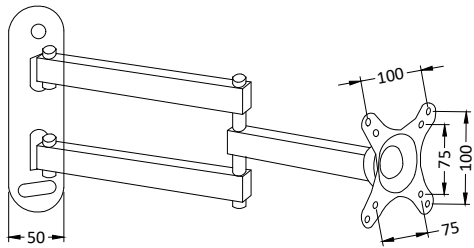
32" TV MONITOR 32HFL3014/12 - WALL MOUNTED TYPICAL INSTALLATION DRAWING



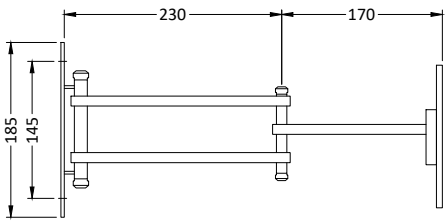
FRONT VIEW



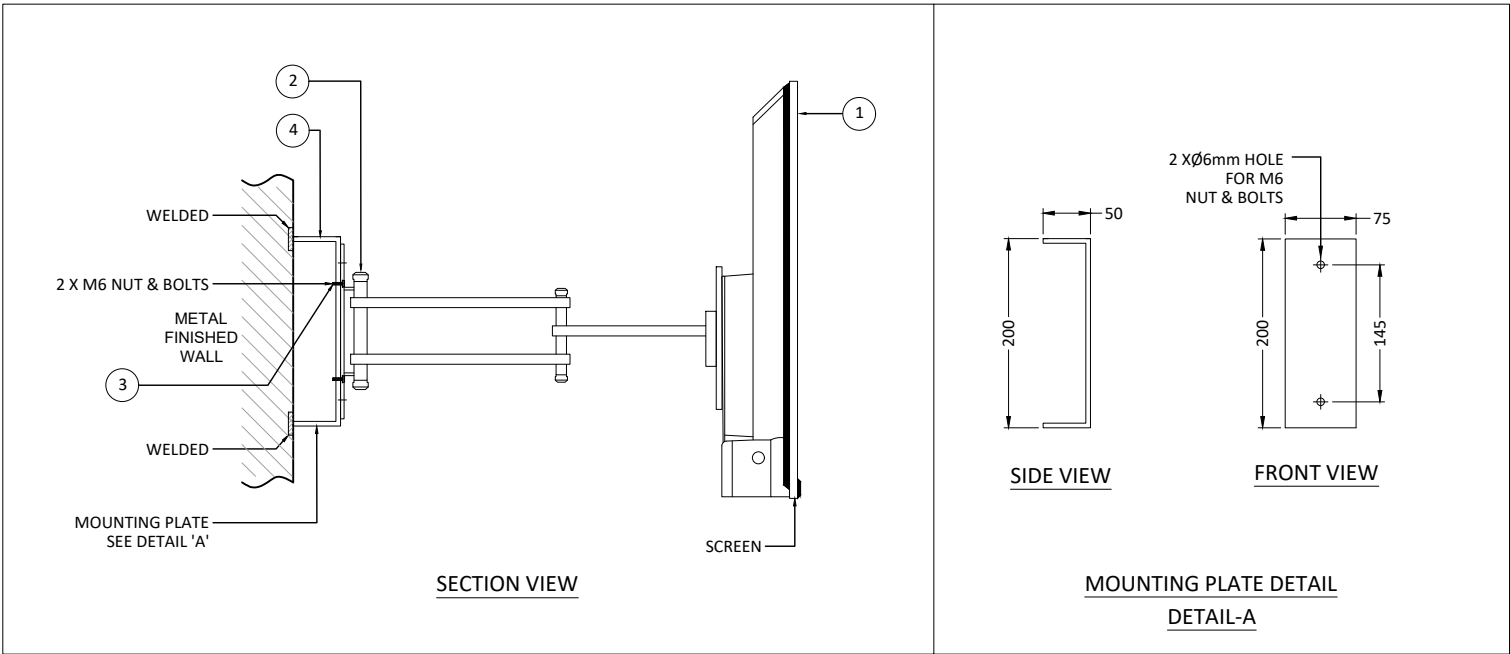
SIDE VIEW



ISOMETRIC VIEW
(TV BRACKET)
SCALE : NTS



TV BRACKET DETAIL



SECTION VIEW

MOUNTING PLATE DETAIL
DETAIL-A

ITEM LIST				
ITEM NO.	DESCRIPTION	MANUFACTURER	SUPPLIED BY	QTY
1	32" TV MONITOR 32HFL3014/12	PHILIPS	VIVO	01
2	SUPTEK TV WALL MOUNT SWIVEL TILT ROTATION FULL MOTION ADJUSTABLE	SUPTEK	VIVO	01
3	M6 BOLT 2 PCS, WASHER 2 PCS NUT 2 PCS	-	VIVO	01
4	12mm THK MS PLATE	-	-	01

REVISIONS

REV.	DATE	DESCRIPTION	DRWN	CHKD	APPD	CLIENT
0	23.12.2024	ISSUED FOR CONSTRUCTION	VM	HS	TN	
A	29.10.2024	ISSUED FOR REVIEW	VM	HS	TN	

COMPANY:



CONTRACTOR:



VENDOR:



TP : +65 6826 2525
Head Office : 21 Toh Guan Rd E, #06-06A, Singapore 608609
Web : vivoasia.com
E-mail : contract@vivoasia.com

PROJECT:

OKWOK-FPSO
TOPSIDES PRODUCTION FACILITIES

LOCATION:

OKWOK FIELD OFFSHORE NIGERIA-OML 67

TITLE:

TELECOM TYPICAL INSTALLATION DRAWING – IPTV SYSTEM

SCALE:

1:8

DRAWING NO.

2024-VIVO-EMEM-TS-TY-012

WCP DRAWING NO.

WCPQKWTP-450-TE-(VAE)DG-014

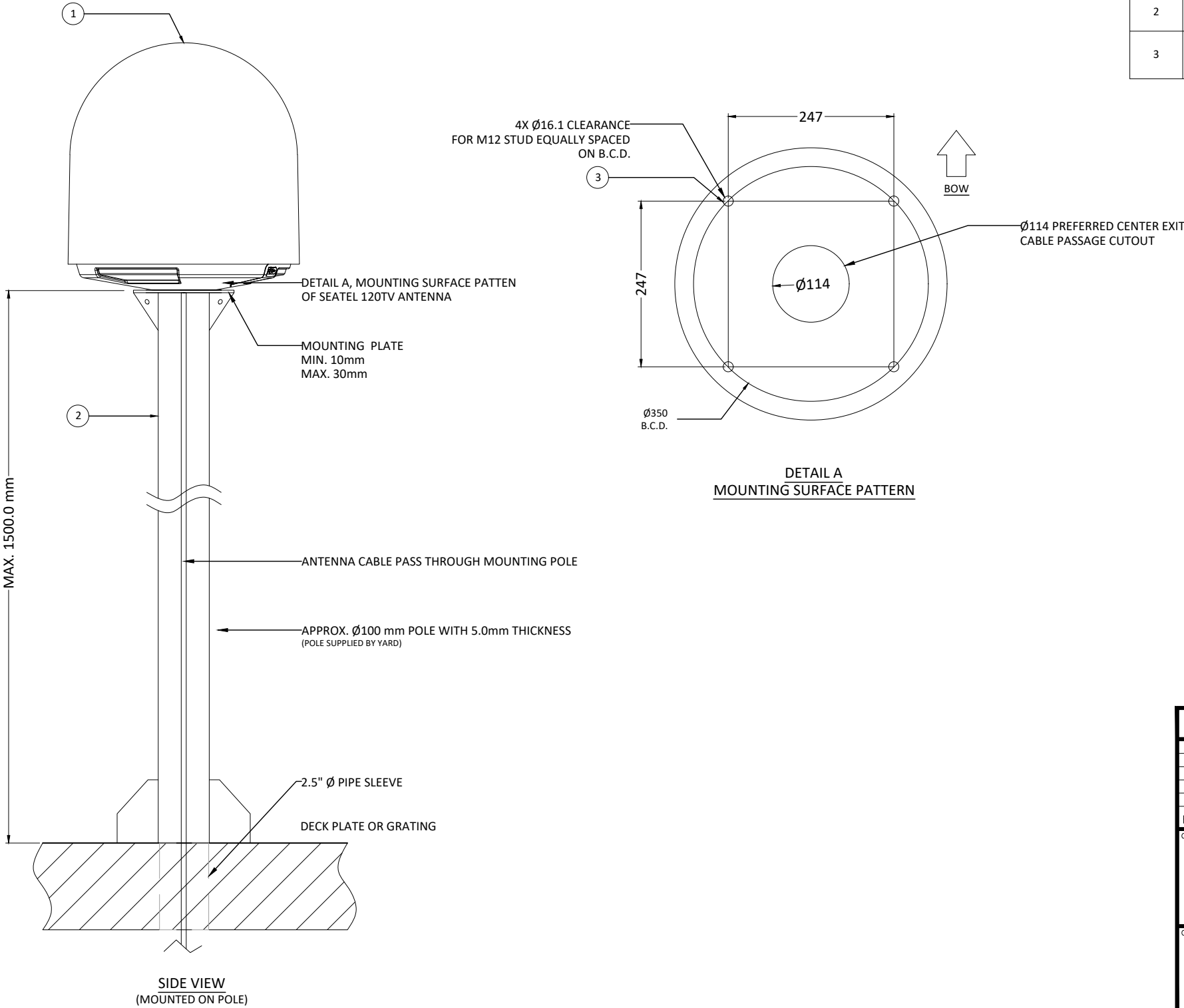
REV.

0

SHEET: 08 of 09

SEATEL 120TV ANTENNA - POLE MOUNTING TYPICAL INSTALLATION DRAWING

ITEM LIST				
ITEM NO.	DESCRIPTION	MANUFACTURER	SUPPLIED BY	QTY
1	SEATEL 120TV ANTENNA	SEATEL	VIVO	01
2	ANTENNA MOUNTING POLE	BY YARD	BY YARD	01
3	M12 STUD NUT 4 PCS WASHER 4 PCS	BY YARD	BY YARD	01



REVISIONS						
0	23.12.2024	ISSUED FOR CONSTRUCTION	VM	HS	TN	
A	29.10.2024	ISSUED FOR REVIEW	VM	HS	TN	
REV.	DATE	DESCRIPTION	DRWN	CHKD	APPD	CLIENT
COMPANY:						
 ORIENTAL OKWOK  ADDAX PETROLEUM						
CONTRACTOR:						
 WORLD CARRIER CORPORATION  HBA Offshore						
VENDOR:						
 VIVO ASIA ENGINEERING & TRADING TP : +65 6826 2525 Head Office : 21 Toh Guan Rd E, #06-06A, Singapore 608609 Web : vivoasia.com E-mail : contract@vivoasia.com						
PROJECT:						
OKWOK-FPSO TOPSIDES PRODUCTION FACILITIES						
LOCATION:						
OKWOK FIELD OFFSHORE NIGERIA-OML 67						
TITLE:						
TELECOM TYPICAL INSTALLATION DRAWING – IPTV SYSTEM						
SCALE:	DRAWING NO.	WCP DRAWING NO.	REV.			
1:30	2024-VIVO-EMEM-TS-TY-012	WCP0KWTP-450-TE-(VAE)DG-014	0			
SHEET:		09 of 09				